Responses of Avian communities to cloud line along an elevation gradient

A Dissertation

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by

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DECLARATION

I A. M. A.S. Adikari, hereby confirm that this dissertation submitted as a partial fulfillment of the degree of Bachelor of Science, represents my own work and is expressed in my own words. External contribution to the work is acknowledged at the point of their use. A full list of the references used has been included. None of the material used in this dissertation violate copyright laws.

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Content Page

List of Figures

List of Tables

Abbreviations

R : Resident

M : Migrant

CR : Critically Endangered

EN : Endangered

VU : Vulnerable

NT : Nearly Threatened

PCA : Principal Component Analyses

GBH : Girth at Breast Hight

ABSTRACT

The elevational gradient is a striking geological phenomenon that avian communities respond. A cloud line is created connecting climatic elements, resulting in a narrow cloud band. Associating with mountains cloud lines are created with the contribution of moisture, wind, and thermal actions. Avian communities of high elevations could respond to these cloud zones. Very few studies attempted to understand the effects of cloud line and elevation on avian communities. To fill this gap, this study was conducted with the aim of identifying avian community responses along the elevation gradient and associated cloud lines. In addition, climatic, vegetation, and habitat variations along the elevation gradient and the avian community responses to them were studied.

Issengard Biosphere Reserve, in Belihuloya (6°42'50.93"N, 80°45'6.39"E) is a submountain forest with nearly a 1 km elevation gradient, was used as the main study site.

Elevational ranges of the avian communities and their activity were also examined from August 2022 to April 2023, along a 480 m to 1420 m gradient. Bird species and abundance were recorded using the 11 horizontal line transects spanning 100m apart along the gradient. Lowest elevation (480m) was at the Samanalawewa Reservoir, the highest elevation (1420m) was the Haagala Peak.

Here, for the first time in Sri Lanka, I showed that the avian communities responding to the cloud line, where the species richness has declined away from the cloud line. The bird species richness, and diversity indicated in a hump-shaped variation along the elevation gradient,

adding evidence to the mid-domain effect, while a higher species evenness value had been maintained throughout the elevation gradient. Canopy cover, tree height, and habitat complexity have indicated similar hump-shaped variation as avian community parameters along the elevation gradient. Therefore, the responses of avian communities along the elevational gradient are supported by vegetation and habitat topography. The leeward side forest patch at the peak has resulted to increase the tree height and contributed to enhancing the faunal and floral community assemblages, somewhat confounding the patterns observed along the elevation gradient. Along the climatic variations, the avian community was found to be more active at around 25 ºC temperature, above 80% relative humidity, less than 25 kLux light intensity, less than 1.2 ms^{-1} wind speed, and non or less than 50% cloud cover. While endemic, migratory, and threatened species of birds exist throughout the elevation gradient, habitat specialist montane species were found prominently from mid to higher elevations. Rapid and frequent climatic variations are present at elevations above 1000 m. Higher percentages of endemic, migratory, montane, and threatened species recorded to occupy at higher elevations, indicates the need for targeted conservation activity at upper elevations of the Issengard Biosphere Reserve and adjacent montane forests of the Haagala Forest Reserve.

The occurrence of endemics, montane species, migrants, and threatened species leads to implement the conservation strategies to protect this sensitive ecosystem.

CHAPTER 1- INTRODUCTION

1.1 Avian Communities

A community is defined as an interacting group of different species in a particular location (Enquist et al. 2002). The factors which determine the structure of the community are (a) species available in the location, (b) the number of individuals in each species, (c) interactions among species, and (d) the ability of a community to return to normal after an unprecedented influence (Enquist et al. 2002; Mutshinda et al., 2009). The alteration of biological communities over time is referred to as ecological succession (Hubbell, 2001; Mutshinda et al., 2008; Rafferty, 2020). Different species of a community occupy their ecological niche which comprises of all their interactions with the other members of the community (Horn & Arthur, 1972).

Members in a community are positioned at different levels of the food chain, which are known as trophic levels. The first level is reserved for the primary producers who are photosynthetic plants that convert the sun's energy into nutrients which can be consumed by other members of the community. Herbivores who are positioned in the second trophic level feed on plants, and carnivores who represent the third trophic level feed on herbivores to gain the energy stored in herbivores. Once the individuals of any of these trophic levels die, bacteria, and fungi-like decomposers break them down and release essential nutrients back to ecosystem (Hubbell, 2001)

Individuals of the same species or different species with equal needs and wants could compete with each other in a community (Hubbell, 2001; Rafferty, 2020). Since several species with the same requirement cannot occupy the same niche, the number of similar species in a habitat is limited driving to lessen the competition through evolution (Hubbell, 2001; Rafferty, 2020). Studies by Hogstad (1976) on three Tit species in a coniferous forest showed that expansion of the feeding niche reduces the degree of interspecific competition (Hogstad, 1978). Geographic separation, segregation between habitats and within the habitat are various ways to reduce competition.

1.2 Biogeographical influence on Avian distribution

Biogeography is molded by both disciplines of biology and geography. It comprises the study of patterns of the geographic distribution of organisms and the factors which govern those patterns (Leduc, 2009; Muller-Hohenstein, 2001) such as geography, climatology, ecology, geology, evolution, etc.

Following the demise of non-avian dinosaurs and pterosaurs during the Cretaceous -Paleogene time boundary, most of the modern orders of birds evolved as well as diversified with various adaptations (Cornell et al., 2007; Benton et al., 2014). The diversity of the avifauna is vast in tropical regions near the equator and decreases towards the poles (Gaston, K. J. (2000). This is discussed by evolutionary, ecological, and historical hypotheses. The geographical variation of resource availability causes a decrease in the species carrying capacity with the latitudinal increment (Fischer, 1960). The evolutionary hypothesis implies that speciation, extinction, and immigration are different among different latitudes (Mittelbach et al., 2007; Rabosky, 2009).

Whilst historical hypotheses believes more diversity is seen with older ecosystems since the availability of ample amount of time for species accumulation and age of the ecosystem with less repeated recolonization at low latitudes causes high species diversity. (Cornell et al., 2007; Santacruz & Weir, 2016).

Formulation of the species richness in a particular area happens through dispersal, extinction, and speciation. Speciation begins with the accumulation of genetic divergences among populations resulting in reproductive isolation among them (Gavrilets, 2004). Allopatric speciation is established by the geographic separation of populations with no contact. Among populations, speciation with ongoing gene flow is called parapatric speciation (Coyne, 1992). In addition, a sympatric variation which is rare in birds occurre through the evolution of reproductive isolation without geographical separation (Bellemain & Ricklefs, 2008) (Sætre, 2008).

The split of a species due to a geographical barrier such as the formation of oceans by tectonic rifts or ancient incidents like montane upliftment is defined as vicariance (Lomolino et al., 2011). This leads to allopatric speciation forming identical biogeographic patterns among them. The dispersal ability of bird groups across the geographical barriers causes accumulation of genetic differences among populations (Schweizer & Liu, 2018).

Different bird species inhabit diverse types of habitats. In order to make the maximum use of their habitat birds have anatomical adaptations like beaks, feet, and plumage patterns, physiological adaptations such as salt excretion, and behavioral adaptations like migration and diving. These physiological adaptations facilitate them to exploit extreme environmental conditions like cold regions and deserts. Anatomical adaptations help them to consume ample amounts of their food. Apart from that, the plumage pattern supports birds to blend with nature and avoid being detected by predators (Eduardo et al, 2010).

1.2.1 Elevation

Elevation is defined using the sea level as the reference point, as the vertical distance from sea level to a point on the land surface. Altitude is the vertical distance between an object like a bird or aircraft and a reference point or a stratum where the object is not always in direct contact with (Mcvicar & Ko, 2013)

1.2.1.1 Impact of elevation on avian communities

Species richness is usually higher at low elevations, which declines with increasing elevation (Adolfo & Navarro, 2015). In addition the species assemblages change at higher elevations (Jankowski et al., 2009). However, when a continuum of elevation from the sea level is considered, the number of species become highest at mid-elevations because of the mixing effect of species from both low and high elevation assemblages (Kozak & Wiens, 2010). Thus, the diversity of many clades of birds show their peak in low to midland tropical areas. According to one of the studies of Jankowski et al (2009), bird species richness declines continuously above 800m a.s.l. Furthermore, the tree growth rates as well as fruit crop production has been recorded to decrease with the increasing elevation (Girardin et al., 2010). Tropical montane birds show narrow elevation distributions driven by various abiotic and biotic factors/processes.

In birds along the elevation gradient, closely related species have been reported to constrain their rangers due to interspecific competition, while species show aggressive territorial behaviors as a response to congeners' songs at such replacement zones. Jankowski et al. (2010) have shown that, Neotropical birds are restricted to narrow ranges, due to the interspecific competitive interactions. Due to the warming of montane landscapes, the upslope range is expanded by generalist competitors, therefore high elevation specialist species are forced into smaller mountaintop habitats (Jankowski et al., 2010).

1.2.2 Habitat

Habitat is a geographical area used by a species to fulfill its one or more essential requirements permanently or temporarily such as water, food, shelter, mates, and ideal conditions for reproduction. Birds have dispersed all over the continents acquiring adaptations in almost every habitat. In distribution patterns sizable variation can be determined in different species (R. MacArthur et al., 1966), spreading out through endemics restricted to a single oceanic island or a specific habitat in a small geographic area to species with cosmopolitan distribution all over the continents.

Even though none of the birds are completely cosmopolitan, Peregrine Falcon (*Falco peregrinus*) breeds across all continents instead of Antarctica (del Hoyo and Collar 2014) and the Osprey (*Pandion haliaetus*) breeds on five continents and sixth (South America) is visited as a migrant.

Endemic species are restricted to a small geographic area or a small oceanic island such as the Smabre Rock Chat which is restricted to rocky slopes with some bushes and grass in the Upper Awash Valley in Ethiopia (Collar and Sharpe 2017).

1.2.2.1 Impact of habitat on avian communities

Diverse avifaunal communities can be found among different levels of habitat subdivisions (R. MacArthur et al., 1966). Among those habitats, the terrestrial habitats can be subdivided into forest and non-forest habitats, while forest habitats contain trees with a canopy where several strata are found. On the other hand, non-forest habitats are like open scrub to grassland and human-constructed habitats. Aquatic habitats comprise water bodies where seasonal or permanent covering by water can be seen. Therefore, these could be categorized as seasonal or perennial aquatic habitats (Fernando & Herath, 2021). Birds are associated with their preferred habitat types and therefore, a subtle change in their habitat cause variations in the composition of birds in that particular area (Chettri et al., 2001). The dynamic movements of the birds in different habitats can be monitored with seasonal changes as well (Green and Catterall, 1998). The habitat parameters and habitat fragmentation affect the distribution of bird communities and they reflect the inter-specific dynamics and population trends related to the habitat (O'Connell et al., 2000). Different kinds of bird communities can be found in different habitats such as dense forests (Flycatchers, Pigeons), grasslands (Cisticolas), shrubs (Wablers), paddy fields (Pipits), streams (Kingfishes), and mountain ranges (Raptors). Forests with a diversity of habitat conditions facilitate a greater variety of bird communities. When shifting the land cover composition from medium to poor ecological conditions as forest to non-forest, bird communities also change coincidently (O'Connell et al., 2000). The study done by Chettri et al. (2001) at Yuksam-Dzongri trekking corridor showed that the heterogeneous habitat conditions posed the diverse bird species.

1.2.3 Vegetation

The species richness of the trees decreases along with the increasing elevation (Jankowski et al., 2013). Vegetation creates space for terrestrial habitats and a path for habitat selection (Lee & Rotenberry, 2005). Nevertheless, vegetation provides food and substrate for foraging and shelter (Robinson & Holmes, 1984). The composition and the structure of the vegetation change with the elevation (Grubb et al., 1963). Therefore, along the gradient, the ranges of some birds limit by the shifting of vegetation associated with habitats (Jankowski et al., 2009).

1.2.3.1 Impact of vegetation on avian communities

When foraging guilds of birds are concerned, the structure of the vegetation plays a big role in the habitat selection of insectivores and omnivores and has less importance (Jankowski et al., 2013). Insectivores birds who have species-specific, inflexible, and time-consuming foraging behaviors to find out cryptic prey, are affected by reduced productivity (Naoki, 2007).

At higher elevations, increasing foraging area to compensate for lower productivity when searching for food is easier for frugivores, granivores, and trap-lining nectivores when considering the energy perspective (Stiles, 1975). The avian species richness and diversity are higher in an open canopy areas than in a closed canopy forest interior (Chettri et al., 2001). To understand the biotic interactions on the distribution of bird species, studying the relationship between birds and vegetation is important (Wiens, 1989). On the other hand, vegetation relies on some birds since they support pollination and fruit dispersal (Schaik et al., 1993). Tree species composition is a significant predictor for avifaunal foraging guilds, especially for frugivores, granivores, and nectivores and there is less significance in vegetation structure compared to the tree species composition (Jankowski et al., 2013). In temperate regions, there is a strong correspondence between the structure and species composition of vegetation and avian assemblages (Lee & Rotenberry, 2005; Fleish-mann & Mac Nally, 2006).

1.2.4 Weather patterns

Weather conditions play an important role in the variation of avian communities in a particular region and their distribution (Mainwaring et al., 2021). Montane birds are generally exposed to extreme weather variations which could lead to reduced survival, altitudinal migration, and reproduction (Morton et al.1972; Stephonson et al. 2011). Temperature, wind speed, and rainfall affect birds directly through heat gain and loss (Root 1988; Petit and Vezina 2014). In contrast, indirect effects on nestling development (Rodriguez and Barba 2016), food availability (Gass and Lertzman 1980; Boyle et al. 2010), or risk of predation (Reyes-Arriagada et al. 2015). In response to adverse weather conditions montane birds generally move to lower elevations (Sereinghaus et al. 2010). Many montane bird species breed from late March to September, whilst summer monsoon and severe tropical storms could affect them (Shiao et al. 2015).

1.2.4.1 Rain

In temperate regions rain comes as a drizzle and does not maintain the same intensity. Temperate birds need to escape wetting more than tropical birds of the colder conditions and huge scarcity of food in winter. Apart from that in the tropics, the physical effect of rain should be considered due to the volume and intensity of the rain can cause to washing away of nests, young, and eggs (Kennedy, 1970a). The rain causes to inhibit song as well (Kennedy, 1970a).

1.2.4.1.1 Impact of rain on avian communities

During the rain, birds inhibit flight and song but do sheltering (Rowan, 1967). Once a bird would catch up to the rain large heat loss takes place from the body. Usually, air is trapped in underwings and between the feathers of birds. But filtering those areas with water causes large heat loss since the conductivity of water is higher. Furthermore, evaporating the water from wet plumage also leads to heat loss, which could result in the death of the bird from hypothermia (Nye, 1964). Thus, most of the birds' plumage is temporarily rainproof. But the ability of shedding water declines with the increasing exposure time to rain. Dr. C.H. Fry who has recorded large number of Red-throated Bee-eaters (*Merops bulockii*) shelter in their burrows during heavy rain, considers overcast weather preceding rainstorms triggers birds to take cover (Kennedy, 1970a). Swifts shelter in the nests during rain, and if heavy fall comes most of the day, they may not emerge (Lack and Lack 1952). In addition, they avoid local storms by flying in front of and away from them. Nonetheless, they have been seen in torrential rain, soaring under thunderclouds (Pounds 1947). By rainfall, migration is lessened, and also widespread and continuous rain has an inhibiting effect on migration (Parslow 1969). Reed

Warblers cease singing and Purple Heron stop calling with the start of a thunderstorm. Redback Shrikes (*Lanius collurio*) also stop singing in rain (Kennedy, 1970b)

1.2.4.2 Wind

Wind patterns have a well-defined impact on the distribution of low and high-pressure centers (Liechti, 2006). Winds get more varied in the northern hemisphere than in the southern. Wind speeds are usually higher above the sea than above the land and generally, the wind does speed with the increment of altitude. With time and space small scale winds are highly variable, among mid-latitudes of the north and south where high-pressure and low-pressure centers pass often (Liechti, 2006). To determine wind aloft, a bird should be able to recognize its direction and speed concerning the ground, and then it keeps into a relationship with an expected direction and speed in wind (Liechti, 2006)

1.2.4.2.1 Impact of wind on avian communities

When the wind rises terrestrial passerines stop feeding and become restless and make for cover. Wind speed can be halved avian speed or doubled easily, and birds have the same order of capacity as the birds' airspeed. Lateral wind float should be bared immediately by heading into the wind. If not the flight route of birds will be shaped by the wind conditions. Therefore, migratory birds should know to detect the varying wind forces and react accordingly. Along the migratory routes, the potential displacement a bird could experience can change considerably. Depending on the time and altitude selected for migration, there is a possibility of encountering favorable winds. Depending on the birds' orientation skills and based on their common power consumption and preferred flight direction, birds have an idea of their airspeed

(Wiltchko and Wiltschko 2003). In wings, the mechanoreceptors near or at the feather follicles are sensitive to the magnitude of airflow over the wings. Thus, when the bird is sitting on the ground these mechanoreceptors provide information about wind speed (Brown & Fedde, 1993).

1.2.4.3 Sun

Sun is the source that gives light. The amount of exposure to the light of the sun per day and the angle of sunlight that encounters the region are important factors to determine the light intensity and temperature available in a particular region.

1.2.4.3.1 Impact of the sun on avian communities

Although there are nocturnal birds in the avian community who expose to the environment for active behaviors at night when sunlight is not available, most of the birds start their day with the sunrise. With the dawn cores, they break the silence of nature, and foraging, territorial, and courtship behaviors are started. While they are active in the diurnal time period, most of the birds are highly active mostly in the dawn and dusk. Some birds can tolerate high temperatures whereas some can tolerate cold temperatures. Lower temperatures contain a negative effect on species richness and bird density and this is stronger in highest-elevation habitats (Walther et al., 2017).

1.2.5 Migration

Migration defines the seasonal movement of birds from one geographic location to another. We can find both latitudinal migration and altitudinal migration of birds. Some migrations reoccur year after year regularly. Each species or group migrate at a particular time of the year or a particular time of the day. Altitudinal migration is a short-distance migration that shows seasonal elevational movements (Hsiung et al., 2018). The migration facilitates birds to seek favorable conditions like ample food and adequate space to live. In addition, birds require a certain amount of territory to carry on their reproductive duties. Long-distance migration enables birds to enjoy the summer at northern latitudes and avoid the hardness of winter. Birds' bodies have numerous adaptations to facilitate migration such as wings, hollow bones, tail, and internal air sacs (Lincoln, 1979)

1.2.5.1 Impact of migration on avian communities

During the annual migration cycle, long-distance migrants and pelagic birds visit a substantial part of the world. This latitudinal migration may have been influenced by both ecological and evolutionary processes (Jablonski et al., 2017). The tendency of avian communities to migrate during the winter is not usually due to seasonal low temperatures. The habituation or disappearance of insects for insect feeders and the mantel of snow or ice preventing the access to seeds and foods on the ground for frugivores and granivores, causes depletion of food. Apart from that, shortened hours of daylight, limits the birds to find their food and withstand the cold winter requires increased energy to maintain body temperature as well (Lincoln, 1979).

1.3 Cloud line as an important feature for the distribution of montane species

A cloud line refers to a narrow cloud band where individual climatic elements are connected and the line is less than one degree of latitude in width (*Cloud Line - Glossary of Meteorology*, 2012.). Clouds are formed when the invisible water vapor in the air condenses into water droplets or crystals of ice which are visible. This parcel of air must be saturated with water vapor for clouds to form. This saturation can take place in two methods; i.e either by enhancing the water content in the air, or by cooling the air, until it reaches its dew point (*Cloud Development*, 2021.).

1.3.1 Impact of cloud line on avian communities

Avian compositional change is highly correlated with moisture which causes to the preparation of the cloud line (Jankowski et al., 2009). Since the environmental conditions above and below the cloud line change, avian communities could also vary with the capability of finding requirements and adaptations. From the Jankowski's (2009) study in a tropical, montane landscape, it was found that zone of rapid transition segregating a distinct community of cloud forest from the rainshadow forest. And on the Caribbean slope shallower moisture gradient was estimated to result in lower beta diversity in the avian community (Jankowski et al., 2009).

1.4 Avian diversity of Sri Lanka

Even though Sri Lanka is a comparatively small island, it holds a very high biodiversity. Sri Lankan avifauna belongs to 23 orders, 89 families, and 266 genera comprising 522 species (Seneviratne and Dayananda, 2021). Of the 244 breeding residents, 34 are endemic to Sri Lanka. Of these 244 species 81 species were found to be threatened as 19 are Critically Endangered (CR), 48 are Endangered (EN) and 14 are Vulnerable (VU) (Seneviratne and Dayananda , 2021). Furthermore, 31 species were found to be Nearly Threatened (NT) while two species, the Glossy Ibis (*Plegadis falcinellus*) and the Blue-eared Kingfisher (*Alcedo meninting*) are kept in the Data Deficient (DD) category, and the rest of 130 species are placed as Least Concern (LC) (Seneviratne and Dayananda, 2021). Of the 19 Critically Endangered species five are found as breeding residents whilst the other 14 species are breeding residents supplemented by migrant populations. In summary, 75% of the breeding residents have been placed in the threatened categories, while 20 of the endemic 34 species are threatened (59%) (c). Nevertheless, the endemicity is 14% of the breeding resident birds. It depicts that the endemicity is relatively high. Finally, Sri Lanka harbours a comparatively high global population for 87% of its bird species, that have therefore been placed as the Least Concerned (Seneviratne and Dayananda, 2021).

A variety of avifaunal habitat types can be found in Sri Lanka, including forest habitats such as wet lowland forests, tropical montane forests, tropical dry mixed evergreen forests/dry monsoon forests, tropical thorn scrubs, non-forest habitats like grasslands, savanna, and urban areas, aquatic habitats such as wetlands, sea-coastal and pelagic waters, where birds can occupy (Fernando and Herath, 2021).

Since Sri Lanka is located between Northern latitudes 5’ and 10’, the northern part of the intertropical convergence zone inside the tropical belt, it obtains high solar radiation throughout the year and monsoonal rainfall leading to great biomass and high avifaunal richness of 522. Sri Lanka as a continental land-bridge island by the colonization of mainland India over the long evolutionary history of the island’s avifauna, it enriched huge bird species richness. In addition, as the southernmost landmass in the Indian Ocean, longitudinally until Antarctica, Sri Lanka has become the final destination of northern hemisphere for 179 migratory birds, using the Central Asia Flyway (Weerakoon & Gunawardena 2012 ; Kotagama

& Ratnavira 2017; Perera, 2021). As Sri Lanka contains approximately 1629 km of coastline surrounded by the vast Indian ocean, a remarkable richness of pelagic sea birds can be found (Kotagama & De Silva 2006; De Silva 2011; Seneviratne et al. 2015; Panagoda et al. 2020) With the monsoon periods and year-round food supply, the tropical aseasonal climate of Sri Lanka allows several breeding times a year for resident birds (Henry 1955; Legge 1983; Kotagama & Ratnavira 2017).

With the influence of southwestern and northeast monsoons, Sri Lanka has been divided into three climatic and six bioclimatic zones (Wijesinghe et al. 1993). Isolated species and populations promoting speciation resulted with climatic filtering. The diverse climate, three marked peneplains in the island's topography, and various types of soils found in different areas have led to various vegetation types on the island serving an array of niches for an extremely diverse avifauna (Ashton & Gunatilleke, 1987; Perera, 2021).

1.4.1 Avifaunal zones of Sri Lanka

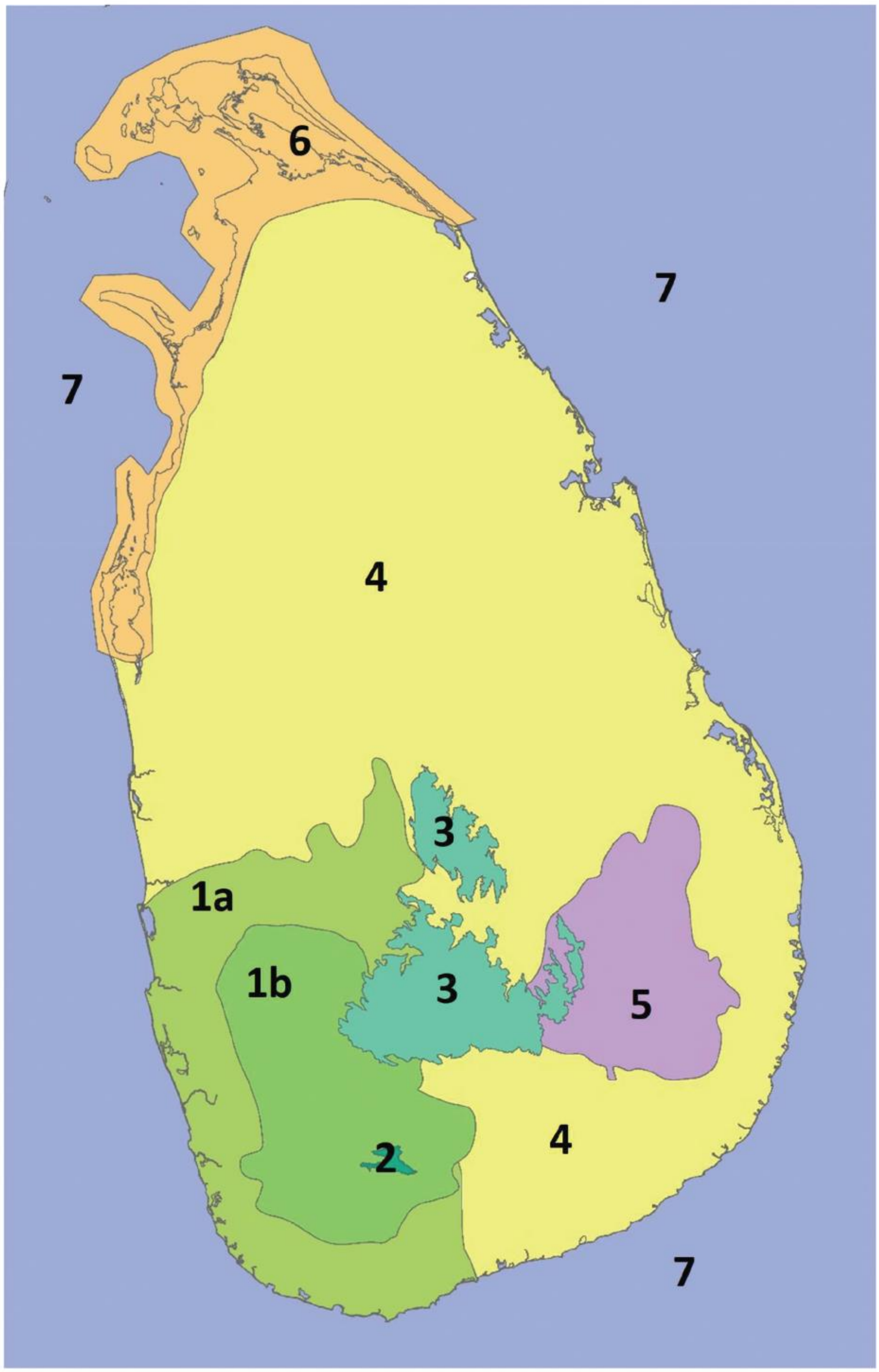

Figure 1.1: 1a – Wet Lowlands, 1b – Wet Lid-hills, 2-Rakwana Hills, 3-Highlands, 4- Dry and Intermediate, 5- Uva, 6- Palk Bay Coastal, 7-Marine (Perera, 2021).

The avifaunal zonation is conducted based on climatic and topographic variations. These zones were determined, using the published distributional data (Kotagama 1986). In the future, using GPS coordinates and GIS applications more descriptive distribution patterns can be recognized.

1.4.2 Probable reasons for the zonation of birds in Sri Lanka

This zonation supports recognizing the avian distribution of Sri Lanka along the climatic and topographic variation. It leads to proper conservation planning for birds and to draw extreme attention to the bird species in need of conservation, helping to carry out proper management of avifauna.

1.5 Current knowledge of avian community responses

Alpha diversity is the local diversity of a location whereas gamma diversity refers to the regional diversity. Beta diversity is the diversity among habitats on a landscape. When the alpha diversity is a small fraction of the gamma diversity, the beta diversity shows a substantial increment (Jankowski et al., 2009). Hence, above avian diversity indices provides a good indicator to understand the avian community structure.

In the Tilaran mountains where a strong moisture gradient exists, high beta diversity has been revealed with the patterns of avian species distributions between 1000m and 1700m elevation (Jankowski et al., 2009). In contrast on the Caribbean slope with low moisture gradient was predicted to result in low beta diversity (Jankowski et al., 2009). Hence, a low speed of compositional change and more continuous species turnover was found. The beta diversity of the tropical mountainside is twice that in the temperate mountainside (Jankowski et al., 2009). Change in species composition is largely correlated with the epiphytic cover and the moisture. In the Montane landscape as a strong ecological force, high beta diversity is generated due to the habitat specialization of birds. Therefore, relatively reasonable zones of composition could be recognized, instead of rapid rate of species turnover (Jankowski et al., 2010).

The species richness of Latin America is six times higher than North America. But the multiplier on the local scale is considerably less. Thus the tropical beta diversity is expected to be twice than that of the temperate landscapes (Macarthur, 1969). In the tropics, beta diversity is high which eases the conservation strategies of that land as the tendency of climatic changes would threaten altering the species composition of the community among habitat specialists.

Narrower environment tolerance appears due to limited distribution along environmental gradients coupled with the small geographical ranges. Within climatic zones or habitat types, diverse faunas are harbored and large species turnover between habitats and zones can be found in tropical montane landscapes. Along environmental gradients, high beta diversity is reflected in habitat specialization by the elemental species (Jankowski et al., 2009). Quantification of beta diversity along environmental gradients could brief strategies to conserve the biological diversity of a landscape. In a high beta diversity context, even little effort to protect areas could embody new species whilst enriching viability of barely adapted local populations. Pattern diversity which refers to the spatial variation in species composition within habitat types, though seldom measured, is of much significance as it indicates the area required to protect species in their habitat types. Species occupied in areas with higher pattern diversity have patchy local distribution and require large areas for viable populations ('pattern diversity' *see* Magurran 2004).

Beta diversity is usually expressed as the ratio of gamma diversity (regional) and alpha species richness. This ratio expresses either habitat specialization among species or the degree to which competitive interactions may have generated saturation of communities (Caley & Schluter, 1998). In analysis at arbitrarily nested analysis scales, the beta diversity is defined as a variety of cells in a grid (Lennon et al., 2001). This analysis assists in locating areas of compositional uniqueness or high diversity but not suit to reveal the biological mechanism. Even though the beta diversity is mostly measured as species turnover without considering the relative abundance of species, the involvement of the relative abundance of the species unfolds more biologically informative assessment, whilst abundances alter greatly (Koleff et al., 2003).

Some analyses of compositional variation of animal communities along altitudinal gradients have been done and most of them have shown gradual compositional turnover with altitude (Lake & Oiselle, 2000) (Young et al., 1998) (Adolfo & Navarro, 1992) (Terborgh & Weske, 1975) (Terborgh et al., 1990) (Lieberman et al., 1996) (Brehm et al., 2003).

Increased bird turnover may be given from the high diversity of tree communities, combined with specialization of environmental conditions such as temperature, rainfall, and indirect effect on the geological substrate and soil nutrient cycling (Aiba & Kitayama, 1999).

According to Jankowski et al., (2012) at Eastern slope of the Andes and adjacent to Manu National Park study has shown, 1000 – 1250 m a.s.l. elevation peaks of birds and tree species turnover are notable. There is a 10% greater dissimilarity between the adjacent plots compared with the values between adjacent plots above and below this elevation (Jankowski et al., 2012). The 2nd elevated turnover occurs between 1750 and 2000m for trees in the cloud forest, while for birds, it is 2000 and 2250m (Jankowski et al., 2012) . These differences in species turnover are influenced by soil differences for trees and habitat structure for birds. In addition, the presence of persistent clouds at these elevations is highly influenced by birds and trees even during the dry season. The lower range boundaries of high-elevation species have concentrated in these cloud zones (Jankowski et al., 2010). It proves that species respond to the position of this cloud base or else the associated changes in vegetation (Jankowski et al., 2009). The 3rd higher turnover of birds occurs at the tree line at 3400m elevation, and above this elevation, it is puna grassland, indicating a rapid vegetational transition (Jankowski et al., 2012). For the bird dissimilarity, the contribution of nestedness was higher above 2100m. Therefore, to a greater extent, the variation of species composition by differences in richness can be seen at high elevations. Multiple regression on distance matrices has shown that for the bird community composition vegetation structure, tree species composition, and elevation composition are significant predictors (Jankowski et al., 2013).

1.6 Objectives

Main objective

To understand the contribution of the elevation and associated cloud lines for the segregation of the avian community structure.

Sub objectives

1. To study climatic, vegetation and habitat parameters along the elevation gradient in the southeastern slope of the central highlands.

2. To study the bird community variation in response to the climatic, vegetation and habitat parameters along the elevation gradient, and

3. To identify elevation ranges for each bird species.

4. To segregate avian communities with respect to elevation and habitat

5. To quantify specific avian communities such as endemics, montane, and threatened species along the elevation gradient to implement conservation strategies.

1.7 Research questions

1. How do avian communities respond to the placement of cloud line along the elevation gradient?

2. How avian communities respond to the elevation gradient?

3. How do avian communities respond to variations of the climatic, habitat and vegetation parameters along the elevation gradient?

4. What are the ranges of the species along the elevation gradient?

5. How do avian communities segregate along the elevation gradient?

6. How do endemic, montane, and threatened species distribute along the elevation gradient?

CHAPTER 2: METHODOLOGY

2.1 Study site

Biogeographical transitional zones are extremely important since they act as ecotones with overlaps of geographical ranges occupying climatic or elevational zones on either side of the region (Morrone, 2020). Belihuloya, which is a significant transitional zone marginalizing the climatic wet zone and intermediate zone and also a geographical hill and low country, are located in the south-central intermediate zone (Wijesooriya et al., 2022). In addition, Belihuloya is a middle topographic peneplain, molded by environmental features and physical characteristics of both wet and dry climates and also the lowlands (Perera & Fernando in press). A privately owned protected area, Issengard Biosphere Reserve (Figure 2.1) is located in Impulpe, Belihuloya, in the Ratnapura District of the Sabaragamuwa province of Sri Lanka (6°42'40.30"N - 6°43'2.54"N, 80°45'7.51"E - 80°44'59.79"E). Centralizing the Issengard Biosphere Reserve, the whole study area (Figure 2.2) spread from 480 m to 1420 m elevation keeping around 1 km elevation gradient. The basis of the study area is laid along the Samanalawewa reservoir. The topmost point of the study area is located at the summit of Haagala peak which is surrounded by several mountain ranges.

This study area covers several habitat types, including dense forests, grasslands, streams, and rocks. Thus, the total study area touches the lowest elevation from the Samanalawewa reservoir, where birds can be encountered, and the highest elevation of the Haagala peak long the same valley.

Figure 2.1: The dropped pin has been showed Issengard Biosphere Reserve in the Sri Lankan map

Figure 2.2: The Study Area

2.2 Site selection

In order to fulfill one of the major objectives, determining the avian community responses along the elevational gradient, the highest range of the elevation was obtained by taking the lowest elevation of the study area as 480 m (MSL) at the Samanalawewa reservoir and the highest elevation as 1420 m (MSL) at the summit of the Haagala peak.

2.2.1 Setting up transects

From the elevation of 480 m to 1420 m, the total elevational gradient was examined establishing 10 line transects keeping around 100 m elevational gaps among each horizontal line transect. The elevation of the land was obtained using a Garmin eTrex waterproof Hiking Global Positioning System (GPS) receiver (screen size 2.2 inches, map type – city tour, battery life – 25 hours, resolution – 480 x 272) which has serial connectivity technology, and continuously tracks and uses up to 12 satellites (Figure 2.3).

A line transect was 50 m in length, and 25 m to either side of the line transect was accounted for in the examination. At the starting and the ending point 25 m radius, the half circle was taken without overlapping the previously mentioned line transect area. Thus, at each sampling elevation an area of 4462.5 m^2 was covered (Figure 2.4). Most of the transects are demarcated along forest patches as the main habitat while including several minor habitats. Habitat types masking the sampling locations are mentioned in the table no 2.1. The beginning and the ending points of each line transect are marked by placing a label which includes the details such as transect number, elevation, and stating or ending mark. As an example, T1 – 480 m, -→ denotes that this is the 1st line transect where the elevation is 480 m and the starting point which extends towards the right side or else, a red filled circle depicts the end point. (Figure 2.5 and 2.6)

Figure 2.3: The etrex Geographical Positioning System (GPS)

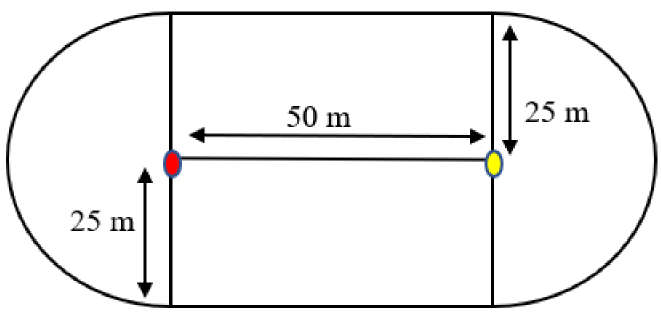

Figure 2.4: The line and point transect layout of a sampling site. Yellow filled circle depicting the starting point of the line transect. Red filled circle depicting the ending point of the line transect.

Figure 2.5: The label of the starting point of the 700 m elevation line transect. T-4 named that this is the 4th transect line. Arrow shows that the T-4 transect line

Figure 2.6: The label of the ending point of the 700 m elevation line transect. Red filled circle represents that it is the ending point of the 50 m line transect of the T-4 line transect.

Total area of a sampling site =	Area of the triangular area where line transect was done	+	Area of the two half circles where point transects were done
=	50 m x 50 m	+	π x 25 m x 25 m
=	4462.5 m^2		

Figure 2.7: Calculation of the area associating one transect

2.3 Data collection

Field work was conducted from August 2022 to April 2023 in the Issengard Biosphere Reserve and the surrounding study site. Each day, at the beginning of the data collection, the elevation of the cloud line was recorded. At each transect, birds were identified while walking along the transect line for 15 minutes. As the environment parameters, humidity, light intensity, wind speed, wind direction, visibility, and cloud cover were measured at each transect. Field work was carried out both at dawn and dusk in the time ranges of 5.45 am to 10.30 am in the dawn and 15.00 pm to 18.30 pm in the dusk. The aforementioned time frames were decided as they were the established time frames that show the highest bird activity.

Figure 2.8: Recording data at 580 m

Figure 2.9: Observing birds at 700 m

2.3.1 Mist netting data

Along the same transects from August 2022 to March 2023, mist netting was done using 6 m, 9 m, and 12 m length mist nets. One transect mist net was laid for about one hour in the same time periods as 5.45 am to 10.30 am in the dawn and 3.00 pm to 5.30 pm in the dusk. Before proceeding to the mist netting, focus was paid to, identifying the bird calls and songs of each transect which were identified at the time. Those birds calls and songs were played back to capture the birds. When specific calls and songs were not occupying common calls like Iora, Chestnut backed Owlet were play backed to gather the hidden birds of the place.

Figure 2.10: Mist netted White-browed Bulbul at 480 m

Figure 2.11: Mist netted Common Iora at 480 m

2.3.2 Camera trap data

The browning trail cameras were set up from 480 m to 1420 m for each elevational transect from August 2022 to April 2023. The camera trap was set to be activated when detecting a combination of both movement and heat. The detecting signal could be a bird, mammal, an insect, or movement of trees. A video is set to be recorded if it is a bird. Instead, when the cause of trigger is another animal or a tree, the audio will be recorded which would contain the bird calls and songs available at the time. The location to set up the camera trap was determined using the signs of highly used paths of the animals, feces, pug marks, and highly grazed areas of the land. Despite the trigger cause can be another animal, transects were used to set up the cameras as they would capture the bird calls and songs along with the other information. The camera trap was attached to a nearby tree of 30 cm to 100 cm above the ground and directed at the paths, streams, or grasslands where maximum number of animals could occupy. The camera settings were standardized over all cameras as follows: Type = Video, Video length = 20s, and Interval = 5s. Each camera was placed for one month at each location. The sensitivity of the sensor was set to low, medium, high according to the location. If the camera was directed to a highly dynamic place with long grasses, trees or streams the sensitivity was set to low. If the place has few disturbances from other effects, then the sensitivity was set as high. The illumination of recordings in poor light conditions as dark, dawn and dusk were assisted by infrared LEDs, capturing black and white recordings. And all the daytime recordings were taken in color. All movies were saved in the .MP4 format on 16 GB, 32 GB, and 64 GB SD cards. Camera trap data for one month was collected from each camera and data were copied onto a computer and stored in unique folders per camera and location. Utilizing the recorded videos, the birds available at each transect were collected.

Figure 2.12: Browning Trail

Figure 2.13: Setting up the trail camera on a tree

Figure 2.14: Set up trail camera on a tree with it's camouflage

Figure 2.15: Resetting the camera, on the same location

2.4 Identifications

Birds available at each transect within the considered 4462.5 m^2 area were identified visually or acoustically listening to their calls and songs. Birds were identified at the species level and the number of each species present was also recorded. The availability of the bird species is specifically related to the type of the vegetation cover of that place. Hence the vegetation cover of each sampling site for each elevation was separately identified.

2.4.1 Species Identification

For the visual identification of birds, observations were done with the naked eye if they were clearly visible at short distance. Most of the time, a pair of binoculars (Nikon 8x40) were used for the visual identification of birds. When identifying a bird visually following criteria were utilized: Shape, size, beak pattern, leg pattern, tail pattern, color, unique features of the body, flying pattern, behaviors, and the habitat which they were occupying. If the species was not recognized at once, a rough sketch was done in the field notebook including all visually observable data, size, place, and the time of the observation. Then it was recognized using the bird guide (Warakagoda et al., 2012) or by obtaining the assistance of an expert. At the same time the bird species were identified from acoustically the calls and songs which could be directly identified within the sampling site. If the calls or songs were not clearly identified, they were recorded by the voice recording app from Samsung Galaxy mobile phone. Those recordings were saved renaming the transect elevation, time, and date. Unfamiliar calls and songs were also identified with the assistance of an expert and checking similar calls and songs from the Xeno-canto website.

2.4.2 Vegetation Identification

Plant species that occupied the transects of the elevational levels were identified along with the respective transect. Since the features of the vegetation at each elevation affects the bird community, vegetation parameters like habitat complexity, tree height, girth at the breast height (GBH) of trees, tree density and canopy cover were also recorded. For each elevation associating the transect site five 10 m x 10 m plots were marked. These vegetation and habitat parameters were recorded for each plot. Average value for each parameter was calculated using the data collected from five plots at each elevation.

2.4.2.1 Habitat complexity

A method of describing changes in vegetation biomass is provided by habitat complexity scores (Coops & Catling, 2000). In comparison to homogenous forests, structurally complex forests provide more diverse conditions since greater variety of microhabitats and trees. To assess the association of bird species richness, abundance, and distributional pattern with habitat complexity in each transect, six habitat features related to the Habitat Complexity index were computed and scored from 0 to 3. For each elevation mean value was taken.Finally, from 1-6 habitat complexity score was given to each elevation. The six conditions which addressed the habitat complexity are the canopy layer, tall shrub layer, low shrub layer, ground layer, logs, fallen branches and rock, and litter layer (Watson et al., 2001).

2.4.2.2 Tree height

To characterize the forest vegetation and to obtain baseline data on the species composition, and forest structure these height measurements are important. At a particular height in the forest canopy some birds and cavity nesters prefer to do nesting and feeding. Therefore, assessing tree height nesting suitability of forests can also be determined (Deyoung, 2022). To measure tree height tangent method can be used. It provides measuring the horizontal distance to the tree and angles from the horizontal to the top and base of the tree (Larjavaara & Muller-Landau, 2013). This can be measured easily using the clinometer. Hence, the tree height of each transect line were measured utilizing the clinometer.

2.4.2.3 Girth at breast height (GBH)

Using measuring tape, GBH was measured for all the trees at each plot. Then only more than 15 cm GBH values were recorded.

2.4.2.4 Tree density

For each plot number of trees with 15 m > GBH, were counted and recorded.

2.4.2.5 Canopy cover

Canopeo mobile app was used to read the canopy cover of the sampling site. Canopeo which is developed as an Automatic Color Threshold (ACT) image analysis tool in the Matlab programming language (Mathworks, Inc., Natick, MA), utilizes color values in the red-green-blue setup. All the pixels in the image are analyzed by Canopeo. The analysis is implemented according to the R/G, B/G ratios and the excess green index selection of pixels for the analysis is implemented. A binary image is the result of the analysis (Liang et al., 2012;) (Azevedo et al., 2012)(Paruelo et al., 2000). The selection criteria or else, the green canopy corresponds to the white pixels whereas, the pixels which did not meet the selection criteria, correspond to the black pixels. Thus, fractional green canopy spreads from no canopy cover as 0 to 100% green canopy cover as 1 (Chen et al., 2010; Richardson et al., 2007). Three readings of the canopy cover from the three places of the sampling site at beginning, middle, and the end were taken, and the mean value was calculated for each site for each sampling session.

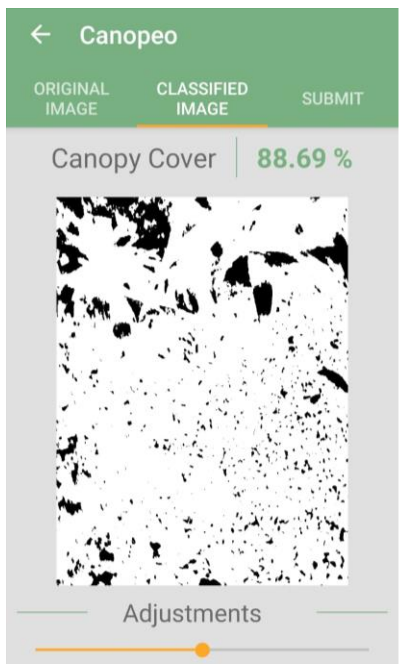

Figure 2.16: Obtained canopy cover data using Canopeo app

2.5 Measuring environmental parameters

The occupying bird species, the number of a particular species, and the number of different kinds of species could change according to the environmental conditions. Hence, to recognize those relationships, the environmental parameters were recoded for each sampling site. Thus, environmental parameters: light intensity, temperature, wind speed, wind direction, humidity, cloud cover, and visibility were measured using specific equipment and methods.

2.5.1 Light intensity

The light condition of each sampling site was recorded utilizing a 1333 Metravi digital lux meter which measures light levels ranging from 0.01 lux – 0.1 Klux / 0.01 fc – 0.01 kfc, repeatedly (high accuracy +/- 5% rdg +/- 0.5 % f.s.) For each sampling site, four light intensity readings were recorded in the field. Finally, the mean value of each sampling site per sampling session was calculated on the excel data sheet.

Figure 2.17: Metravi digital lux meter

2.5.2 Temperature

The temperature of each sampling site was recorded utilizing the reading of the temperature from the Sper Scientific 800015 large display indoor/outdoor thermometer. Four changing temperatures within around 30 seconds were recorded in the field and the mean value was calculated in the excel data sheet.

2.5.3 Humidity

Humidity was recognized by Sper Scientific 800015 humidity monitor, which contains an outdoor sensor that is mounted on a 4-foot cable. When the sensor of the humidity meter was exposed to the environment of the sampling site at the starting point, the humidity is obtained as a percentage. The sensor was kept for around 10 seconds to reset according to the environment. Then, within around 30 seconds, varying readings for the humidity were recorded in the field and their mean value for a sampling site per session was calculated in the excel sheet.

Figure 2.18: Sper scientific 800015 humidity/temperature, indoor/outdoor monitor

2.5.4 Wind speed and wind direction

The wind speed of the sampling site was measured by VA-8020 digital anemometer which had an 80-4000 fpm (0.4-20 m/s) wind velocity range. The direction of the wind was recognized before measuring the wind speed. Then the fan of the anemometer was directed towards the wind. Wind direction was noted using the compass of the Huawei Y9 mobile phone. Wind direction and the speed both were measured at the starting point of the sampling site. The three highest wind velocity values within around 30 seconds were recorded in the field and then the mean wind speed was calculated on the excel sheet.

Figure 2.19 : VA – 8020 digital

2.5.6 Cloud cover

The available cloud cover was measured as a percentage from the three places of the transect line as the beginning, middle, and the end. An 8 cm x 6 cm mirror was divided into four rectangles by demarcating lines marked on the mirror. It was directed toward the sky and the cloud cover percentage of the rectangular parts of the mirror was estimated and the total percentage per place was obtained. Then, the mean value of the three places was calculated.

2.5.7 Visibility

To obtain a value for the visibility, a visibility chart (Figure 2.20) was created including a three-digit numbering system. Then the maximum visibility and the lowest visibility were determined, and the range was converted into the percentage scale. Within the transect line, 20 steps away from the visibility chart, the 1st visible digit trio from the end of the visibility chart was recognized. The adjacent two-digit trios from the 1st visible digit trio along the visibility chart were also read to ensure the visibility starting point.

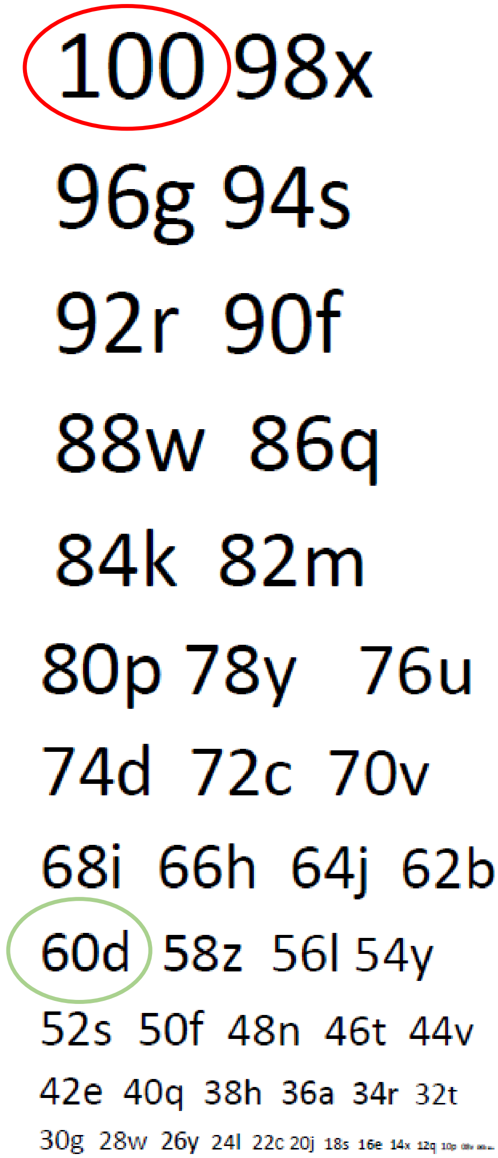

Figure 2.20 : Visibility chart. Light green circled trio is the most visible trio in the highest visibility. Red circled trio is the

2.5.8 Habitat type

The ten elevational sampling sites consist of varying habitat types. Some sampling sites contain more than one habitat type as well.

2.5.9 Habitat and Locality data

Using the Global Positioning System (GPS) receiver, the GPS location of each sampling site and the elevation were recorded. The GPS location of each elevation level and the major and minor habitat types of those sampling sites are mentioned in the table below.

Table 2.1 Details of the ten transects

Transect Name	**Elevation**	**GPS coordinates**	**Major habitat**
T - 1	480 m	6°41'43.58"N 80°45'44.10"E	Forest
T - 2	580 m	6°42'24.52"N 80°45'19.19"E	Forest
T - 3	700 m	6°42'40.30"N 80°45'7.51"E	Forest
T - 4	780 m	6°42'47.39"N 80°45'10.21"E	Forest
T - 5	870 m	6°42'49.64"N	Forest

		80°45'9.81"E	
T - 6	1000 m	6°42'55.76"N 80°45'14.07"E	Forest
T - 7	1100 m	6°43'0.69"N 80°45'10.59"E	Grassland
T - 8	1250 m	6°43'2.54"N 80°44'59.79"E	Forest
T - 9	1360 m	6°43'9.62"N 80°44'50.64"E	Grassland
T – 10	1420 m	6°43'23.45"N 80°44'32.46"E	Forest

2.6 Data Analysis

For the feasibility of further analysis, all the data collected in the field were stored in a Microsoft Excel software spreadsheet.

2.6.1 Avian community indices calculation

Using Microsoft Excel software functions abundance, species richness, diversity and evenness were calculated. The number of individuals per species and per elevation, was tabulated in columns and elevations in rows in the excel spreadsheet.

2.6.2 Methods of calculating community indices

Species richness - This was calculated by counting the number of species.

Abundance – This was calculated by counting the total number of individuals.

Shannon-Weaner Diversity Index – This index was used to measure the species diversity. This depicts the information on commonness and rarity of the species and the evenness of species in the sampling site. To compare the species diversity Shannon Diversity index was calculated using the following formula.

Shannon Diversity Index = H

$$H = -\Sigma \left(\frac{n_i}{n}\right) [\ln(n_i/n)]$$

Where:

n_i = number of individuals of species

n = Total number of species per elevation

n_i/n = the relative abundance of each species; proportion of Shannon Weaner made up of the i^{th} species.

Evenness – This is calculated, by dividing the Shannon Weaner diversity, by the logarithm of a number of taxa.

$E = H/H_{max}$

$H_{max} = \ln(N)$

Where;

E = Evenness

H'= Simpson Diversity Index

Hmax = Maximum diversity possible

N = Number of species

Percentage Species Endemism – This is calculated as follows,

$$\frac{Number\ of\ endemic\ species\ in\ the\ particular\ elevation}{Total\ number\ of\ species\ in\ the\ plot} \text{ x } 100\%$$

2.6.3 Analyze responses of avian communities to the cloud line

During the study period with the presence of the cloud line species richness at each elevation was calculated. To plot the variation with respect to cloud line R studio software was used. Elevation of the cloud line was taken as 0 on the x axis. Species richness is on y axis. Species richness at each elevation was marked as the distance away from the cloud line.

2.6.4 Analyze relationship between avian community response variabilities and elevation gradient

Total species richness, abundance, diversity, evenness were calculated. Statistical analysis was done using R studio software. According to the variations of each response variable, the Spearman or Pearson test was done. Curved plots were plotted, and the smooth dotted lines were obtained.

2.6.5 Analyze species richness variations along the elevation gradient

Using the data collected for 9 months species richness of endemics, threatened species, montane species, and migratory species were calculated. In addition percentage of species endemism, percentage of threatened species were obtained. Bar graphs were plotted to seek the patterns of variation. Variations of types of species richness along the elevation gradient were tested using R studio 4.2.3 software. Considering the patterns of variation Pearson or Spearman test were tested to obtain the relationship. Then, curved dash line graphs were plotted on scatter plots.

2.6.6 Analyze relationships of vegetation and habitat variabilities along the elevation gradient.

Using the ten transects trees girth at breast height (GBH), tree density, canopy cover, and habitat complexity were measures using five plots for each transects. Then, for each elevation average value for all these variables were obtained. Using R studio 4.2.3 software relationship of vegetation and habitat parameters were tested. Pearson and Spearman correlation tests were used and curved dash line graphs were plotted on the scatter plot.

2.6.7 Analyze effect of the climatic parameters on the avian communities

Species richness variation with the climatic parameters such as temperature, relative humidity, wind speed, light intensity, visibility and cloud cover were examined using the data collected throughout the study period. Using R studio 4.2.3 software scatter plots were done between species richness and climatic variabilities separately. With respect to the patterns of scatter plots Pearson or Spearman correlation tests were tested to see the relationship between species richness and climatic parameters.

2.6.8 Analyze avian community response variables variation during the study period

Variation of the Total species richness, species richness of migrants, and species diversity throughout the study period were plotted on bar graphs using R studio 4.2.3 software.

2.6.9 Species' elevation ranges identification

To determine the species elevational ranges, Poynton and Boycott (1996) method was used. The abundance of each species was recorded from a 480 m to 1420 m elevation gradient. When a species was present in a particular transect , it was marked using the symbol "1". The symbol "x" was used to mark the elevations where the species was not recorded but there is a possibility to being them in that elevation since the species have been recorded either side of the elevations. Some elevation ranges were not marked as "x" to combine the elevation ranges because there is a possibility of appearing species at 1420 m leeward side forest patch without appearing at adjacent elevations on the windward side.

Further, the elevational range profile of each species was plotted by using Python statistical software acquiring the minimum and maximum elevational values.

2.6.10 Determination of associations among avian communities along the elevation gradient

In order to explore the affinities of birds from different elevations, Cluster Analysis was performed by using R studio 4.2.3 statistical software.

2.6.11 Further analysis of associations among avian communities along the elevation gradient

To identify the indicator species, Principal Component Analysis (PCA) was performed. This was implemented using the collected data of each species in each sampling site. (Appendix 1)

CHAPTER 3 – RESULTS

From August 2022 to April 2023 along 940 m elevation gradient at Issengard Biosphere Reserve, Belihuloya 97 avian species were recorded by 752 observations. This avifauna comprises of 21 endemic species, 7 montane species, 19 threatened species, and 11 migratory species.

3.1 Avian community responses to the cloud line

Resulted cloud lines throughout the study period were placed on the zero position of the x axis. Species richness presented at each elevation was plotted on y axis as distance away from the cloud line.

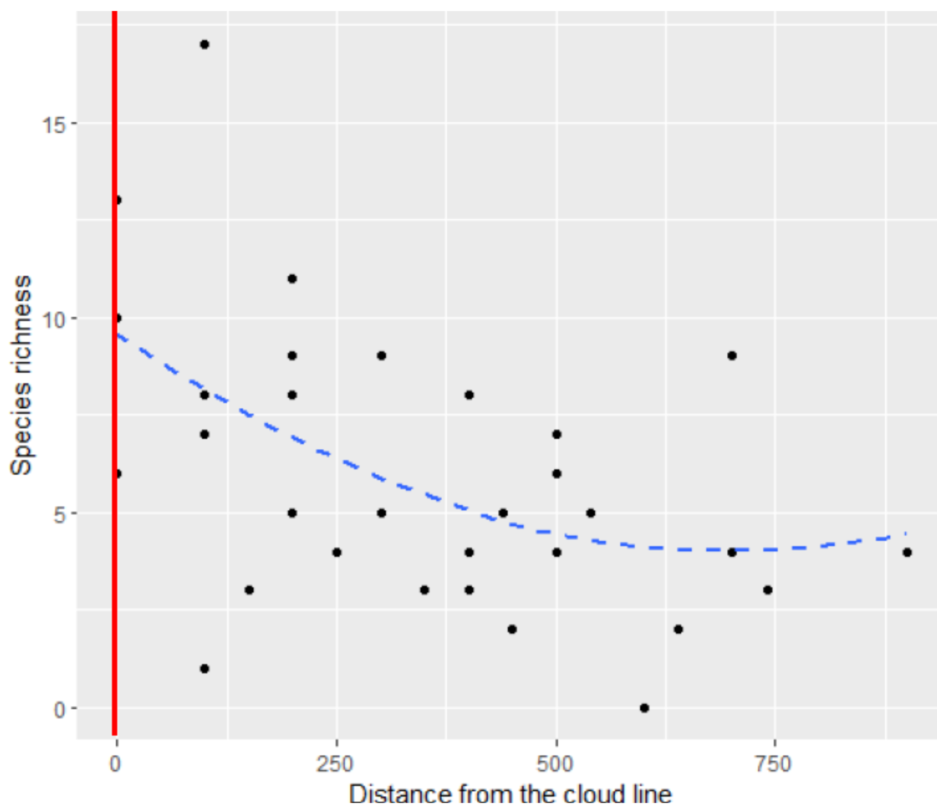

Figure 3.1 – Species richness variation with the presence of cloud line. The red line has indicated the cloud line.

Species richness has declined away from the cloud line. Highest species richness values have indicated at or towards the cloud line.

3.2 Variation of avian community parameters along the elevation gradient

From the data collected throughout the 9 months, for all 10 elevations total species richness, average abundance, Shannon Weiner diversity index, and Shannon Weiner evenness index were calculated. Bar graphs were plotted for all avian community parameters along the elevation gradient.

Table 3.1 – Values of avian community parameters along the elevation gradient

Elevation (m) (MSL)	Main Habitat type	Total Species Richness	Average Abundance	Species Diversity	Species Evenness
480	Forest	41	11.7	3.33208	0.897271
580	Forest	40	12	3.23762	0.87767
700	Forest	47	8.55	3.45185	0.89655
780	Forest	36	4.72	3.35747	0.93692
870	Forest	28	4.19	2.91287	0.874157
1000	Forest	37	7.65	3.21401	0.890081
1100	Grassland	16	4.47	2.54742	0.918788

1250	Forest	29	5.18	3.06604	0.910535
1360	Grassland	12	4.46	2.25109	0.905905
1410	Forest	36	7.87	2.94402	0.821544

Since the main habitat type of the 1100 m and 1360 m elevations are different from other habitat types, these two elevational data have been omitted for some comparisons.

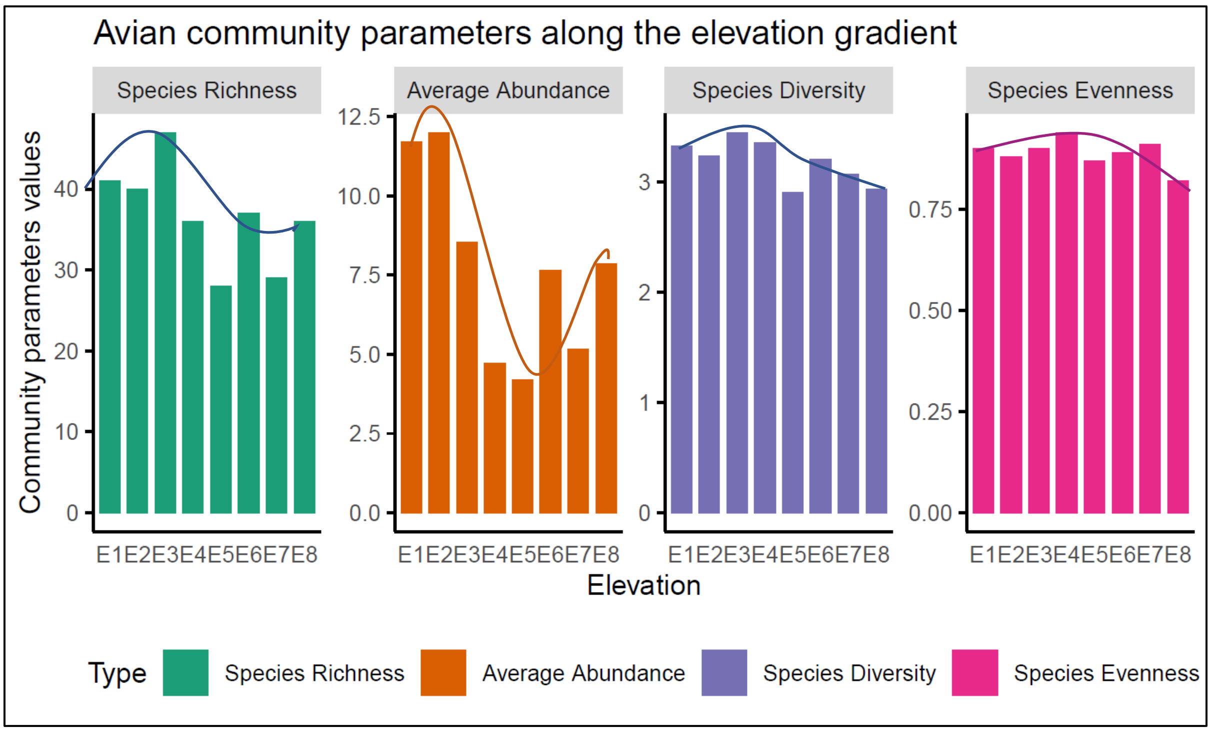

Figure 3.2 – Comparison of the variation patterns of avian community parameters along the elevation gradient. Elevation in meters: E1-480, E2-580, E3-700, E4-780, E5-870, E6-1000, E7-1250, E8-1410

3.2.1 Correlations of avian community response variables along the elevation gradient

Variations of avian community response variables (species richness, average abundance, species diversity and species richness) along the elevation gradient were tested. Considering the variations of the figure 3.2 Spearman or Pearson correlation tests were done to obtain correlation and P values for the relationships. Curved lines were plotted on the scatter plot.

Hypotheses

Hypothesis 1 :	There is a correlation between species richness and elevation gradient.
Hypothesis 2 :	There is a correlation between average abundance and elevation gradient.
Hypothesis 3 :	There is a correlation between species diversity and elevation gradient.
Hypothesis 4 :	There is a correlation between species evenness and elevation gradient.

Table 3.2 Correlation coefficients and P values of avian community response variables along the elevation gradient

Response Variable	Correlation Coefficient	P value
Species Richness	-0.6706707	0.06869
Average abundance	-0.547619	0.171
Species Diversity	-0.6878969	0.05932
Species Evenness	-0.4467224	0.2672

The Spearman correlation test was done for species richness and abundance since the figure 3.1 trend lines show a more curved relationship between species richness and abundance along the elevation gradient. In addition, since the trend line for species diversity and species evenness is less curved and more linear, Pearson correlation was done for species diversity and species evenness along the elevation gradient. These tests have resulted in the four avian community parameter relationships to not be equal to zero but, containing a negative correlation with elevation. Therefore, the alternative hypotheses were accepted. But non of the community variables give a significant linear relationship as the P values are not less than 0.05. Therefore, the curved relationships of the avian community parameters along the elevation gradient were evaluated.

3.2.2 Curved relationship of avian community response variabilities along the elevation gradient

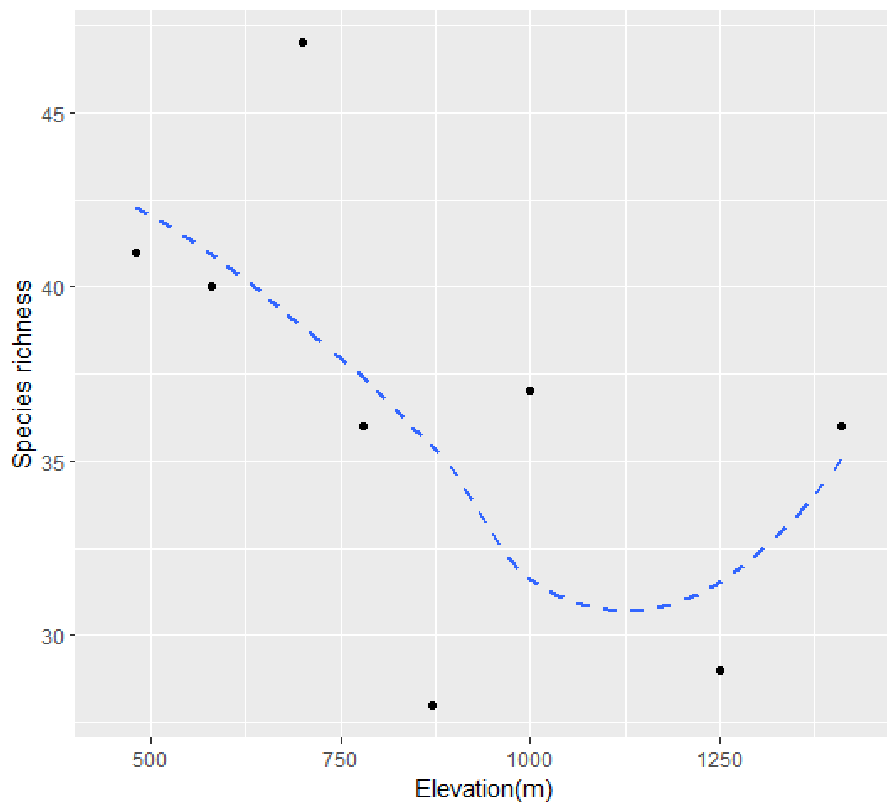

Figure 3.3 – Curved pattern of the Species Richness variation along the elevation gradient

Species Richness is higher at initial elevations, followed by declines at mid-elevations, and finally, at 1420 m, it has shown a little incline again.

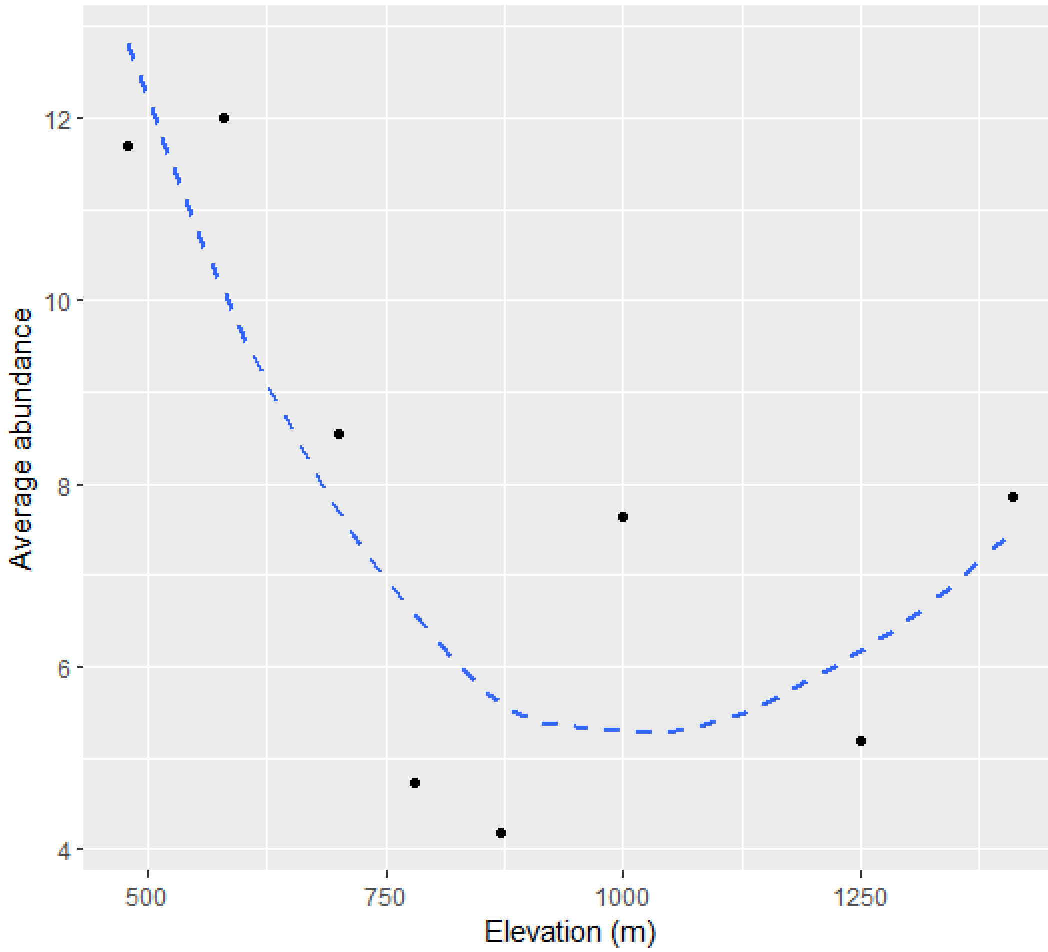

Figure 3.4 – Curved pattern of the Average abundance variation along the elevation gradient

The average abundance of individual species is higher at lower elevations and, declines during mid-elevations, resulting in few increments at higher elevations; 1000 m, 1250 m, and 1410 m.

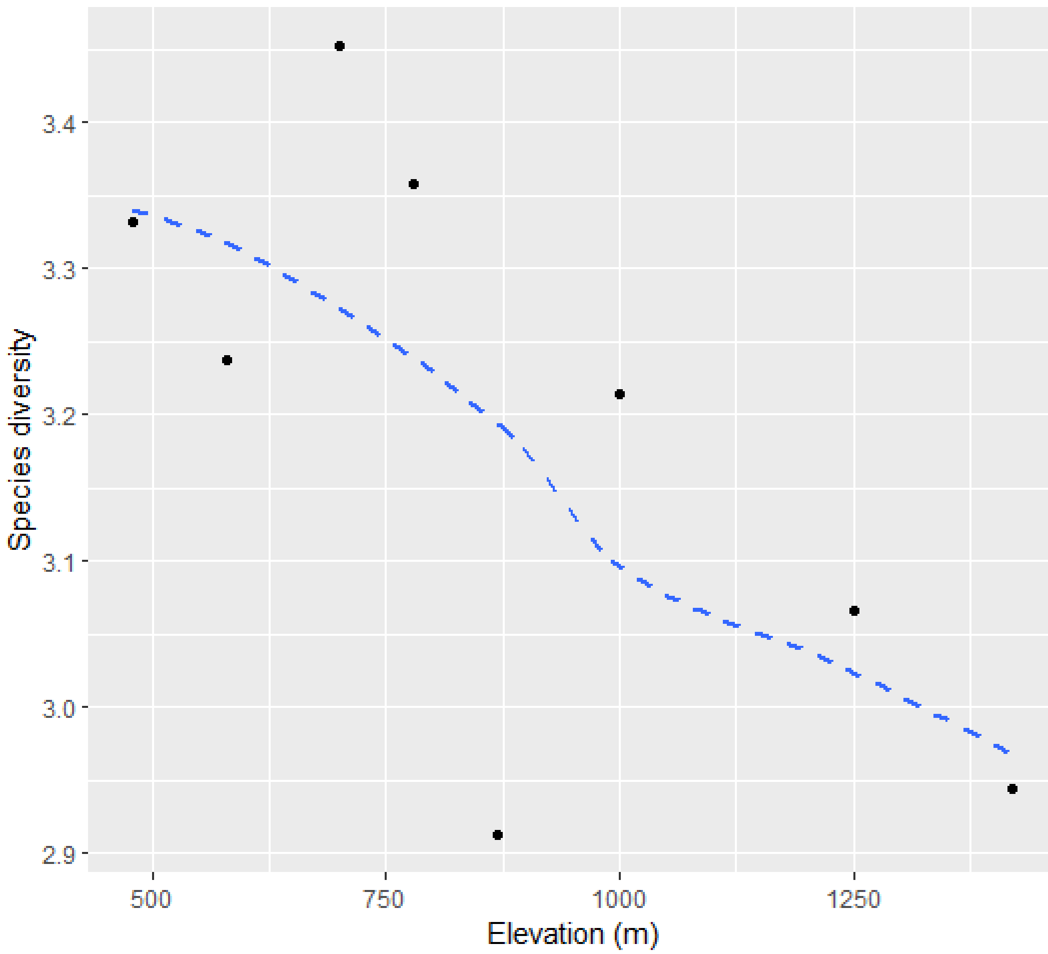

Figure 3.5 – Curved pattern of the Species diversity variation along the elevation gradient

Shannon Weiner species diversity is higher at lower elevations and declines towards higher elevations while resulting in the lowest diversity at 870 m elevation.

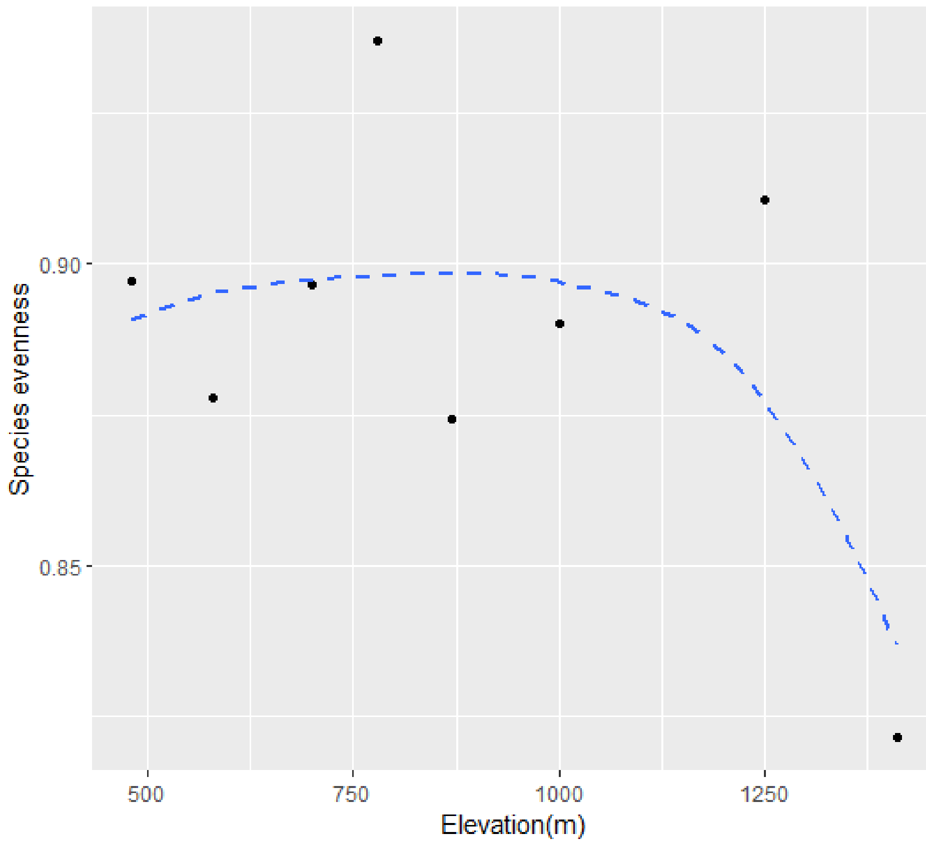

Figure 3.6 – Curved pattern of the Species evenness variation along the elevation gradient

Shannon'Weinerer evenness index values resulted more than 0.8 for all the elevations. In addition except for 1420 m elevation the other elevations showed more than 0.86 value for the species evenness in their habitats. Therefore, the variation of evenness along the elevation gradient has shown a decline at 1420 m elevation.

3.3 Species richness variation along the elevation gradient

Using the data collected for 9 months species richness of endemics, threatened species, montane species, and migratory species were calculated. In addition percentage of species endemism, percentage of threatened species were obtained. Bar graphs were plotted to seek the patterns of variation.

Table 3.3 – Values of different species richness types

Elevation (m) (MSL)	Species Richness - Total	Species Richness - Endemic	Species Richness - Migratory	Species Richness - Montane	Species Richness - Threatened	Percentage of species endemism	Percentage of threatened species
480	41	8	5	0	4	19.51	9.76
580	40	9	4	0	2	22.50	5.00
700	47	10	7	0	4	21.28	8.51
780	36	11	6	1	5	30.56	13.89
870	28	8	4	0	1	28.57	3.57
1000	37	13	5	3	8	35.14	21.62
1100	16	2	3	0	4	12.50	25.00
1250	29	7	5	4	10	24.14	34.48
1360	12	3	2	1	3	25.00	25.00
1420	36	13	3	4	11	36.11	30.56

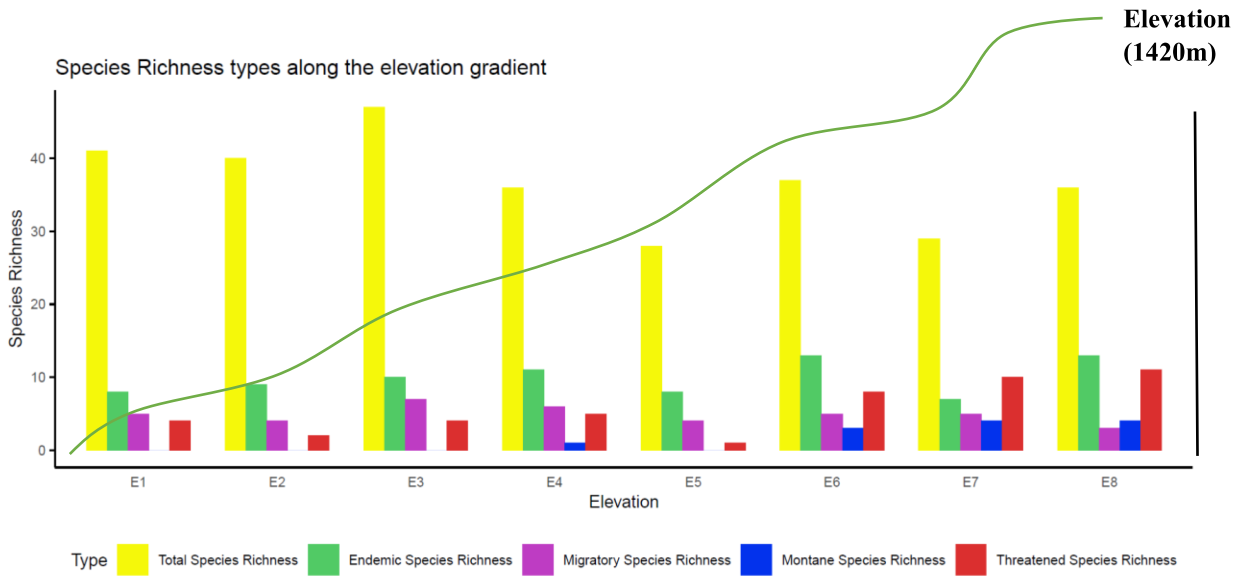

Figure 3.7 – Types of the species richness variation along the elevation. Elevation in meters: E1-480, E2-580, E3-700, E4-780, E5-870, E6-1000, E7-1250, E8-1420

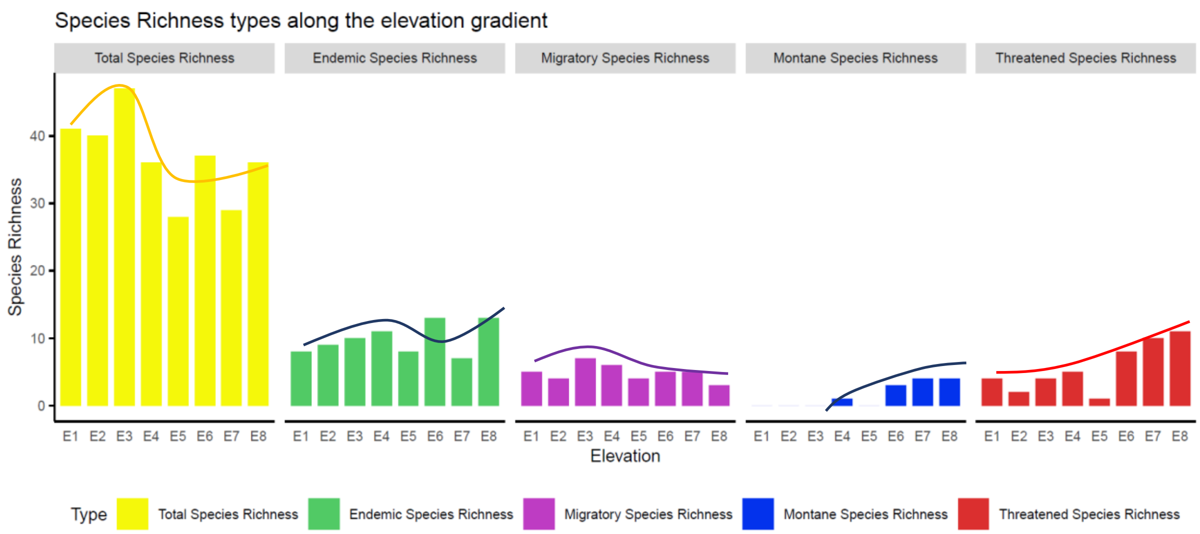

Figure 3.8 – Comparison of the variation patterns of different types of species richness along the elevation gradient. Elevation in meters: E1-480, E2-580, E3-700, E4-780, E5-870, E6-1000, E7-1250, E8-1420

3.3.1 Correlations of Species richness types along the elevation gradient.

Variations of types of species richness along the elevation gradient were tested. Considering the patterns of variation in figure 3.6 and 3.7, Pearson or Spearman test were tested to obtain the relationship. Then, curved dash line graphs were plotted on scatter plots.

Hypotheses

Hypothesis 5 :	There is a correlation between endemic species richness and elevation gradient.
Hypothesis 6 :	There is a correlation between migratory species richness and elevation gradient.
Hypothesis 7 :	There is a correlation between montane species richness and elevation gradient.
Hypothesis 8 :	There is a correlation between threatened species richness and elevation gradient.

Table 3.4 Correlation and P values of species richness response variables along the elevation gradient

Response Variable	Correlation	P value
Species Richness of Total species	-0.6706707	0.06869
Species Richness of endemic species	0.2891776	0.4873
Species Richness of Migratory species	-0.3559957	0.3868
Species Richness of montane species	0.914833	0.001447

Species Richness of threatened species	0.8355599	0.00979

For species richness of endemics and species richness of migratory species, Spearman correlation test was done since their variations are more curved along the elevation gradient. The Pearson correlation test was applied for species richness of montane and threatened species variation along the elevation. All the species richness variations have shown either a positive or negative correlation with elevation gradient rejecting the null hypothesis 5,6,7 and 8. Species richness of endemics has shown a low positive correlation with elevation gradient, whereas species richness of migratory species has given a low negative correlation with the elevation gradient (Table 3.2). But for both of these species richness types have not resulted less than 0.05 P value (Table 3.2). Thus, there are no significant linear relationships between species richness of endemics and species richness of migratory with elevation gradient.

Both species richness of montane and threatened have shown a high positive correlation with elevation (Table 3.2). Resulting in P values of 0.001447 and 0.00979 respectively for species richness of montane and threatened species. They have proved that there are significant correlations with elevation (Figure 3.8 and 3.9).

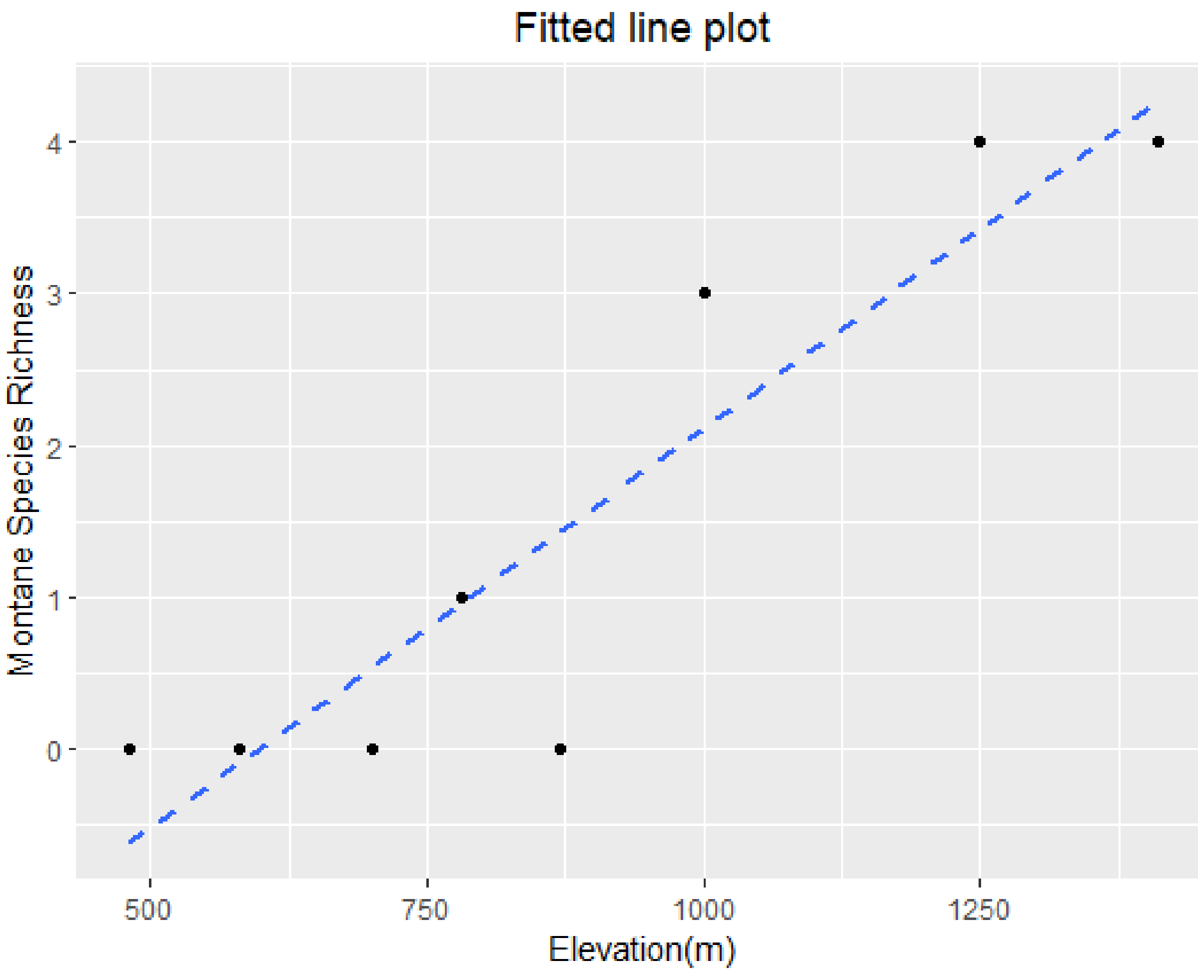

Figure 3.9 - Fitted line plot of Pearson correlaion of the species richness of montane species along the elevation

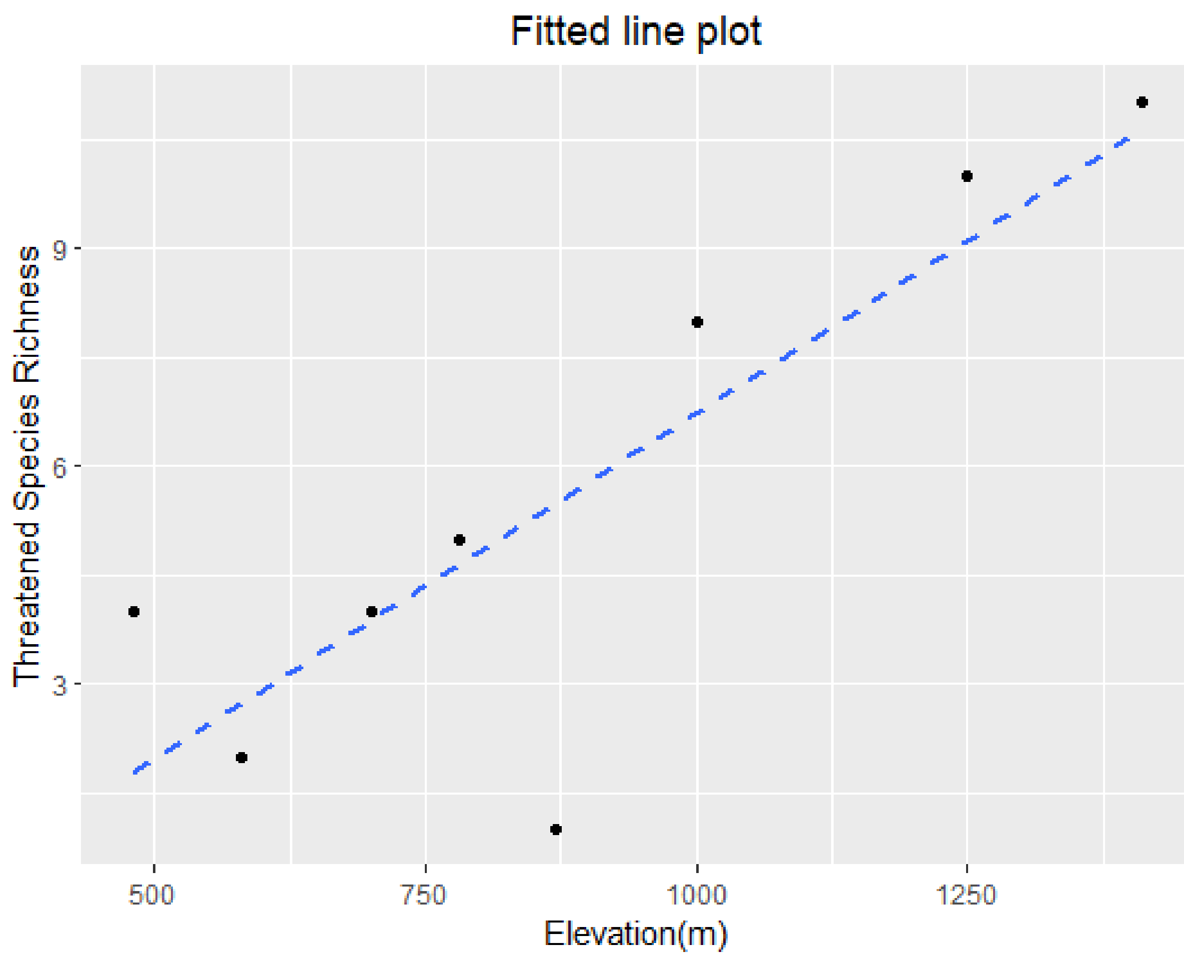

Figure 3.10 - Fitted line plot of Pearson correlaion of the species richness of threatened species along the elevation

3.3.2 Curved relationship of types of species richness along the elevation gradient

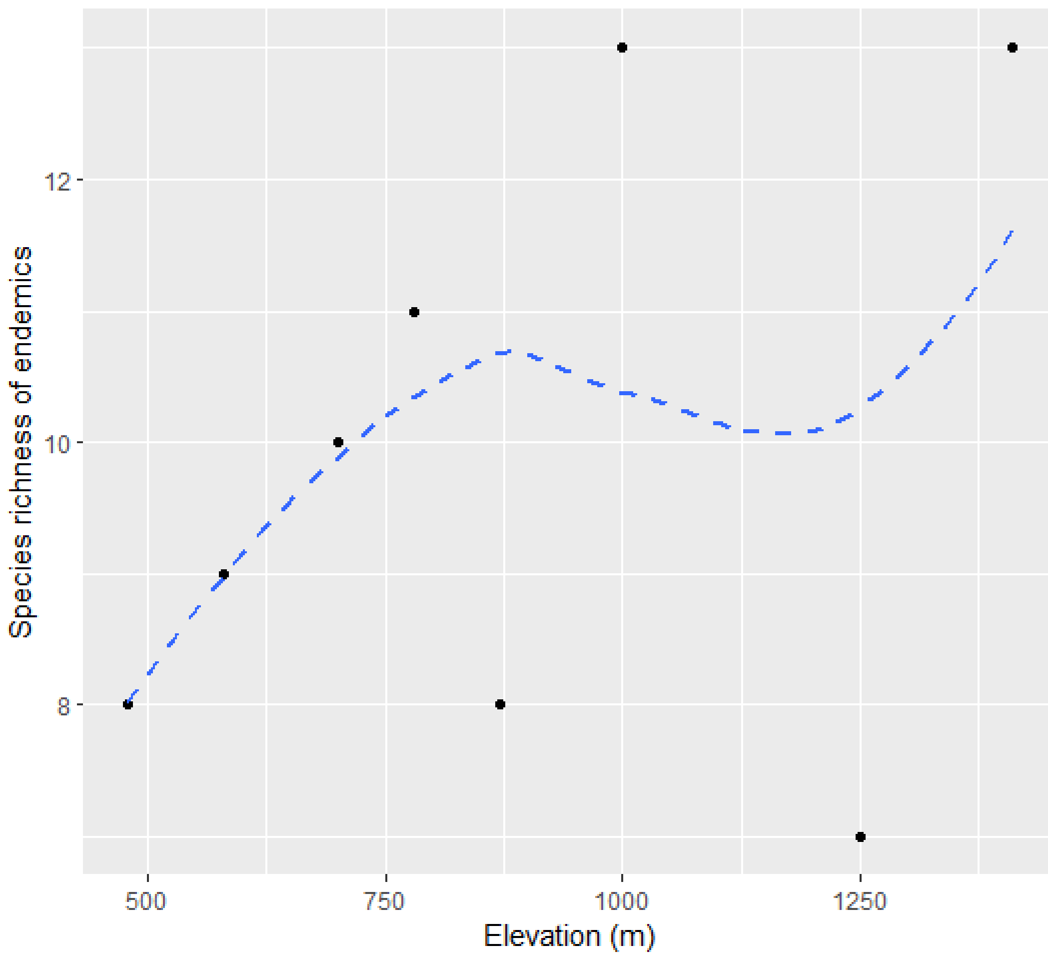

Figure 3.11 - Curved pattern of the endemic species richness variation along the elevation

At lower elevations, species richness of endemics has increased, followed by some mid elevation drop and, finally at 1420 m, species richness of endemics has increased again.

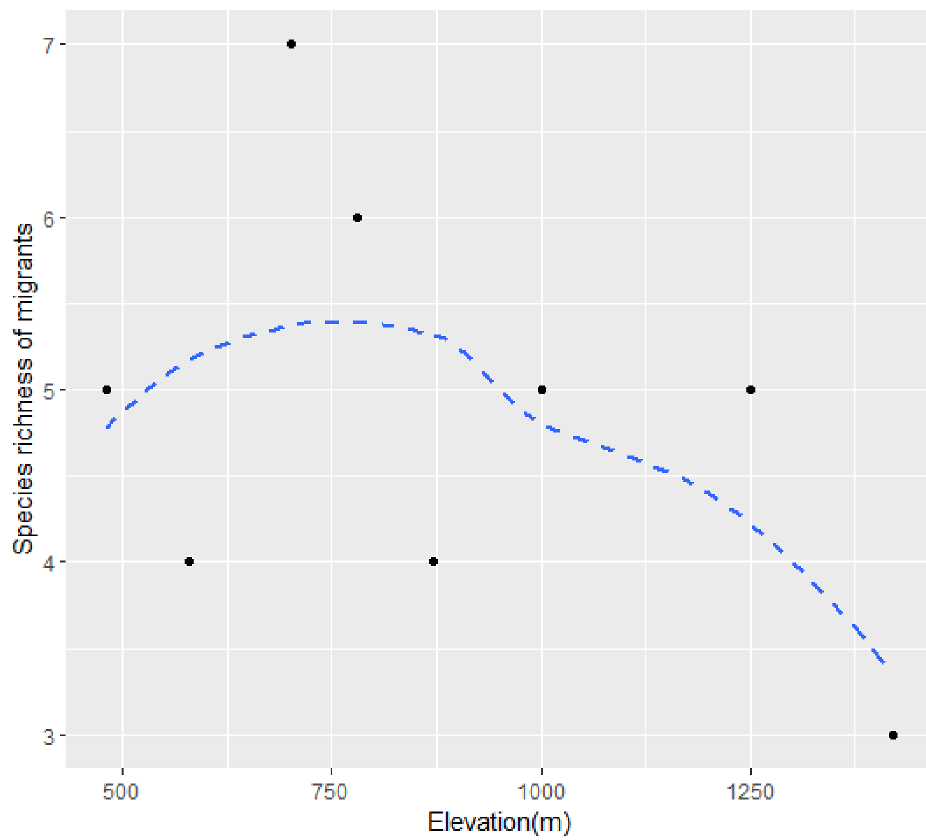

Figure 3.12 - Curved pattern of the species richness of migrants' variation along the elevation

Species richness of migrants has depicted more like a hump-shaped pattern along the elevation gradient, recording the highest migratory species richness at mid elevations and lowest at highest 1420 m.

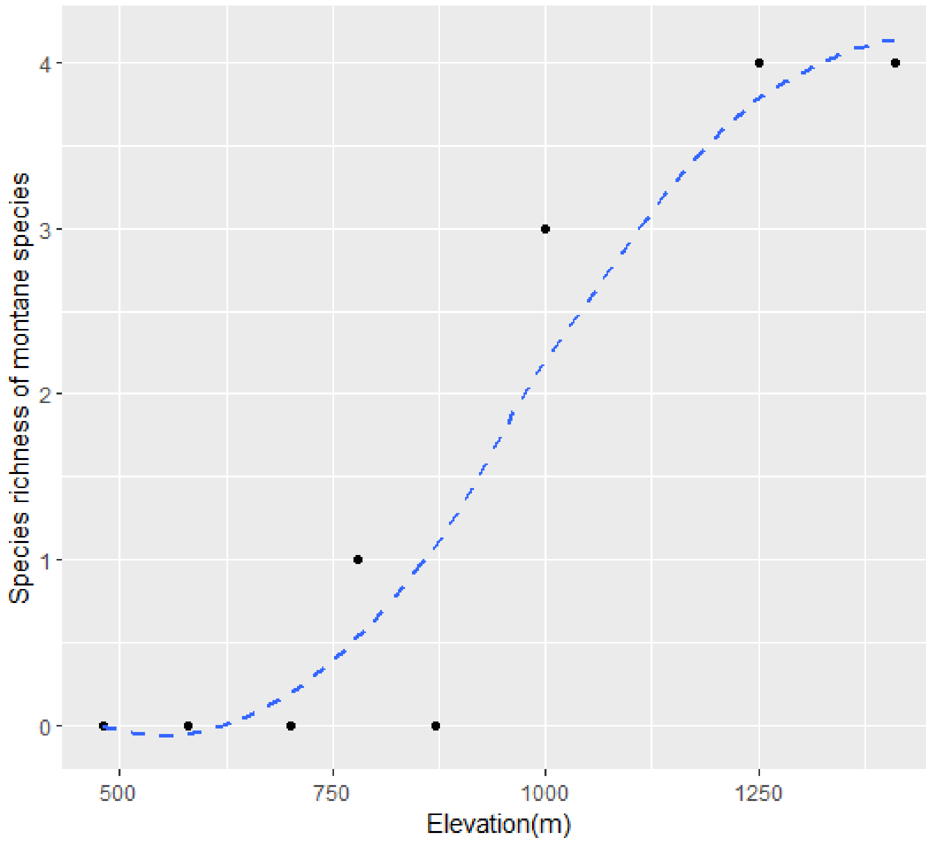

Figure 3.13 - Curved pattern of the montane species richness variation along the elevation gradient.

There is a clear incline of the species richness of montane along the elevation gradient. Specifically, at lower elevations montane species were not recorded. The lowest elevation where the montane species have been recorded is at 780 m elevation whilst, starting from 1000 m elevation and up to 1420 m, the highest elevation, species richness of montane species has increased.

3.4 Vegetation and habitat variabilities along the elevation gradient.

Using the ten transects trees girth at breast height (GBH), tree density, canopy cover, and habitat complexity were measures using five plots for each transects. Then, for each elevation average value for all these variables were obtained.

Table 3.5 Values of vegetation and habitat parameters along the elevation gradient

Elevation(m) (MSL)	GBH Average	Height Average	Tree density	Habitat complexity	Canopy cover
480	84.31	10.68	27	9.40	15.26
580	73.42	10.25	15	13.00	9.02
700	57	14.98	83	14.60	30.44
780	63.49	12.26	154	14.60	23.48
870	64.26	11.37	136	11.50	22.31
1000	58	8.77	181	11.70	20.96
1100	62	4.79	3	4.20	0.3638
1250	60.53	5.64	182	9.30	11.3
1360	32.68	2.85	10	4.90	0.0509
1410	61.21	17.51	193	10.70	10.81

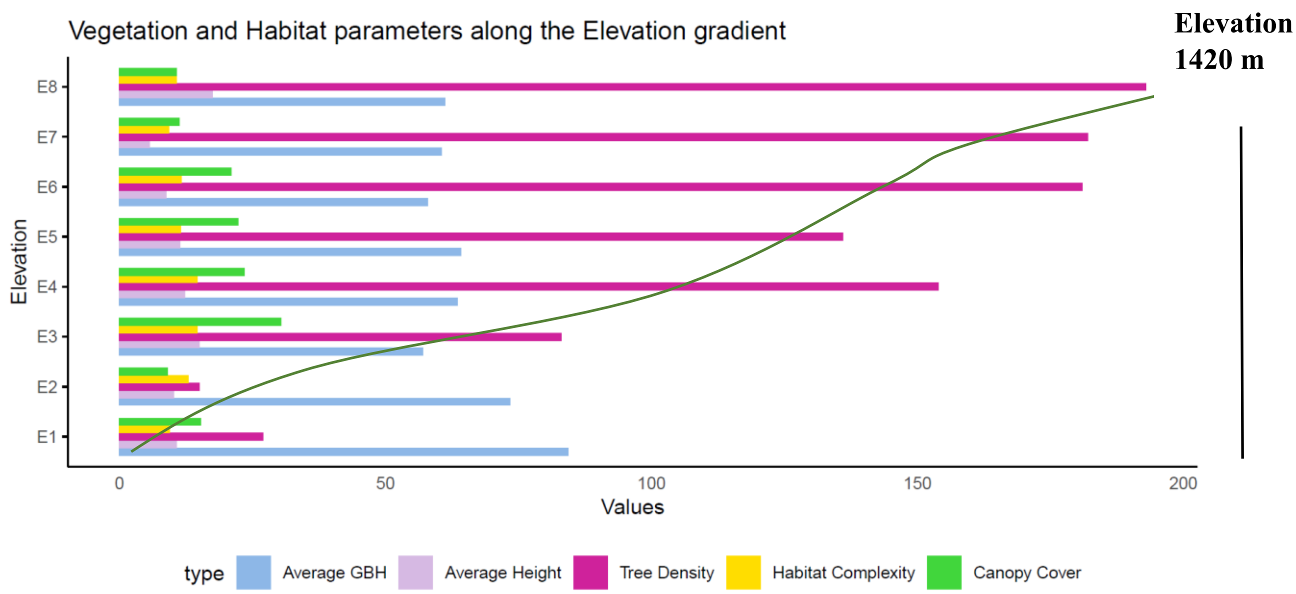

Figure 3.14 – Vegetation and habitat type variation along the elevation. Elevation in meters: E1-480, E2-580, E3-700, E4-780, E5-870, E6-1000, E7-1250, E8-14120

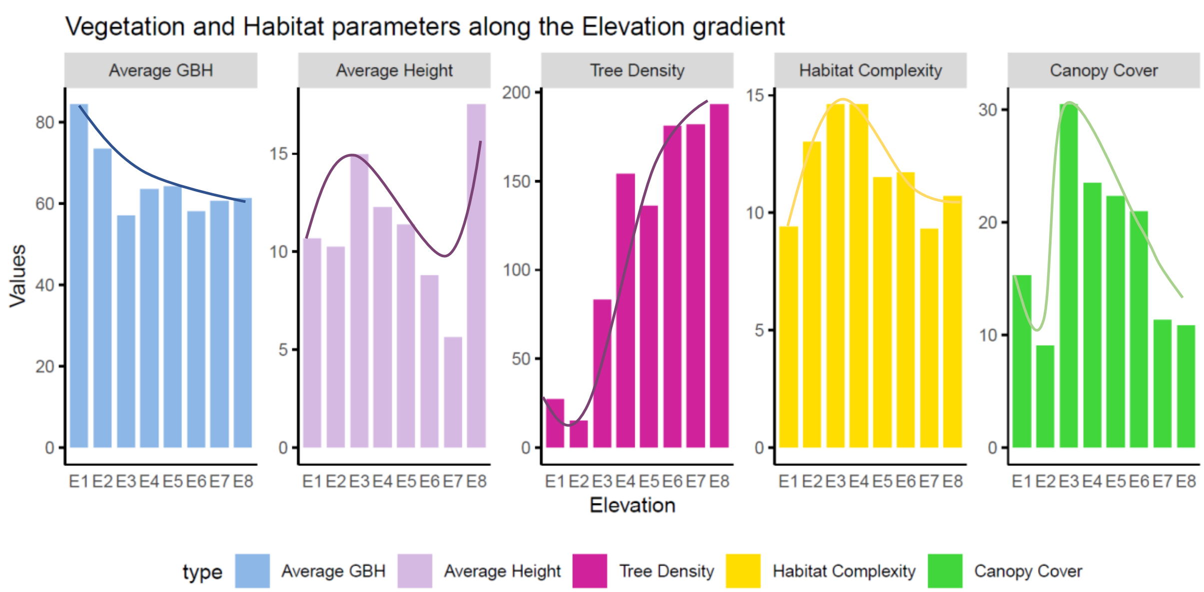

Figure 3.15 – Variation of separate vegetation and habitat parameters variation along the elevation. Elevation in meters: E1-480, E2-580, E3-700, E4-780, E5-870, E6-1000, E7-1250, E8-1420

3.4.1 Correlations of Vegetation and habitat parameters along the elevation gradient

To check the relationship of vegetation and habitat parameters considering the patterns of variations in figure 3.14 and 3.15 Pearson and Spearman correlation tests were done. Then curved dash line graphs were plotted on the scatter plot.

Hypotheses

Hypothesis 9:	There is a correlation between girth at breast height of trees and elevation gradient.
Hypothesis 10:	There is a correlation between tree height and elevation gradient
Hypothesis 11 :	There is a correlation between tree density and elevation gradient.
Hypothesis 12 :	There is a correlation between habitat complexity and elevation gradient.
Hypothesis 13 :	There is a correlation between canopy cover and elevation gradient.

Table 3.6 Correlation and P values of vegetation and habitat parameters along the elevation gradient

Response Variable	Correlation	P value
Average GBH	-0.5238095	0.1966
Average Height	-0.2606061	0.4697
Tree Density	0.879387	0.003999
Habitat Complexity	-0.3473116	0.3993
Canopy Cover	-0.4787879	0.1661

As depicted in Figures 3.14 and 3.15, vegetation and habitat parameters show increasing, decreasing and some hump-shaped variations along the elevation gradient. In order to detect the correlations between these parameters and elevation gradient, average tree height, habitat complexity and canopy cover were tested using the Pearson correlation test. The average GBH and tree density were tested using the Spearman correlation test.

All these parameters resulted in negative or positive correlation with elevation gradient, leading to reject the null hypotheses 9, 10, 11, 12 and, 13.

Both tree height and habitat complexity have shown weak negative correlation coefficients whereas average GBH and canopy cover have shown negative correlation coefficients of around 0.5 along the elevation gradient (Table 3.3.2). Even though, all these four parameters have a negative correlation with elevation gradient, x these were not statistically significant correlation with elevation gradient, since the P value is not less than 0.05 (Table 3.3.2).

Tree density has displayed a strong positive correlation with elevation gradient along with a small P value of 0.003999, proving that there is a significant linear correlation between them (Figure 15).

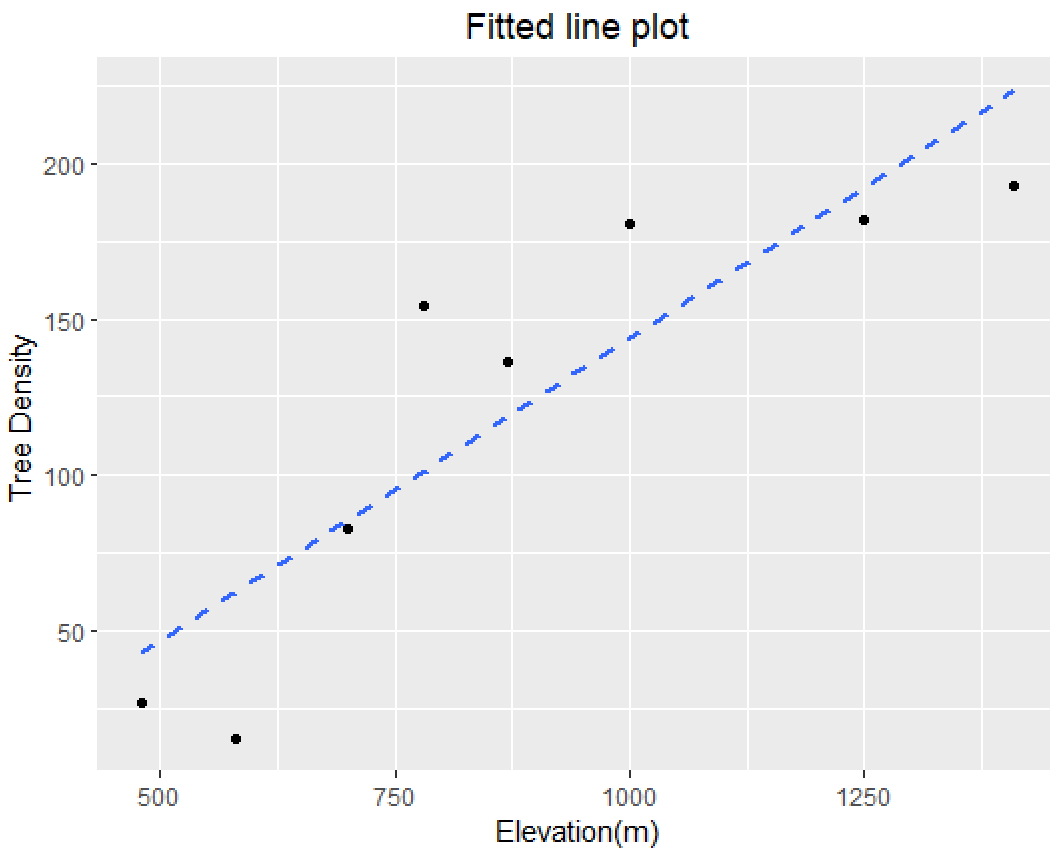

Figure 3.16 - Fitted line plot of Pearson correlaion for the tree density along the elevation

3.4.2 Curved relationship of vegetation and habitat parameters along the elevation gradient.

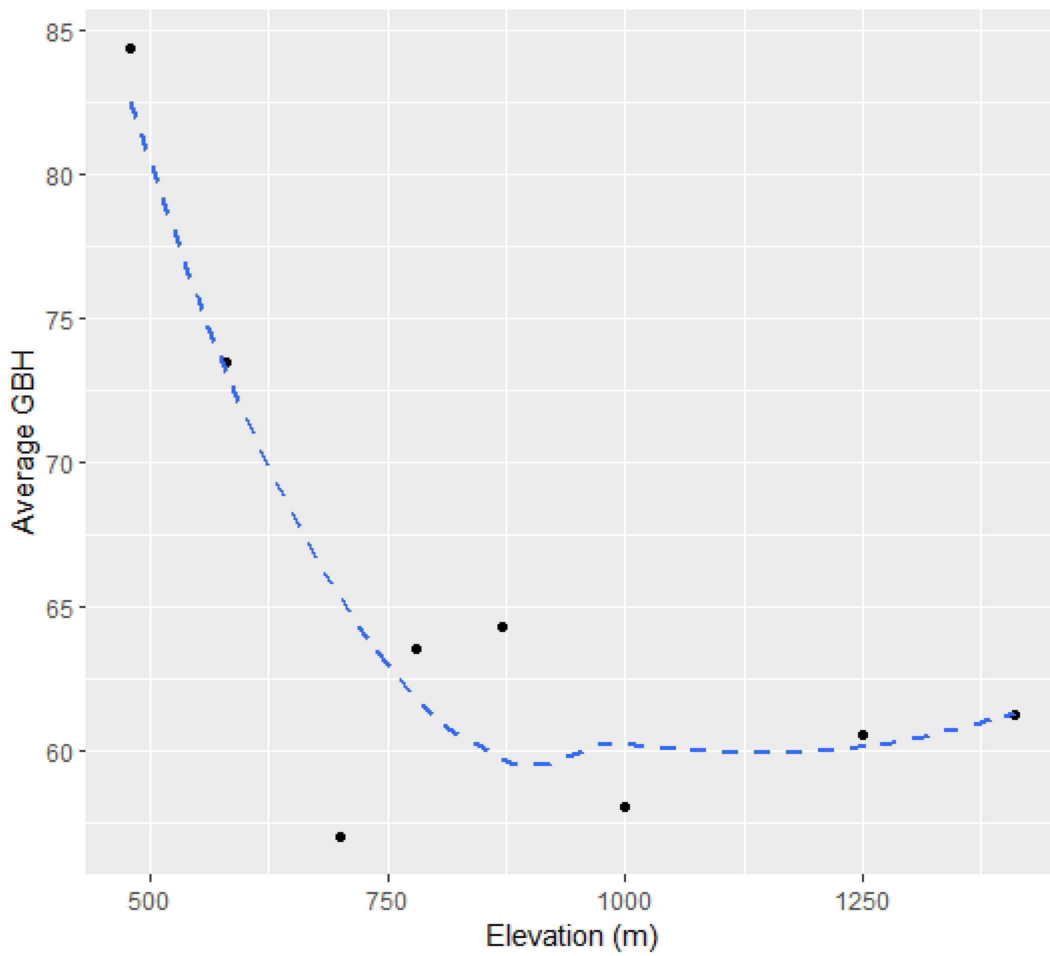

Figure 3.17 - Curved pattern of the average GBH variation along the elevation.

The average GBH of the trees has indicated a higher average GBH at lower elevations, followed by lower values at mid elevations and a little incline at higher elevations.

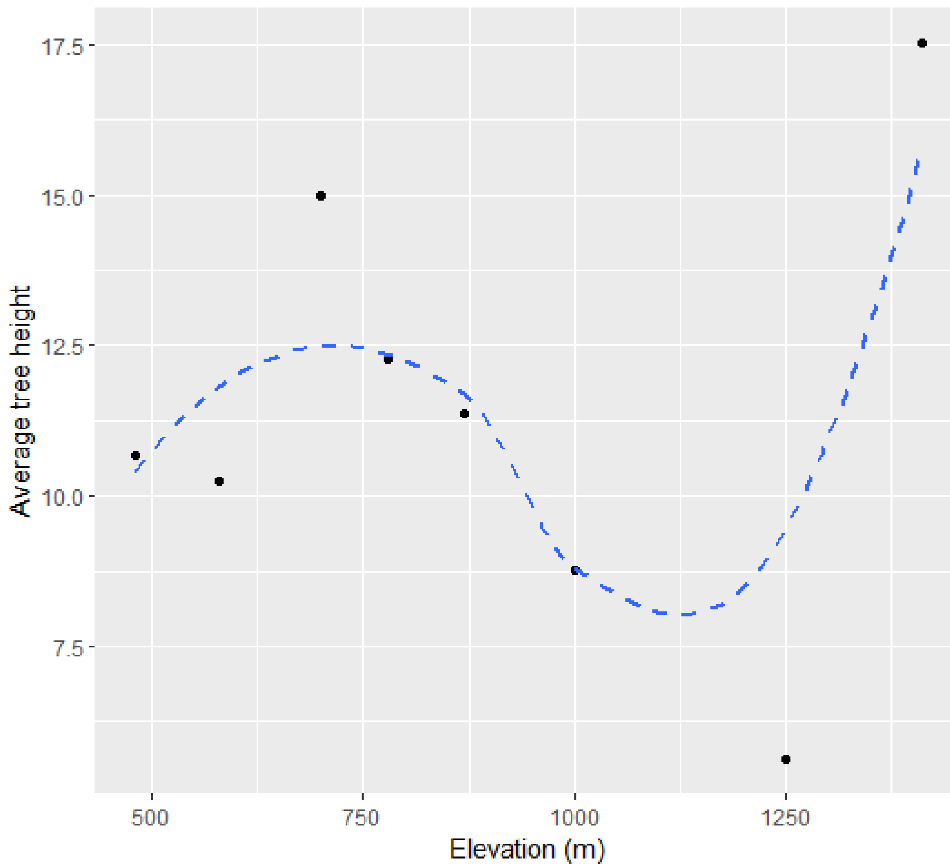

Figure 3.18 - Curved pattern of the average tree height variation along the elevation.

Except for the 1410 m highest elevation, the tree height variation has indicated a hump-shaped variation along the elevation gradient, whereas the highest elevation has resulted in the highest value for the average tree height.

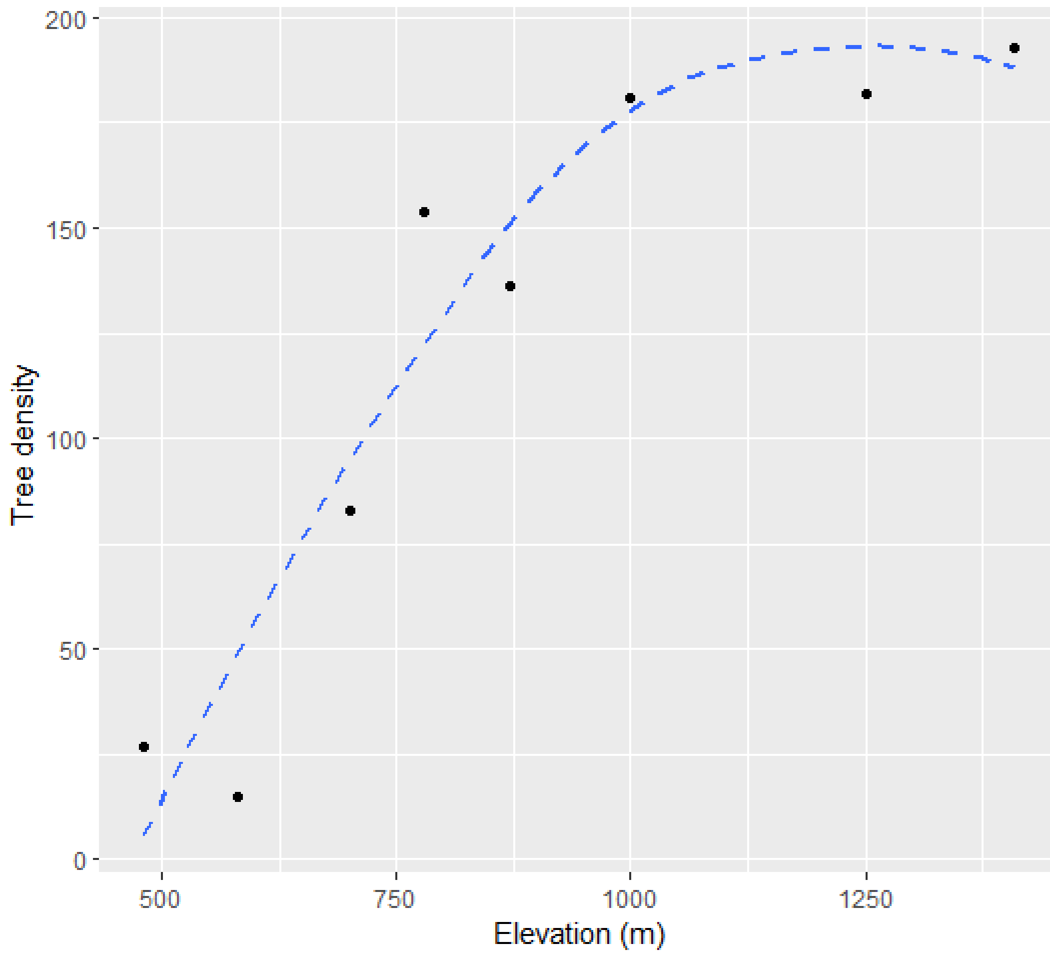

Figure 3.19 - Curved pattern of the average tree density variation along the elevation gradient.

The variation of tree density along the elevation gradient has shown a clear increment from lower elevations to the higher elevations reaching an asymptote after the elevation of 1000 m.

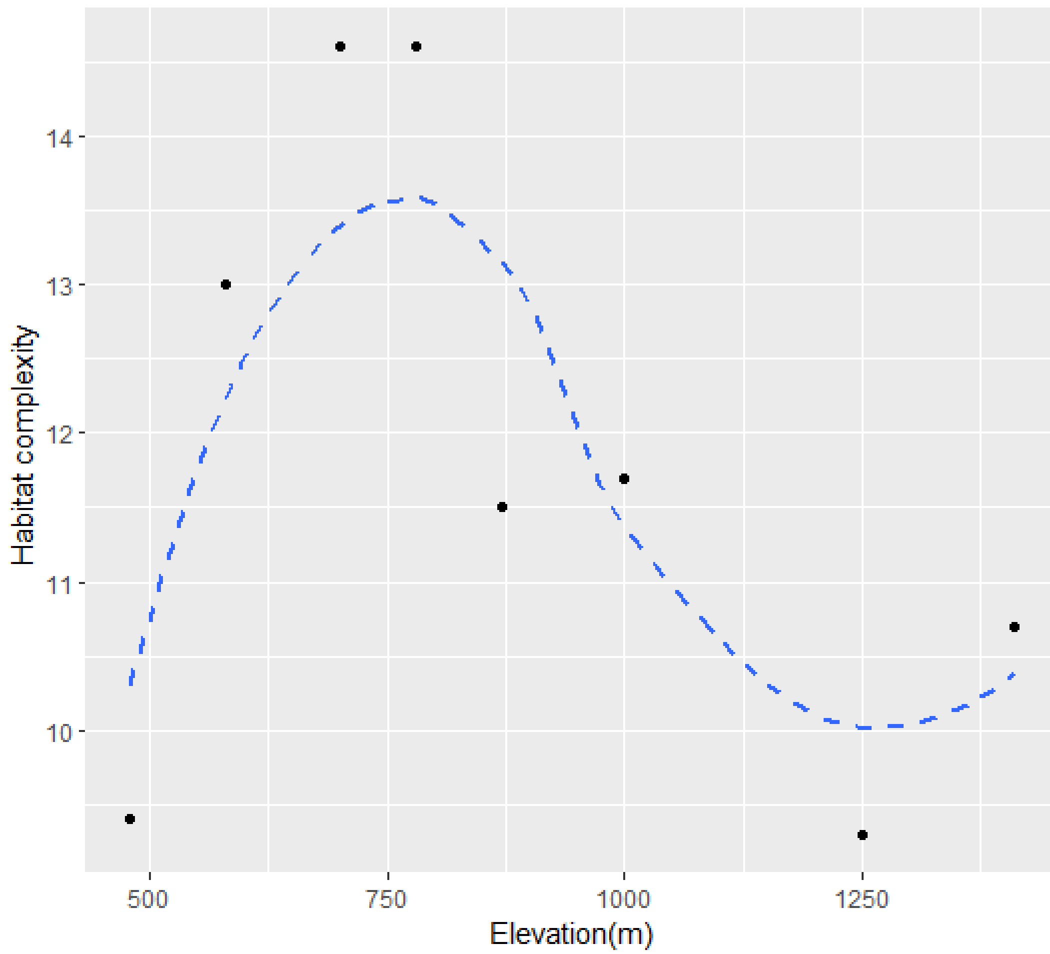

Figure 3.20 - Curved pattern of the habitat complexity variation along the elevation gradient.

Habitat complexity has displayed lower values for lower elevations, higher values for mid elevations, and lower values again for higher elevations resulting in a hump-shaped variation.

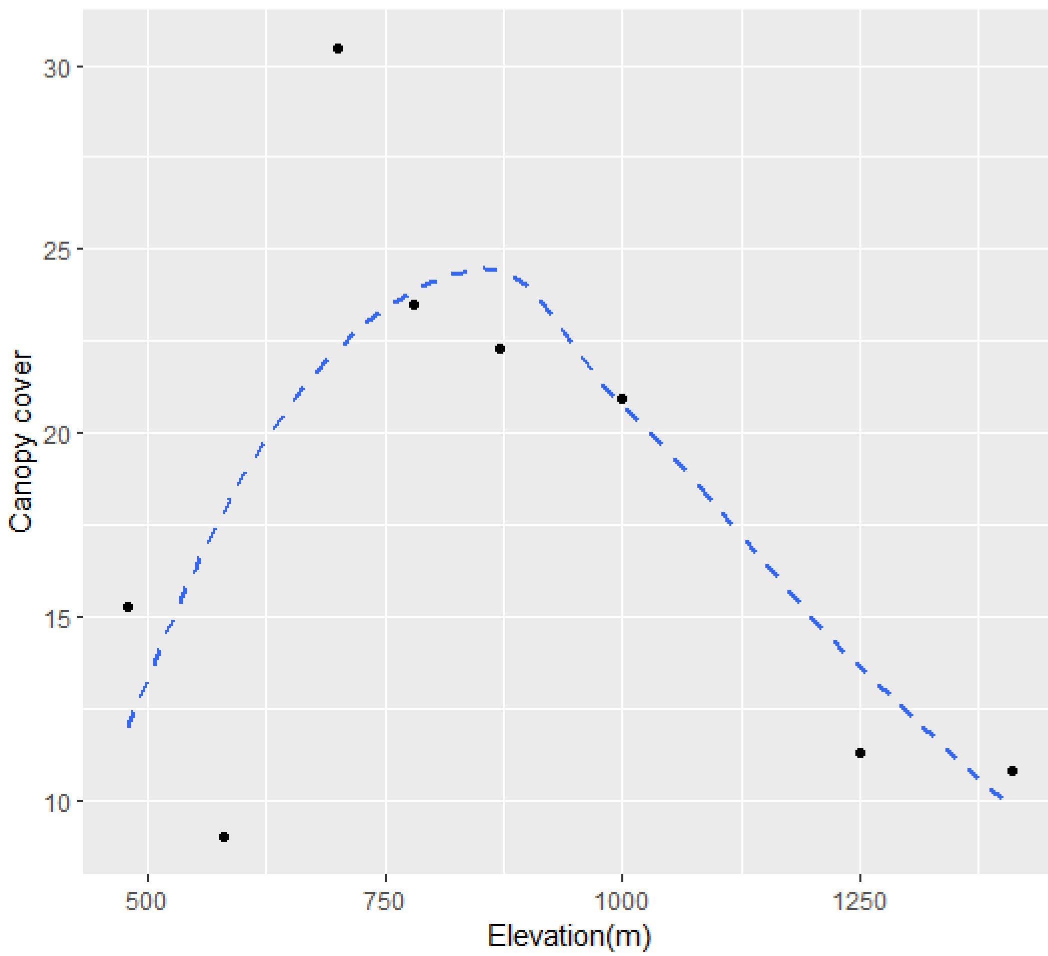

Figure 3.21 - Curved pattern of the canopy cover variation along the elevation gradient.

With the highest values for the canopy cover at mid elevations and lower values for the lower and higher elevations, canopy cover also shows a hump-shaped variation along the elevation gradient.

3.5 - Effect of the grasslands for the variabilities of avian community species richness and habitat parameters

All curved plotted were plotted including grassland habitats and not including grassland habitats. This helps to see the decline of the pattern of grassland.

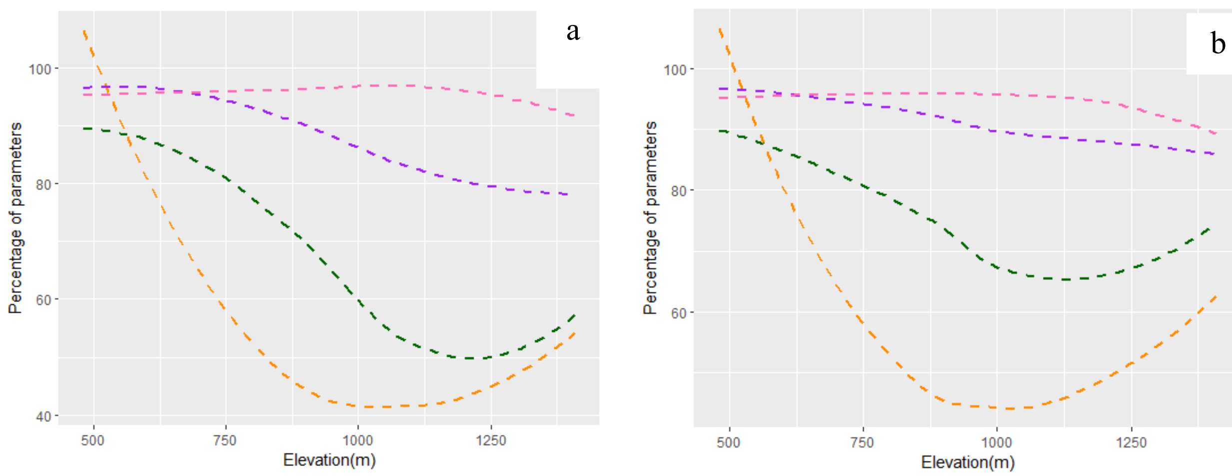

Figure 3.22 : Avian community parameters variation along (a) all studied elevations and (b) without grassland elevations. ---- Species richness, ----- Average abundance, ----- Species diversity, -----Species evenness

The variation patterns of the avian community parameters of both the graphs (a) and (b) are mostly similar. But there are little declines in all four parameters in between 1000 m and 1420 m due to the presence of the grasslands.

3.5.1 Effect of the grasslands for the different types of species richness variation

Different types of avian richness variation along the elevation gradient with grassland and without grassland were plotted to understand the affect of grassland.

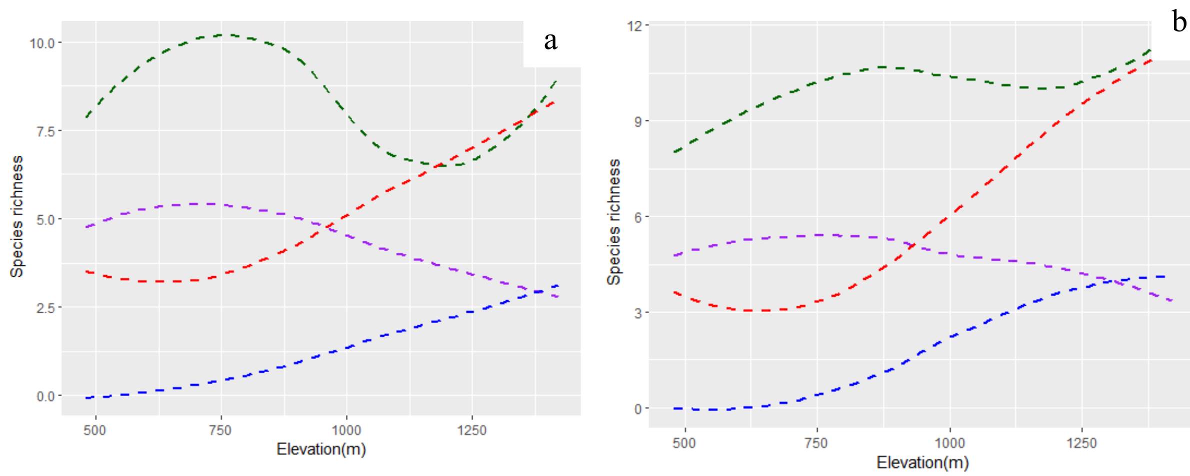

Figure 3.23 – Different types of species richness variations along (a) all studied elevations and (b) without grassland elevations. ---- Endemic species richness, ----- Migratory species diversity, ----- Montane species evenness, ----- Threatened species richness

Compared to the adjacent elevations of the grasslands all four species richness types have little declines at the 1100 m and 1360 m elevations. A high decrement has shown in the endemic species richness.

3.5.2 Effect of the grasslands for the vegetation and habitat parameters variation

Vegetation and habitat parameters variation with and without grasslands along the elevation gradient, to see the effect of grasslands.

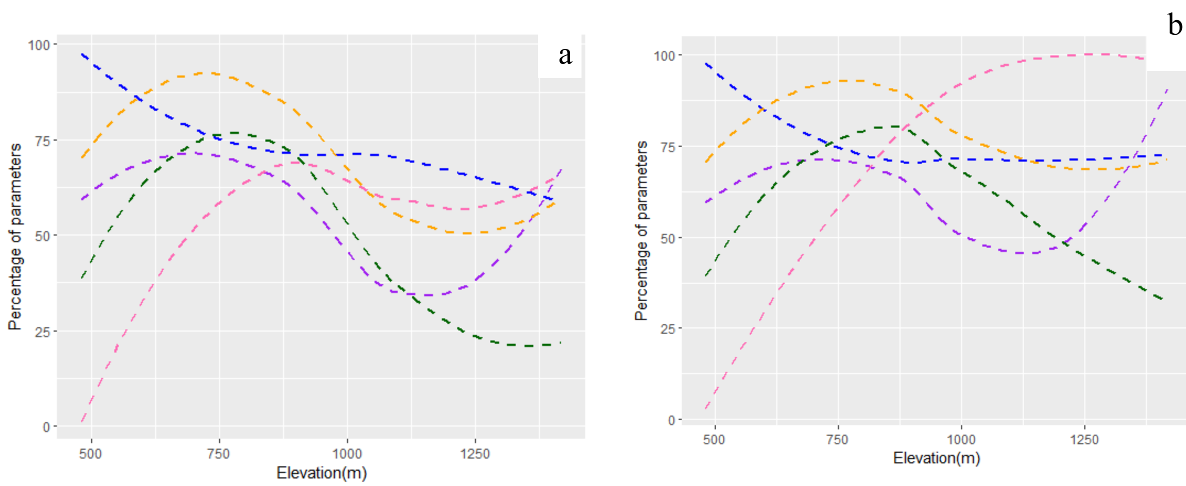

Figure 3.24 – Vegetation and habitat parameters variation along (a) all studied elevations and (b) without grassland elevations. ---- Average GBH, ----- Average tree height, ----- Tree density, ----- Habitat complexity, ----- Canopy cover

All vegetation and habitat parameters have declined at grassland elevations providing clearly visible declines in the graph (a) compared to those in graph (b). There is a greater variation in the shape of the trend for tree density.

3.6 Contribution of vegetation and habitat parameters for the avian community variations, along the elevation gradient

More similar patterns were plotted together to understand the effect of vegetation and habitat to shape avian community response variables, along the elevation gradient.

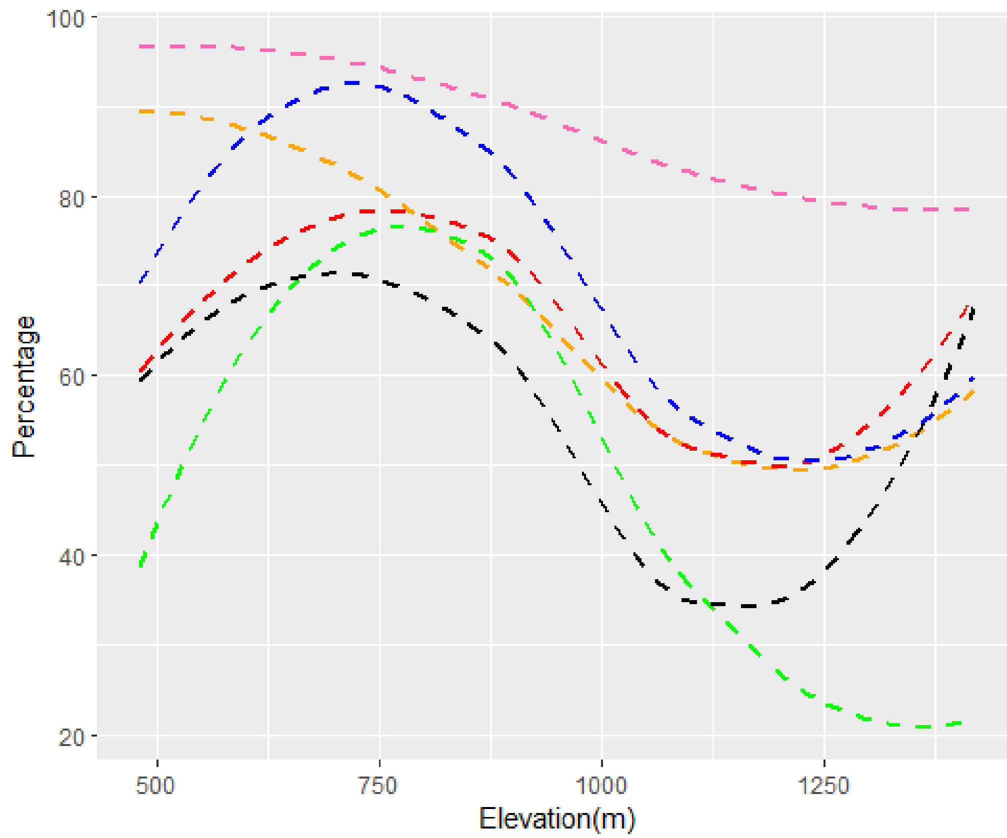

Figure 3.25 – Similar patterns of variation for several avian community, vegetation, and habitat parameters along the elevation gradient.

---- Total species richness, ---- Endemic species richness, ---- Species diversity, ---- Habitat complexity, ---- Canopy cover, ---- Tree height

The similar hump-shaped variations of the total species richness, endemic species richness, and species diversity from the avian community parameters, with the habitat complexity, canopy cover, and tree height from the vegetation and habitat parameters have shown that the avian community variations, are not only determined by the elevational gradient but also confounded by the contribution of the structure of vegetation and habitat at particular elevations.

3.7 Effect of the climatic parameters on the avian communities

Species richness variation with the climatic parameters such as temperature, relative humidity, wind speed, light intensity, visibility and cloud cover were plotted using the data collected throughout the study period.

3.7.1 Correlations of species richness along with climatic parameters

Scatter plots were done between species richness and climatic variabilities separately. With respect to the patterns of scatter plots Pearson or Spearman correlation tests were tested to see the relationship between species richness and climatic parameters.

Hypotheses

Hypothesis 14 : There is a correlation between species richness and temperature variation.

Hypothesis 15 : There is a correlation between species richness and humidity variation.

Hypothesis 16 : There is a correlation between species richness and wind speed variation.

Hypothesis 17 : There is a correlation between species richness and light intensity variation.

Hypothesis 18 : There is a correlation between canopy cover and elevation gradient.

Hypothesis 19 : There is a correlation between species richness and cloud cover variation.

Table 3.6 Correlation coefficients and P values of species richness along the climatic variabilities

Response	Correlation	P value
Temperature	-0.1789107	0.03577
Humidity	0.2967547	0.0007403
Wind speed	-0.1779531	0.03411
Light intensity	-0.2584327	0.002128
Visibility	0.1883635	0.04022
Cloud cover	-0.2890533	0.002531

In order to check the correlations of climatic variabilities with species richness Pearson correlation was tested for temperature, humidity, light intensity and, cloud cover. The Spearman correlation was conducted for wind speed and visibility.

Since the correlation value for all the environmental parameters have not equal to 0 there are negative correlations between temperature, wind speed, light intensity, and cloud cover with species richness, whereas there are positive correlations between humidity and visibility with species richness, resulting in rejection of the all 14,15,16,17,18, and 19 null hypotheses.

Importantly, all these environmental parameters have less than 0.05 P value, representing that there are significant correlations between these environmental parameters and the species richness.

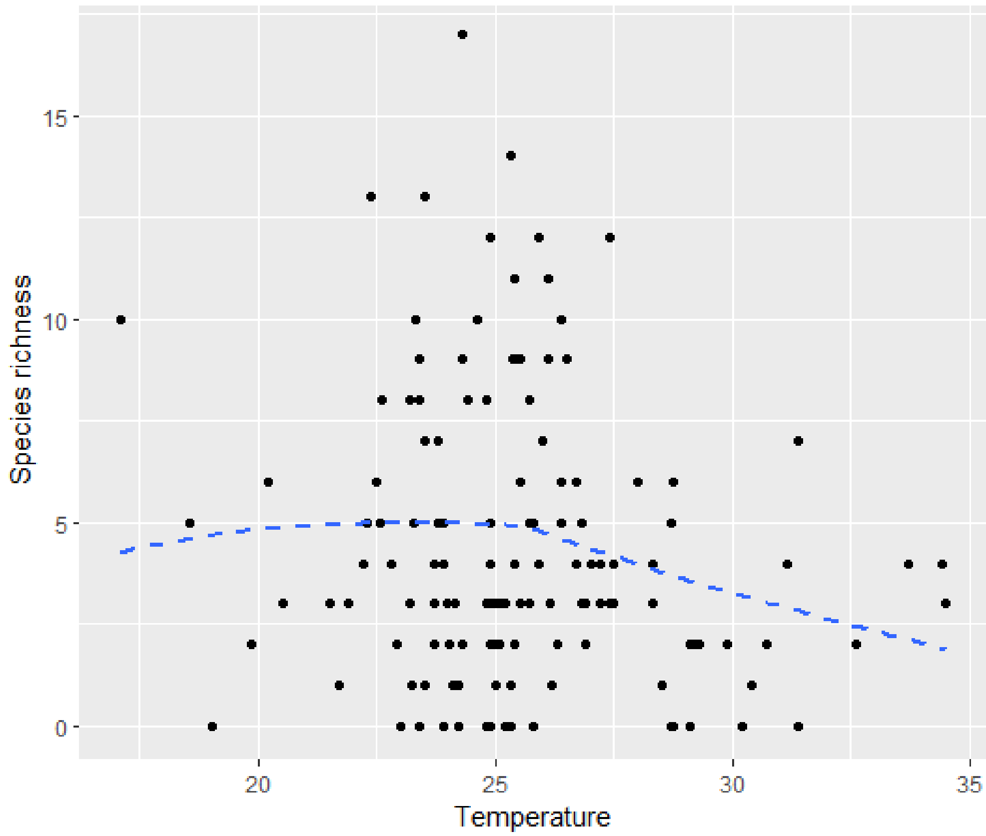

Figure 3.26 – Curved plot of species richness variation along the temperature**.**

During the study period the temperature has been changed between 17 °C and 35 °C. Species richness has declined at higher and lower temperatures. However, mostly the temperature of the study transects has appeared around 25°C temperature as a high number of observations have been recorded at mid temperatures between.

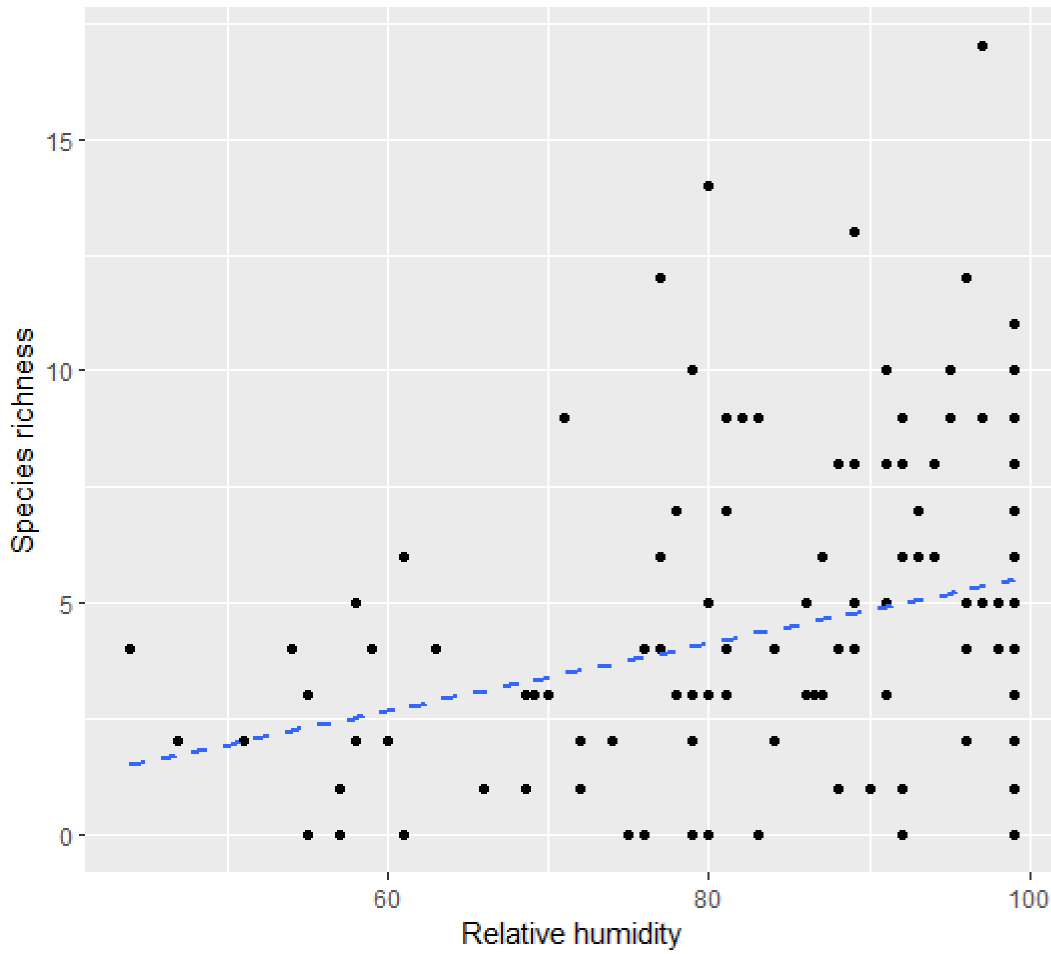

Figure 3.27 – Fitted line plot of the Spearman correlation test of species richness variation along the humidity.

The relative humidity has varied during the study period from around 40 – 100 %. Most of the observations have been recorded at higher humidity conditions, while a higher species richness was recorded at higher humidity conditions compared to lower humidity.

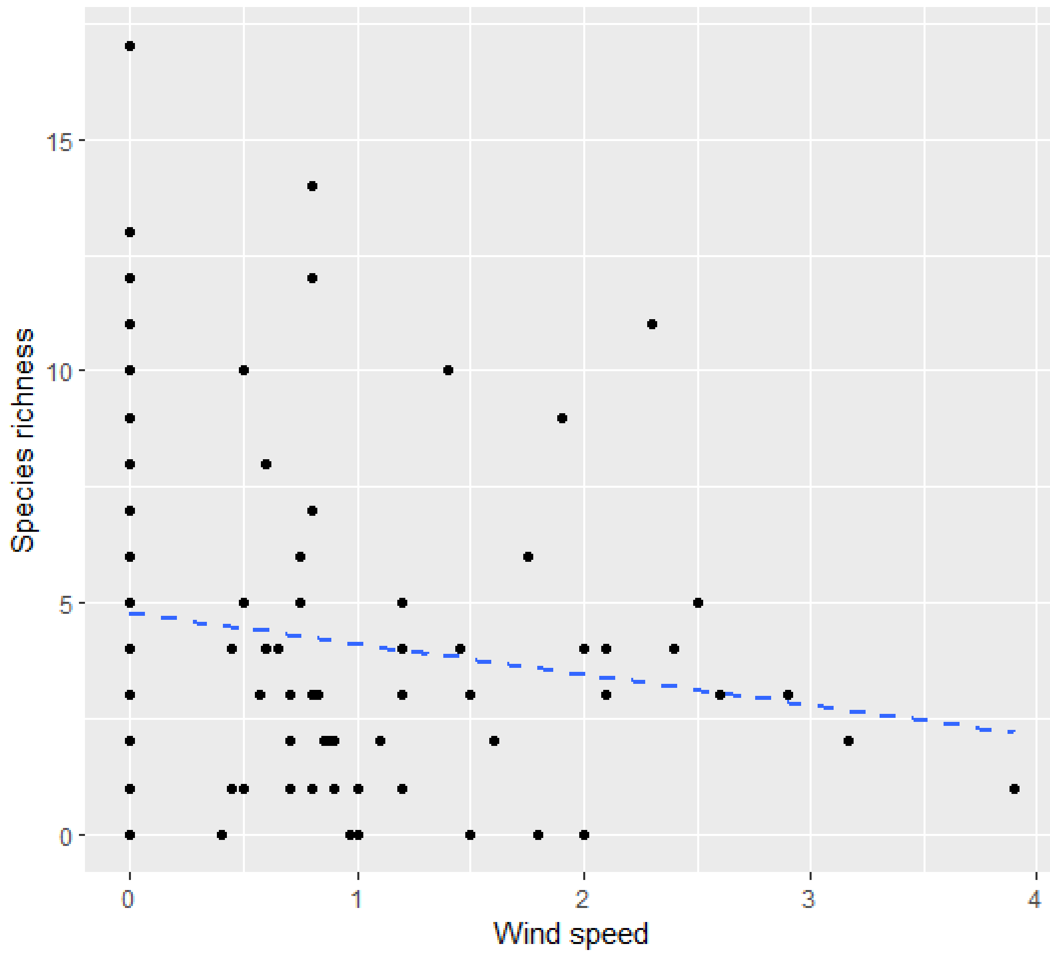

Figure 3.28 – Fitted line plot of the Spearman correlation test of species richness variation along the wind speed.

The wind speed during the study period has changed from 0 – 4 ms^{-1} . Most of the observations have been recorded with lower wind speeds, while the species richness has decreased with increased wind speed.

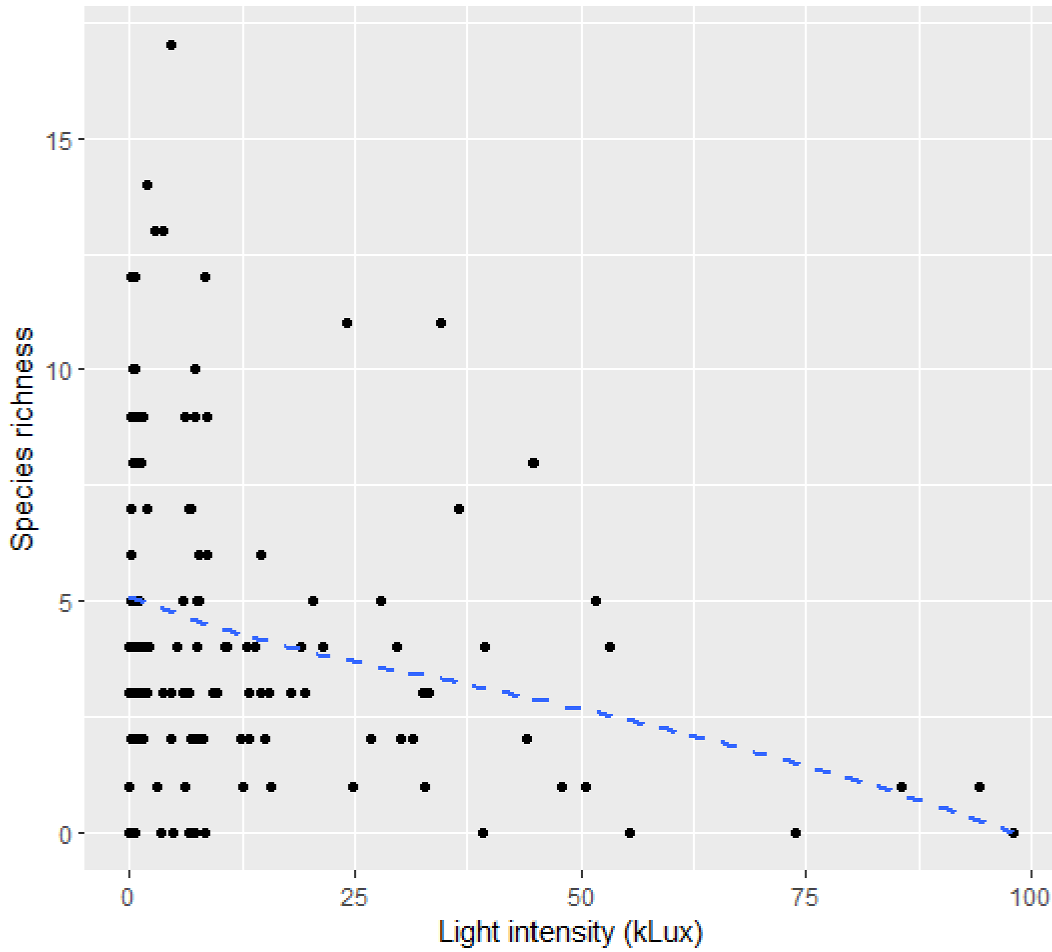

Figure 3.29 – Fitted line plot of the Pearson correlation test of species richness variation along the light intensity.

Most of the observations have been made at less than 25 kLux, while the species richness has decreased with increasing light intensity.

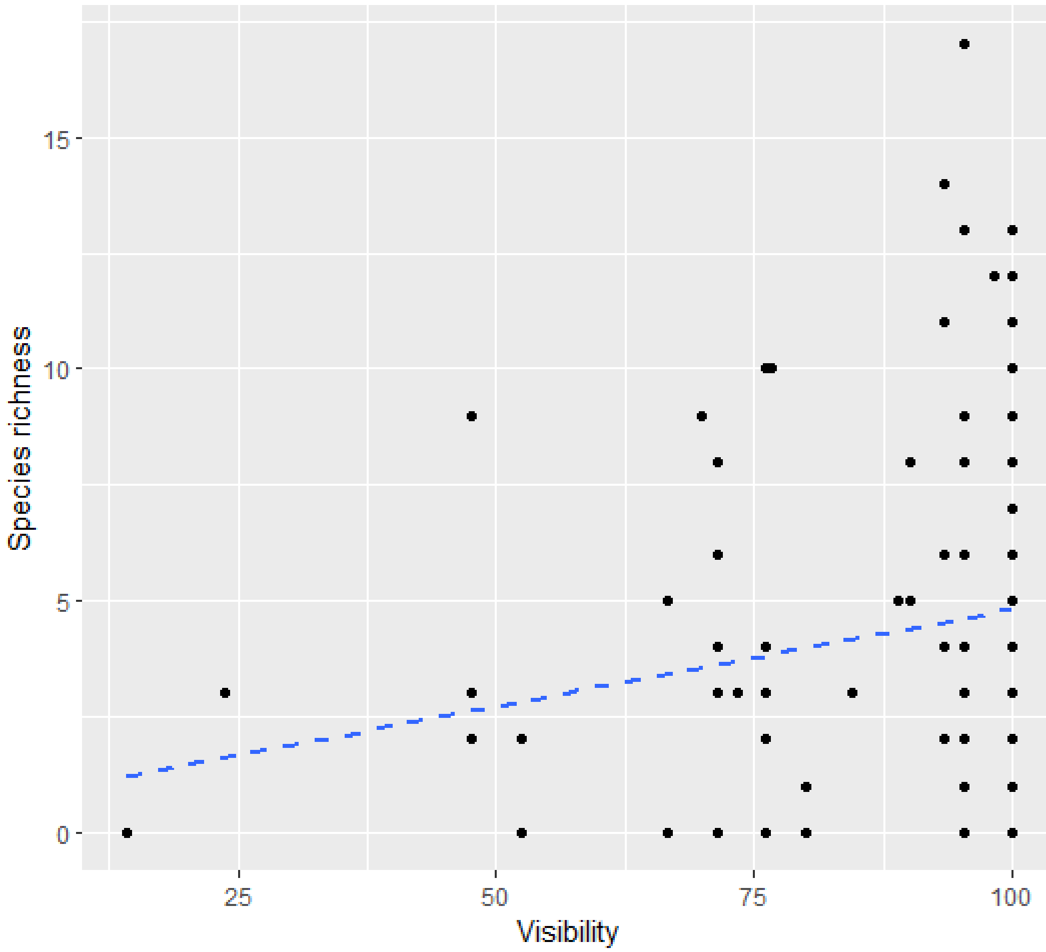

Figure 3.30 – Fitted line plot of the Spearman correlation test of species richness variation along the visibility.

During the study period the visibility has been varied from around 10 – 100%. Low species richness has been recorded at lower visibilities, while a higher visibility has been observed during most observations.

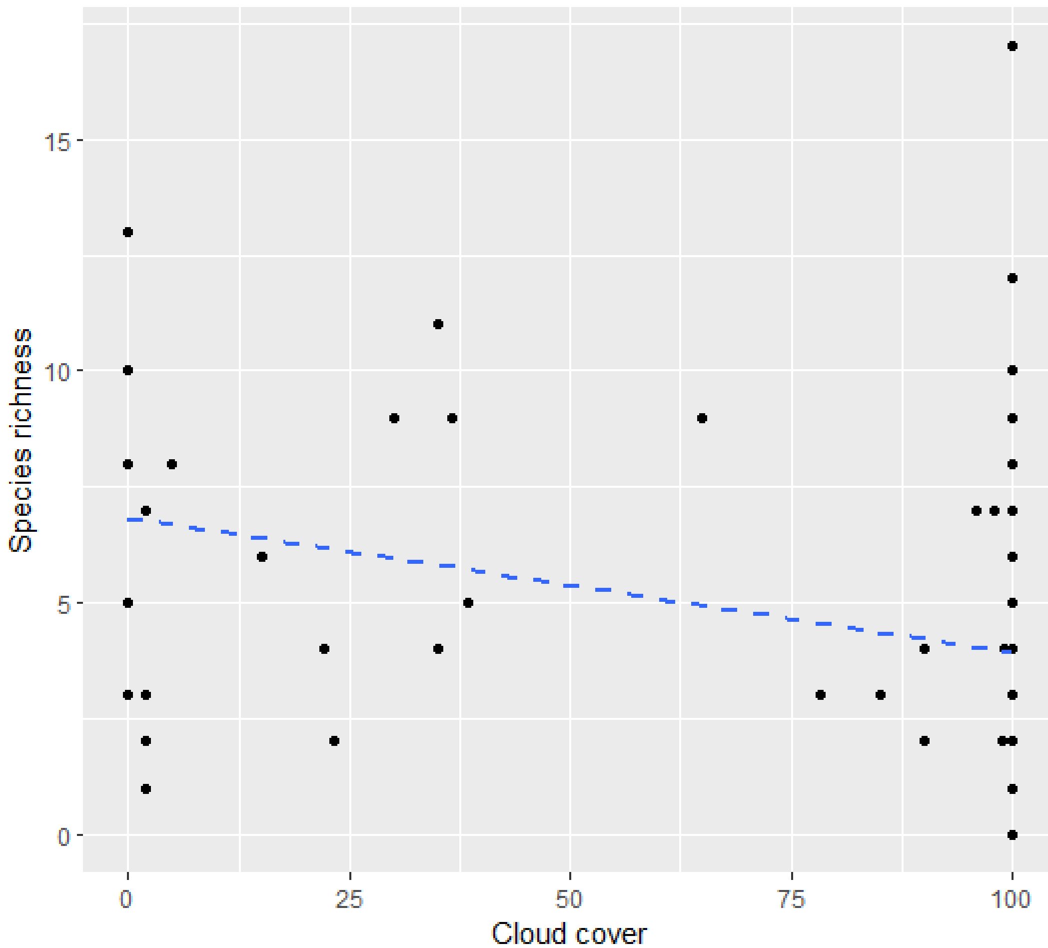

Figure 3.31 – Fitted line plot of the Pearson correlation test of species richness variation along the cloud cover.

Cloud cover has changed from 0-100 % during the study period, while more observations have been made at both the lowest and highest cloud cover values. There is a little decline in the species richness towards the higher cloud cover percentage.

3.8 Avian community response variables variation along the along the study period

Total species richness, species richness of migrants, and species diversity variation throughout the study period (from August to April; migratory season).

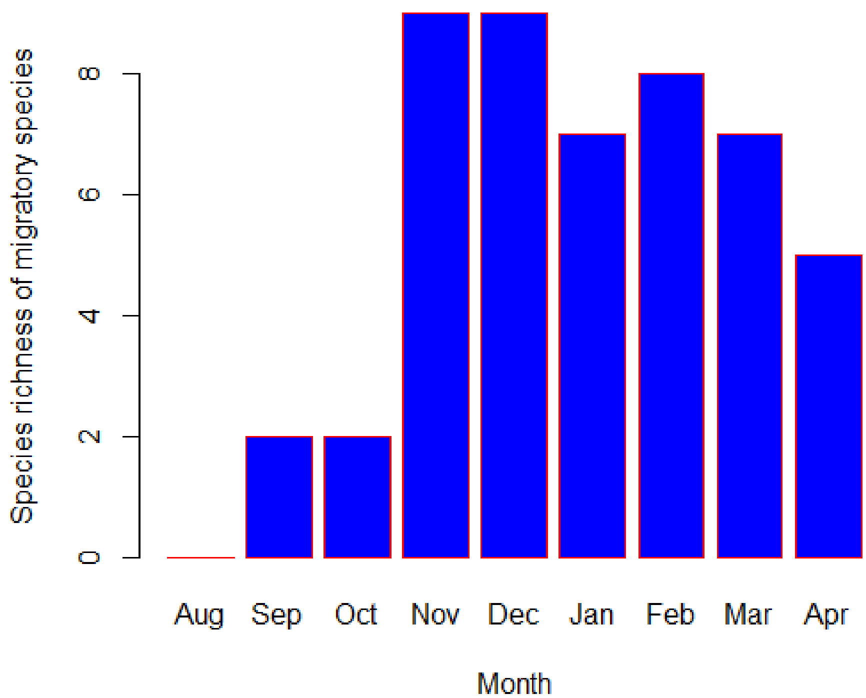

Figure 3.32– Migratory species richness variation along the study period.

There is a clear increment of the migratory species from August to December beginning with zero migrants in August, followed by a gradual decline from December to April.

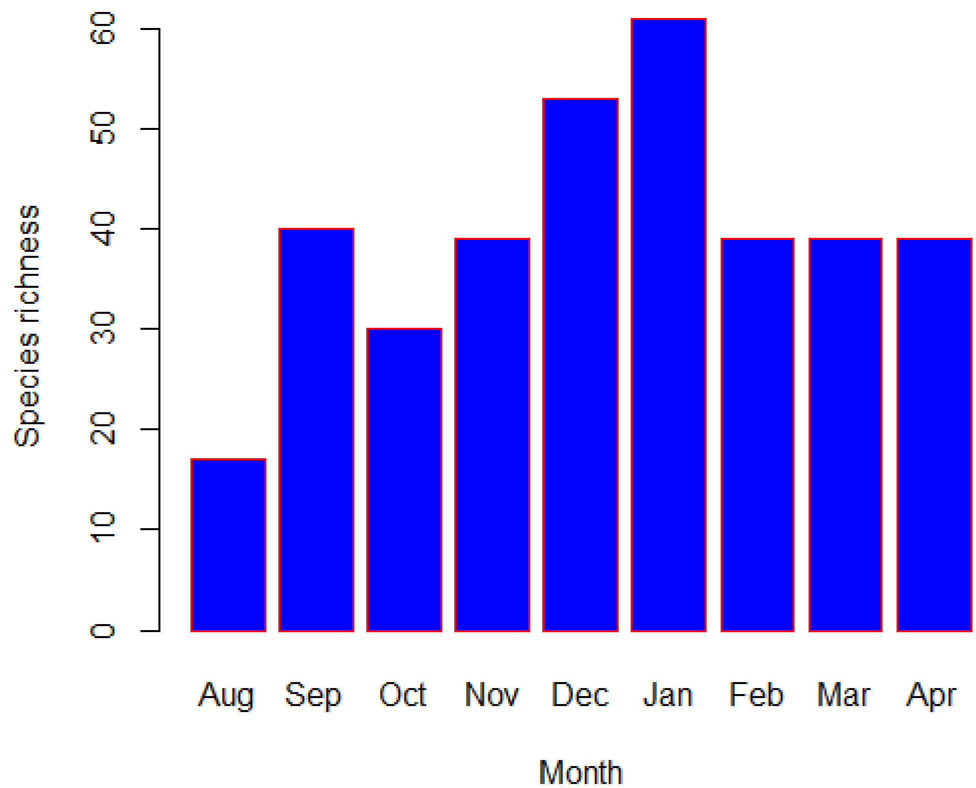

Figure 3.33 – Avian species richness variation along the study period

Species richness has increased from August to January, and then showing a little decline to February March and April, however maintaining a comparatively higher species richness from February to April.

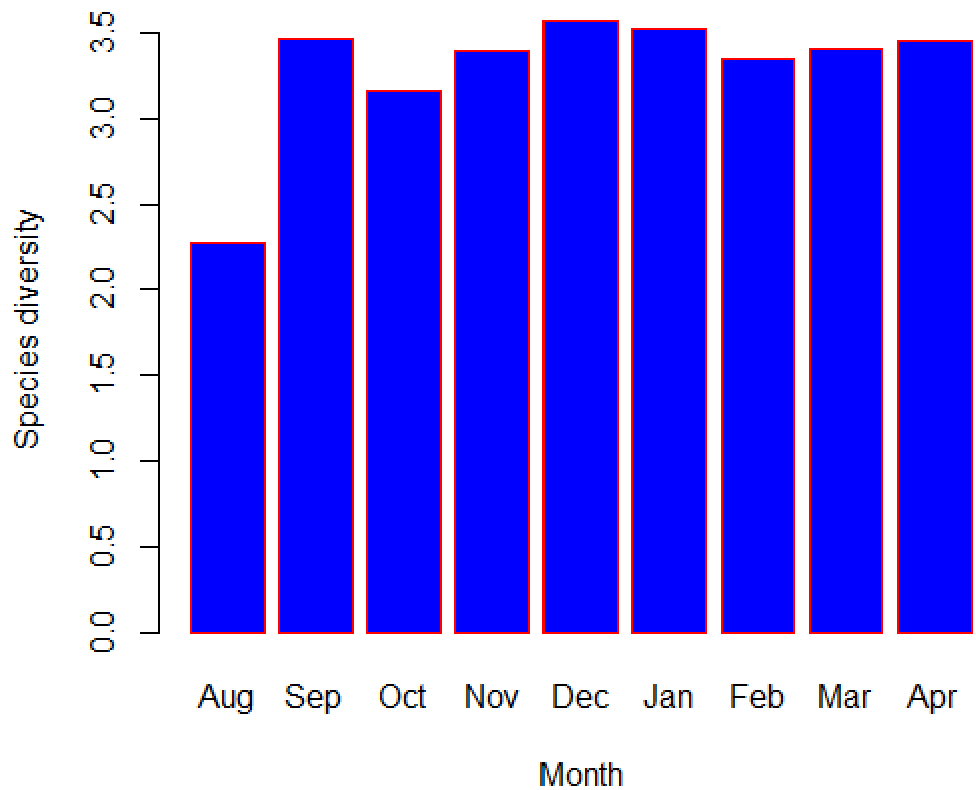

Figure 3.34 – Avian species diversity variation along the study period.

Species diversity depicted by figure 3.24 is mostly similar in all the months except for August. Comparatively there is a considerable decline in the species diversity in August.

3.9 - Elevational ranges of the avian community

Table 3.7 -The presence of the avian species along the elevation gradient.

Species	480	580	700	780	870	1000	1100	1250	1360	1420
Spurfowl	1	1	1	1	1	1	1	1	1	1
Jungle fowl	1	1	1	1	1	1	x	x	x	3
Crested Serpant Eagle	1	1	1	1	1	1	1	1	1	1
Indian swiftlet	1	1	1	1	1	1	1	1	1	1
Yellow fronted Barbet	1	1	1	1	1	1	1	1	1	1
Brown capped Babbler	1	1	1	1	1	1	X	1	1	1
Scimitter Babbler	1	1	1	1	1	1	1	1	1	1
Tickell's blue Flycatcher	1	1	1	1	1	1	1	1	1	1
Pale billed Flowerpecker	1	1	1	1	1	1	1	1	1	1
Spot winged Thrush	1	1	1	1	1	1	1	1	1	1
Blue-tailed Bee-eater	1					1	1	1	1	
Banded bay Cuckoo	1	1	1	1	1	1	1	1		
Common Iora	1	1	1	1	1	1	1	1		
Black capped Bulbul	1	1	1	1	1	1	X	1		
Asian brown Flycatcher	1	1	1	1	1	1	X	1		
Brown headed barbet	1	1	1	1	1	1				1
Brown breasted Flycatcher	1	1	1	1	1	1				1
Orange minivet	1	1	1				1	X	1	1
Barn Swallow	1					1	1	1	1	1
Sri Lanka Hanging Parrot	1	1	1	1	1	1	1			
Red vented Bulbul	1	1	1	X	1	1	1			
Square tailed bulbul	1	1	1	1	1	1	1			
Crimson fronted Barbet	1	1	1	1	1	1				

	480	580	700	780	870	1000	1100			
Asian Paradise Flycatcher	1	1	1	1	1	1				
Loten's sunbird	1	1	1	X	1	1				
Oriental white eye	1	1	1	1	X	1				
Velvet fronted nuthatch	1	X	1	X	X	1				
Sri Lanka Lesser Flameback	1	1	1	1	1					1
Indian Cuckoo	1	1	1	1	1					
White browed Bulbul	1	1	X	X	1					
Greater coucal	1	1	1	1						
Black hooded Oriole	1	1	1	1						
Common Tailorbird	1	1	1	1						
Dark fronted Babbler	1	1	1	1						
Purple rumped Sunbird	1	1	1	1						
Spotted Dove	1	1	1							
Emerald Dove	1	1	1							
Dwaft kingfisher	1	1	1							
White bellied drongo	1	1	1							
Yellow billed Babbler	1	1	1							
Lesser hill myna	1	1	1	1	1					
Oriental magpie Robin	1	1								
Indian Blue Robin	1	1								
Indian peafowl	1									
Common Kingfisher	1									
Jerdon's leafbird	1									
Common mynah	1									
Large billed leaf warbler		1	1	1	1	1	X	1	X	1
Brown shrike		1	1	1	X	X	1	1		
Common Hawk Cuckoo		1	1	1	1	1	1			
Ashy Prinia		1	X	X	X	1	1			
White rumped Shama		1	1	1	1					1
Chestnut backed Owlet		1	1	1						1
Layerd's Parakeet		1	1	1						

	480	580	700	780	870	1000	1100	1250	1360	1420
Sri Lanka Grey Hornbill		1	1	1						
Purple sunbird		1	X	1						
Sri Lanka Frogmouth		1	1							1
Alexandrian parakeet		1	1							
Rose ringed Parakeet		1	1							
Brown wood owl		1	1							
Golden fronted leafbird		1	1							
Grey breasted Prinia		1	1							
Plum headed parakeet		1								
Plain prinia		1								
Black eagle		1	1	X	1	1	1	1	1	1
Great tit		1	1	X	X	X	X	1	X	1
Sri Lanka Green pigeon		1	1	1	1	X	X	1		
Black-naped Monarch		1	1	X	1	1				
Drongo cuckoo		1	1	1						
Wood pigeon			1	1	1	1				
Dull blue Flycatcher			1	X	X	1				
Malabar Trogon			1	1	1					
Coppersmith Barbet			1	1	1					
Green Warbler			1	1						1
Indian pitta			1							
Bar-winged Flycatcher-shrike			1							
Jungle prinia				1	X	X	1	X	X	1
Asian palm swift					1	1	1	1	1	1
Sri Lankan Swallow					1	1	1	1	1	1
Yellow browed Bulbul					1	1				1
Little swift					1	1				
Sri Lanka Woodshrike					1					
Brahminy kite						1	1	1	1	1
Oriental Honey-buzzard						1	1	1	1	1
Sri Lanaka Drongo						1				
Rufous bellied Eagle							1	1	1	1
Legg'sHawk eagle								1	1	1
Hill Swallow								1	1	1

Paddyfield Pipit	1	1	1
Zitting cisticola	1	1	1
Sri Lankan whiteye	1	X	1
Peregrine Falcon	1		
King quail			1
White bellied sea eagle			1
Yellow eared Bulbul			1
Orange billed Babbler			1
Grey-headed Canary Flycatcher			1

x **Particular species was not recorded in these elevations but could be available since the species have been recorded either side the elevation.**

Some elevation ranges were not marked as “x” because there is a possibility of appearing species at 1420 m leeward side forest patch without appearing at adjacent elevations on the windward side.

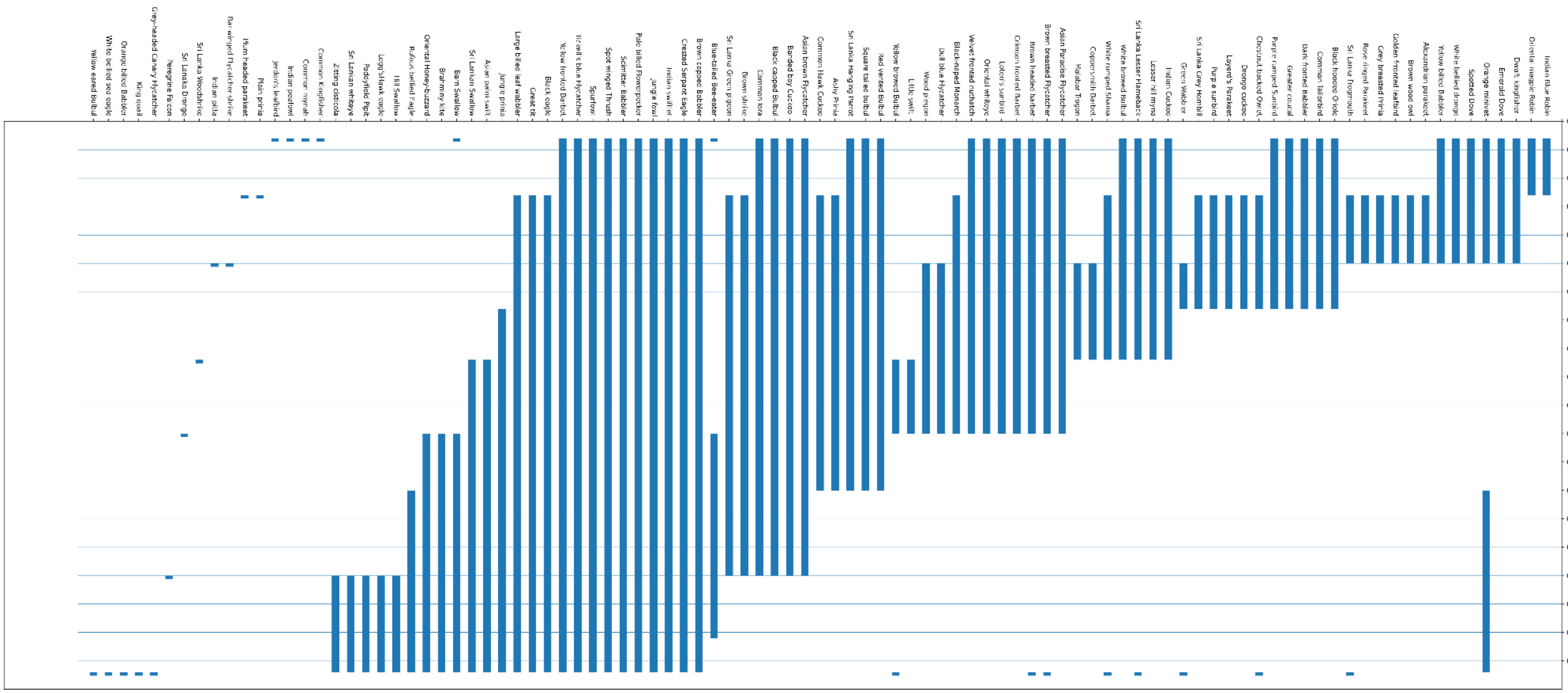

Figure 3.35 – The distribution of the avian community along the elevation gradient. (Graph courtesy; Mr.Janith Weeraman)

3.10 Associations among avian communities along the elevation gradient.

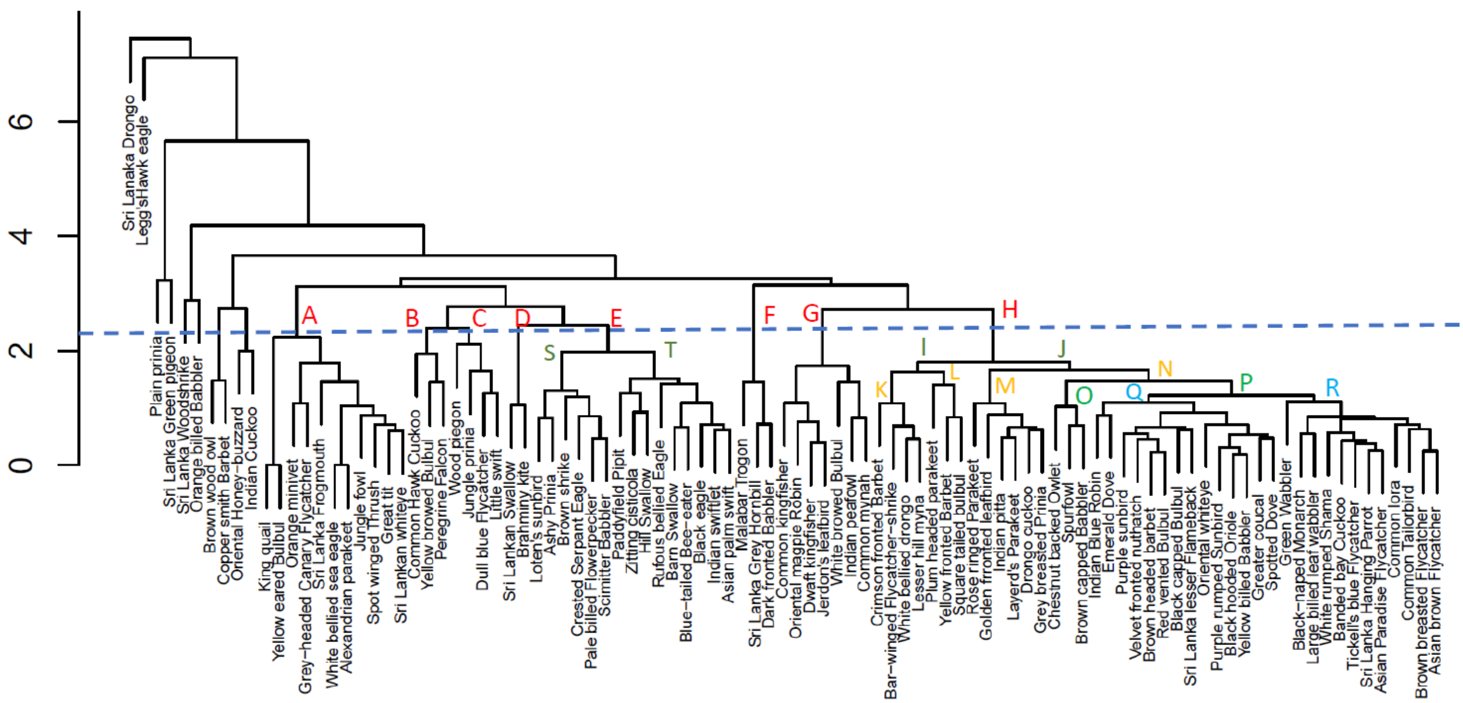

Figure 3.36 - Cluster dendrogram with sensitivity line (dark dashed line), 8 main clades (red), 4 subclades of E and H (dark green), 4 subclades of I and J (yellow), 2 subclades of N (light green), and 2 subclades of P (light blue)

The cluster dendrogram of all recorded species at the study site can be divided into 8 main clusters using 2.5 sensitivity phenon line. Main clade was divided into subclade S and T. Main clade H was divided into I and J subclades. I and J subclades were again divided into K, L and M,N respectively. N was divided into O and P. Finally, subclade P was divided into Q and R.

Along these clades and subclades communities can be recognized with respect to the elevation and habitat type.

Figure 3.37 – PCA of 97 recorded species. With reference to cluster analysis -main clade of A; red ellipse, main clade G; light blue, subclade N; brown ellipse, and one of the outer clade; dash lined eclipse

Species name	**Abbreviation**	**Species name**	**Abbreviation**
Alexandrian parakeet	Alp	Indian Blue Robin	IBR
Ashy Prinia	AsP	Indian Cuckoo	InC
Asian brown Flycatcher	AbF	Indian peafowl	Indnpf
Asian palm swift	Aps	Indian pitta	Indnpt
Asian Paradise Flycatcher	APF	Indian swiftlet	Ins
Banded bay Cuckoo	BbC	Jerdon's leafbird	J'l
Bar-winged Flycatcher-shrike	B-F	Jungle fowl	Jnf

Barn Swallow	BrS	Jungle prinia	Jnp
Black-naped Monarch	B-M	King quail	Knq
Black capped Bulbul	BlcB	Large billed leaf wabbler	Lblw
Black eagle	Ble	Layerd's Parakeet	L'P
Black hooded Oriole	BhO	Legg'sHawk eagle	L'e
Blue-tailed Bee-eater	B-B	Lesser hill myna	Lhm
Brahminy kite	Brk	Little swift	Lts
Brown breasted Flycatcher	BbF	Loten's sunbird	L's
Brown capped Babbler	BrcB	Malabar Trogon	MlT
Brown headed barbet	Bhb	Orange billed Babbler	ObB
Brown shrike	Brs	Orange minivet	Orm
Brown wood owl	Bwo	Oriental Honey-buzzard	OH-
Chestnut backed Owlet	CbO	Oriental magpie Robin	OmR
Common Hawk Cuckoo	CHC	Oriental whiteye	Orw
Common Iora	CmI	Paddyfield Pipit	PdP
Common mynah	Cmm	Pale billed Flowerpecker	PbF
Common Tailorbird	CmT	Peregrine Falcon	PrF
Copper smith Barbet	CsB	Plain prinia	Plp
Crested Serpant Eagle	CSE	Plum headed parakeet	Php
Crimson fronted Barbet	CfB	Purple rumped Sunbird	PrS
Dark fronted Babbler	DfB	Purple sunbird	Prs
Drongo cuckoo	Drc	Red backed Flameback	RbF
Dull blue Flycatcher	DbF	Red vented Bulbul	RvB
Dwaft kingfisher	Dwk	Rose ringed Parakeet	RrP
Emerald Dove	EmD	Rufous bellied Eagle	RbE
Golden frronted leafbird	Gfl	Scimitter Babbler	ScB
Great tit	Grt	Spot winged Thrush	SwT
Greater coucal	Grc	Spotted Dove	SpD
Green Wabbler	GrW	Spurfowl	Spr
Grey-headed Canary Flycatcher	GCF	Square tailed bulbul	Stb
Grey breasted Prinia	GbP	Sri Lanaka Drongo	SLD
Hill Swallow	HlS	Sri Lanka Frogmouth	SLF

Species name	**Abbreviation**
Sri Lanka Green pigeon	SLGp
Sri Lanka Grey Hornbill	SLGH
Sri Lanka Hanging Parrot	SLHP
Sri Lanka Whistling Thrush	SLWT
Sri Lanka Woodshrike	SLW
Sri Lankan Swallow	SLS
Sri Lankan whiteye	SLw

Tickell's blue Flycatcher	TbF
Velvet fronted nuthatch	Vfn
White bellied drongo	Wbd
White bellied sea eagle	Wbse
White browed Bulbul	WbB
White rumped Shama	WrS
Wood piegon	Wdp
Yellow billed Babbler	YllwblB
Yellow browed Bulbul	YllwbrB
Yellow eared Bulbul	YeB
Yellow fronted Barbet	YfB
Zitting cisticola	Ztc

In this PCA analysis it is difficult to display all the clustered groups as in the dendrogram. Therefore, some possible clusters are shown in the figure 3.37. This PCA analysis results support to strengthen the results obtained in the cluster analysis, since both results provide the same clustering pattern.

CHAPTER 4 – DISCUSSION

A total number of 97 avian species belonging to 42 families and 14 orders, were recorded by 752 observations from August 2022 to April 2023 along a 480 m to 1420 m elevation gradient surrounding the Issengard Biosphere Reserve, at Belihuloya in Rathnapura district of Sabaragamuwa province, Sri Lanka. The avian community comprises 21 endemic species which is 61.76 % of the endemic percentage when compared to the Sri Lankan endemic species richness, 11 migratory species, 7 montane species, and 19 threatened species (MoE,2021). Most of the species come under the families Muscicapidae and Accipitridae.

4.1 Avian responses to the cloud line

Cloud line is an interesting feature observed at my study site. Avian community response to the cloud line was recorded for the first time in Sri Lanka in this study. Species richness strongly responds to the cloud line (Figure 3.1). Avian species richness increases at or towards the cloud line, and decreases away from the cloud line. The accumulation of both montane and forest specialist under the cloud line might cause the higher species richness under the cloud line.

4.2 Responses of avian community to the elevation and habitat

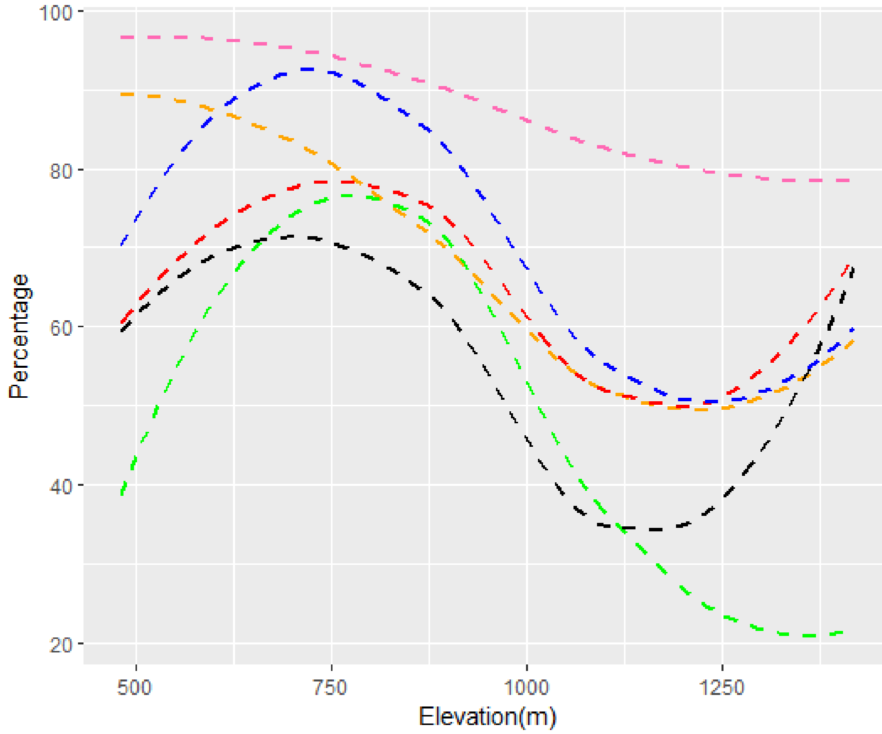

Figure 4.1 – Similar patterns of variation for several avian community, vegetation, and habitat parameters along the elevation gradient.

---- Total species richness, ---- Endemic species richness, ---- Species diversity, ---- Habitat complexity, ---- Canopy cover, ---- Tree height

There is a significant similarity in patterns of the variation in avian community, vegetation, and habitat parameters along the elevation gradient (Figure 4.1). This study proves the contribution of elevation position, vegetation structure, and habitat type to predict the avian community parameters. Jankowski et al., (2013) have also mentioned elevation position, vegetation structure, and tree composition as cardinal factors to determine avian community. The choice of the habitat for birds is influenced by the physical structure of the vegetation

(Able & Noon, 1976), (R. H. MacArthur & MacArthur, 1961), (R. H. MacArthur et al., 1962). The variations of the avian community responses are not only affected by the elevation due to geographical changes but also the vegetation and habitat type deviations along the elevation gradient. Similarly, endemic species richness, habitat complexity, tree density, canopy cover, and tree height have gradually increased and then have a gradual decline until around 1250 m.

The peak of the species richness, and species diversity appear at 780 m (Table 3.1). From the vegetation and habitat variabilities, the habitat complexity and canopy cover are also highest at the 780 m (Table 3.5). When considering all ten elevations, lower values of the species richness, and species diversity are found at 1360 m and 1100 m grasslands (Table 3.1). The habitat complexity and canopy cover also give lower values at 1360 m and 1100 m (Table 3.5).

When the elevations are compared except for grasslands, the lowest value given is 870 m for species richness, species diversity, habitat complexity, and canopy cover (Table 3.1 and 3.5). This proves more profoundly the similar pattern variation of the avian community, vegetation and habitat type parameters as figure 4.1.

In total species richness, endemic species, and habitat complexity there is a slight incline and tree height has a much higher incline towards 1420 m (Figure 4.1). Therefore, the tallest trees are recorded at 1420 m at the Haagala peak (Table 3.5).

At lower elevations the tree height does not vary significantly. But when reaching the higher elevations the tree height decreases with increasing elevational gradient (Mao et al., 2018).

From 480 m to 1360 m, my transects are placed at the windward side of the mountain. In contrast, the 1420 m transect at the peak of the Haagala mountain spreads towards the leeward side of the mountain, as this area gets south-western monsoon rainfall.

The height of the trees on the windward side of the mountain is lower, and on the leeward side of the mountain it is higher (Brandle & Finch, 1914) (Means, 2020). Therefore, the drastic increment of the tree height at 1420 m could be due to the touch of the leeward side forest patch at the peak.

The tree species richness is higher on the leeward side than on the windward side (Diogo et al., 2020). My study also tallies with Diogo et at.,(2020) as showing the highest tree density reporting at 1420 m from the entire elevation gradient (Table 3.5).

At 1420 m the richness of endemic species, montane species, and threatened species have inclined. Some species such as Yellow-eared bulbul, Grey-headed canary flycatcher which are both endemics, and montane species and King quail a threatened species were only observed at 1420 m (Table 3.7 & Figure 3.35).

Many studies on species richness along the elevation gradients have displayed hump-shaped relationships, as the mid domain effect; towards the center of the mountain species ranges increases due to the shared geographical domains (Colwell & Lees, 2000). Bird studies in the Andes mountains (Herzog et al., 2005), trees in Costa Rica (Lieberman et al.,1996), ants in the Western United States (Sanders,2002), and small mammals in the Philippines (Heaney,2001) are some examples. Here, in this study also, we found moderate hump-shape variation in total species richness, more prominent shape in the species richness of endemics and even more profound hump-shape in habitat complexity, canopy cover, and tree height except for the 1420 m highest elevation (Figure 4.1). Therefore, not only species richness but in vegetation and habitat parameters, we can see this hump-shape variation.

The reason for not appearing the initial clear inclining part of the hump-shape model could be that the starting point elevation of the study area where the lowest point at Samanala river basin is 480 m. Therefore, 0 – 480 m lower elevational region does not appear in this study area.

4.3 Avian community variations with the climatic variables

Mountain climate also plays a big role to maintain a unique ecosystem. The temperature of the study area varied from 17 – 35 °C (Figure 3.26). The recorded temperature generally falls around 22-25 °C. Birds are more active at moderate temperatures and their activity decreases at either higher or lower temperatures. Gillman and Wright (2014) have also shown that the species richness increases at moderate temperatures, which tallies with this study's results (Gillman & Wright, 2014).

Relative humidity indicated a positive correlation with species richness (Figure 3.28, $P<0.05$). During the day, strong upslope moisture transport can be seen in the diurnal cycle in humidity. This is counterbalanced by the downslope transport leading to a dry weather at night. Therefore, diurnal humidity enhances towards the mountain summit (Fig 3.27; Duane et al., 2008). In this study also during daytime, greater than 75% relative humidity was generally recorded.

Wind speed has a negative correlation with species richness (Figure 3.28, $P<0.05$). During the study period most observations were recorded at zero wind speed at low elevations and mostly the wind was present at higher elevations. Raptors who use the soaring flight pattern are more available at high elevations where the wind is regularly present in this study area. For different flight strategies of raptors, the effect of wind; wind speed and direction, flight elevations, and thermal convections matter (Vansteelant et al., 2014), (Bruderer et al.,1994), (Kerlinger and Gauthreaux, 1984), (Maransky et al., 1997).

Light intensity has a negative correlation with species richness (Figure 3.29, $P<0.05$). In the forest, light intensity is much less than in an open area. During the observations, less than 25 kLux light intensity was recorded. Light intensity has a significant effect on the physiological and behavioral processors, regulation of growth, and reproduction (Kang et al., 2020). Physiological roles are affected as sight facilitation, releasing the regulated reproductive hormones, and affecting social behaviors (Deep et al., 2010), (Manser, 1996), (Olanrewaju et al., 2018).

Species richness has a negative correlation with cloud cover (Figure 3.31). No cloud cover or presence of a low cloud cover has shown higher species richness. The type and the amount of cloud cover could impact the precipitation conditions. Therefore, the presence of more cloud cover can cause rain. Thus, the birds' activity; movements and calls could reduce, resulting in low avian observations (Craig & Roberts, 2001).

Visibility showed a positive correlation with species richness (Figure 3.30, $P<0.05$). Mostly during the study period, 100% visibility was recorded. With the presence of mist at the transect in the elevation the visibility of species decreases. Therefore, with the mist, the activity of the birds could also reduce (Craig & Roberts, 2001).

This study was conducted throughout the migratory season. As shown in the figure 3.32, at the beginning of the migratory season; August none of the migrants were recorded. Then gradually migratory species richness increased to a peak around December. Finally, a little decline can be observed towards April. The total species richness (Figure 3.33) gradually increases towards December and January. Then it declines to a moderate level by February, March, and April. In the species diversity (Figure 3.34), from September to April generally similar diversity values appear. But the diversity has reduced in August. At the beginning of the migratory season none of the migrants were present. It might be a reason for this decline in diversity.

4.4 Elevation ranges of the species

From the 97 recorded species ten species can be found throughout the elevation gradient. They are, Spurfowl, Jungle fowl, Crested Serpant Eagle, Indian Swiftlet, Yellow fronted Barbet, Brown capped Babbler, Scimitar Babbler, Tickell's blue Flycatcher, Pale billed flowerpecker and Spot winged Thrush. These species belong to family Phasianidae, Accipitridae, Apodidae, Megalamidae, Pellorneidae, Timaliidae, Muscicapidae, Dicaeidae, and Turdidae. There are Galliformes, Accipitriformes, Strigiformes, and Passeriformes. According to the foraging guild, they are Granivores, Carnivores, Insectivores, Fruigivores and Nectarivores. Thus, these foraging guilds can be facilitated at each elevation.

Blue-tailed Bee-eaters appeared at 480 m, 1000 m, 1100 m, 1250 m, and 1360 m elevations. These elevations are more open areas where it facilitates to maintain it's insectivores foraging type.

4.4 Avian community segregation

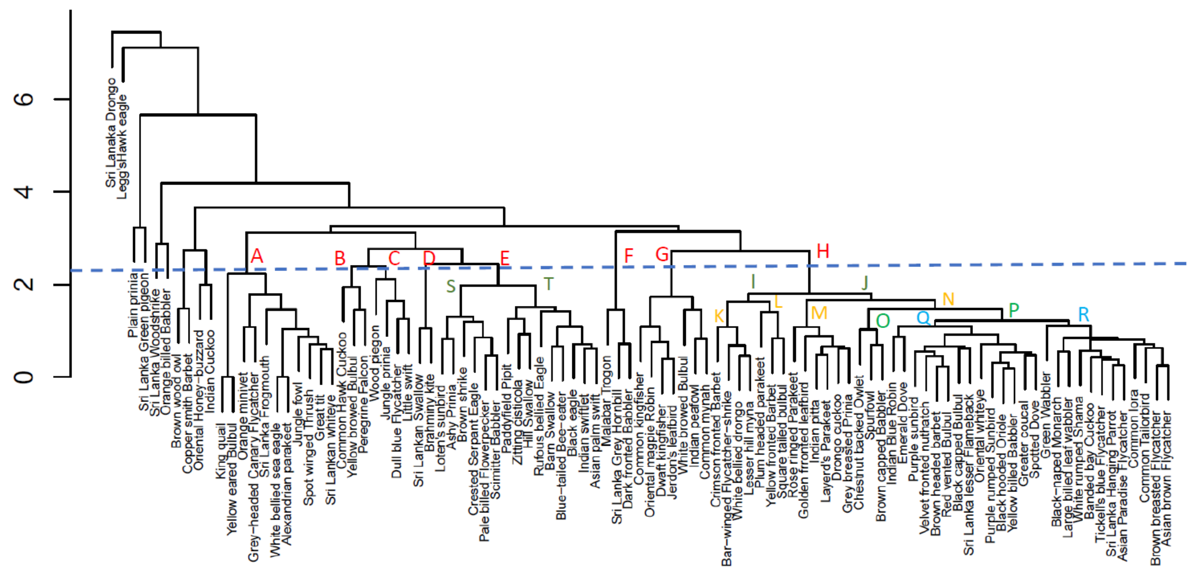

Figure 5.2 - Cluster dendrogram with sensitivity line (dark dashed line), 8 main clades (red), 4 subclades of E and H (dark green), 4 subclades of I and J (yellow), 2 subclades of N (light green), and 2 subclades of P (light blue)

The cluster dendrogram of all recorded species at the study site can be divided into 8 main clusters using 2.5 sensitivity phenon line. Cluster A contains 11 species of the community that associate 1420 m. This community contains species under the order Galliformes, Accipitriformes, Psittaciformes, Stringiformes, and Passeriformes. Considering foraging guild this community comprises granivores; Sri Lankan Spurfowl, Sri Lankan Jungle fowl, omnivores; Kind Quail, Sri Lanka Spot-winged Thrush, carnivores; White bellied Sea eagle, Sri Lankan Chestnut backed owlet, Sri Lankan Frogmouth, frugivores; Alexandrine Parakeet, Sri Lankan Yellow eared Bulbul, and insectivores; Orange Minivet, Great Tit, Grey-headed Canary Flycatcher. The association of different foraging guilds cause this community to exist in the same habitat with less competition.

Clade C species associate 870 m and 1000 m elevations as a community. Wood Pigeon is a granivore whereas others are insectivores. From them Wood Pigeon and Dull blue Flycatcher do not extend beyond 1000 m. Jungle Prinia and Little swift can associate 1420 m as well. Apart from that except for Little Swift others collectively associate 780 m – 1000 m whole elevation range as a community.

Clade E has been divided into subclades as S and T. Subclade S species associate 580 m – 1000 m range as a community. The foraging guild of this community is as follows. Loten's sunbird, and Pale billed Flowerpecker are nectarivores. Ashy Prinia, Brown Shrike, and Scimitar Babbler are insectivores. Crested Serpent eagle is a carnivore. In this community Crested Serpent Eagle, Pale billed Flowerpecker, and Scimitar Babbler act on 480 m – 1420 m whole elevational gradient as a community.

The subclade T represents a community that associates 1250 m and 1360 m elevations. The foraging guild of this community is insectivores; Paddyfield Pipit, Zitting Cisticola, Hill Swallow, Barn Swallow, Blue-tailed Bee-eater, Indian Swiftlet, Asian Palm Swift, and carnivores; Rufous-bellied eagle, Black eagle. As these two elevations partly or completely associate with grasslands more insectivores and birds of prey can be seen. From the species of this community except Blue-tailed Bee-eater others share 1250 m – 1420 m range as a community too.

Clade F contains Malabar Trogon, Sri Lankan Grey Hornbill, and Dark fronted Babbler who associate 780 m as an assemblage. They are omnivores, omnivores and insectivores respectively.

The species of the clade G associate 480 m elevation as a community. The foraging guild of this community is omnivores; Common Kingfisher, Oriental Dwarf kingfisher, Common Myna, insectivores; Oriental Magpie-robin, Jerdon's leafbird, frugivores; White-browed Bulbul, and granivores; Indian Peafowl.

The clade H has been divided into subclades I and J. These sister clades have been branched again as K,L and M,N respectively. K subclade community associates 700 m as frugivores; Crimson-fronted Barbet, insectivores; Bar-winged Flycatcher shrike, White-bellied Drongo, and omnivores; Lesser Hill Myna. In contrast, L subclade community associates 580 m. Plum-headed parakeet, Yellow fronted Barbet and Square-tailed Bulbul all share the frugivore foraging guild.

M subclade community exist at 700 m . Except for Indian Pitta, others associate 580 m elevation as well. Rose-ring Parakeet, and Layard's Parakeet are frugivores. Indian Pitta, and Grey-breasted Flycatcher are insectivores and Drongo Cuckoo is omnivores.

The Main N subclade has further grouped into two branches as O and P. Branch O consists of 1420 m associating endemic species as Chestnut Backed Owlet; carnivore, Spurfowl; granivore, and Brown capped babbler; insectivore. From them Spurfowl and Brown capped Babbler associate 480 m to 1420 m whole elevation gradient together.

Branched P finally sub-branched into Q and R. Q community associates 480 m. Their foraging guild consists of insectivores; Indian blue Robin, Velvet-fronted Nuthatch, Sri Lanka Lesser Flameback, Oriental White-eye, granivores; Emerald Dove, Spotted Dove, nectarivores; Purple Sunbird, Purple-rumped Sunbird, frugivores; Brown-headed Barbet, Red-vented Bulbul, omnivores; Black-hooded Oriole, Yellow-billed Babbler, and carnivores; Greater Coucal. This community contains five foraging strategies.

Finally, the R subclade community comprised species that associate 700 m and 780 m / value 6 habitat complexity region. Their foraging guild can be analyzed as insectivores; Green Warbler, Black-naped Monarch, Large-billed Leafwarbler, White-rumped Shama, Tickell's Blue Flycatcher, Asian Paradise flycatcher, Common Iora, Common Tailorbird, Brown-breasted Flycatcher and Asian Brown Flycatcher, omnivores; Banded Bay Cuckoo, and frugivores; Sri Lanka Hanging parrot. The majority of this community are insectivores. Other than the Green Warbler other species of this community associate 580 m also collectively.

Figure 4.3 – PCA of 97 recorded species. With reference to cluster analysis -main clade of A; red ellipse, main clade G; light blue, subclade N; brown ellipse, and one of the outer clade; dash lined eclipse

In this PCA analysis it is difficult to display all the clustered groups as in the dendrogram. Therefore, some possible clusters are shown in the figure 4.3. This PCA analysis results support to strengthen the results obtained in the cluster analysis, since both results provide the same clustering pattern.

4.5 Montane range

There are rapid climate variations throughout the elevation gradient of this study area but profoundly it varies at the above 1100 m elevations as sudden appearance of a thick mist and sometimes followed by rain.

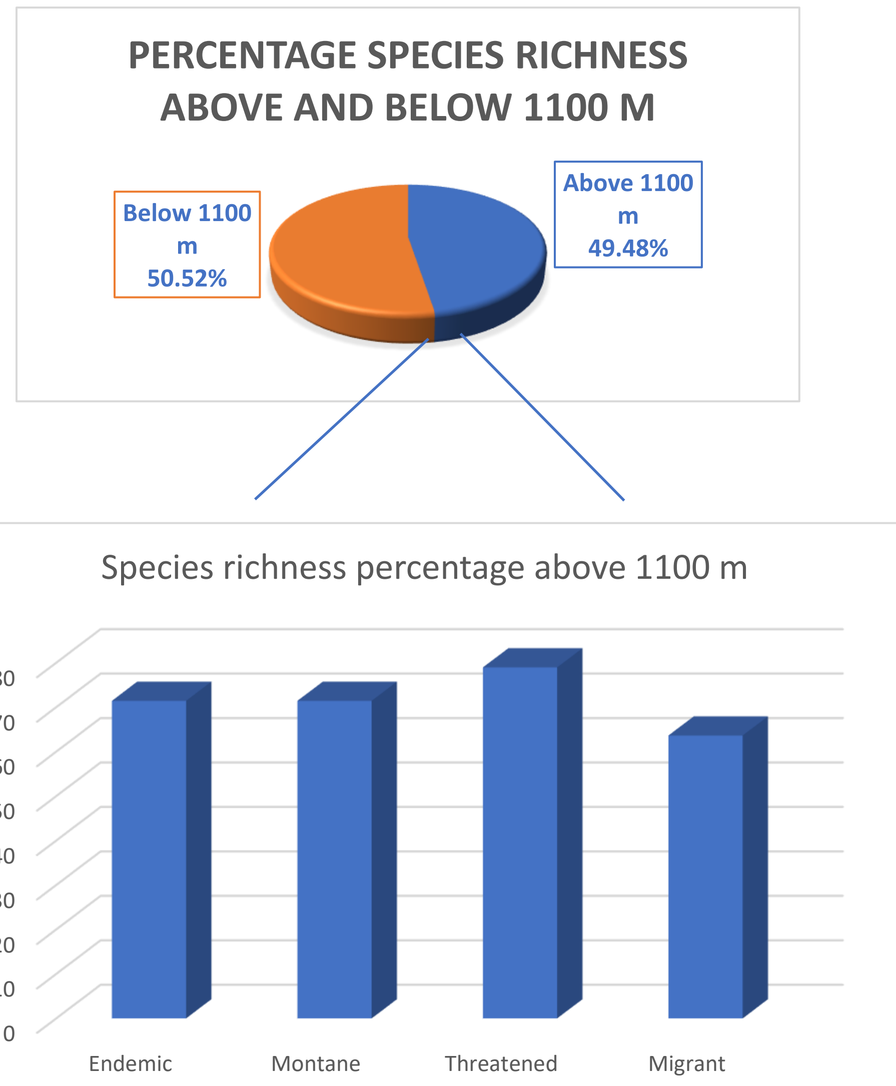

Figure 4.4: According to species recorded, species richness above and below the 1100 m (Pie chart). From them endemic, montane, threatened, and migratory species percentages above 1100 m

From the total species richness, 49 species; 49.48 % survive above 1100 m. It proves that nearly 50% of species have the adaptations to withstand rapid climatic variations of the higher elevations at Issengard Biosphere Reserve. From a total of 21 endemics, 15 species; 71.43%, 5 out of 7 montane species; 71.43%, 15 threatened species; 78.95%, and 7 migrants; 63.63% are available above 1100 m. This makes it evident that mostly endemic, montane, threatened, and migratory species of this study area have nucleated to this specific ecosystem evolving their adaptations. Thus, conserving the prevailing ecosystem of upper elevations of Issengard Biosphere Reserve is vital.

Endemic species are restricted to a specific geographical region while, montane species occur at mesoscale climatic variations (Ruggiero & Hawkins, 2008). Threatened species limited to some areas as their area of occupancy could be less than 2000 km^2. Migratory species also depend upon the properties of interconnected regions (Runge et al., 2015). Due to these reasons, these specific avian community assemblages occupy the most fitted space for them to occupy along the elevational gradient. Thus, specially upper elevations of the Issengard Biosphere Reserve deserve a high conservation priority with the presence of these avian communities.

When considering the movement of the six endemic species those who were not recorded in the upper Issengard region, Sri Lankan Grey Hornbill (*Ocyceros gingalensis*), Crimson Fronted Barbet (*Psilopogon rubricapillus*), Sri Lankan Drongo (*Dicrurus forficatus*), and Sri Lankan Woodshrike (*Tephrodornis affinis*) spread up to mid hills (Warakagoda et al., 2012). Wood pigeon is uncommon in lowlands and it is a rare breeding residence (Warakagoda et al., 2012). In addition, Dull-blue Flycatcher (*Eumyias sordidus*) is rather uncommon in mid hills and above but occasionally descends to lower elevations (Warakagoda et al., 2012). The study conducted at Horton Plains National Park by Dharmarathne and Mahaulpatha, found that the preferred habitat of Dull-blue Flycatcher is montane Forest (Dharmarathne & Mahaulpatha, 2017). But they have noticed during their study that the habitat characteristics such as having considerable canopy cover had played a huge role in the habitat selection of Dull-blue Flycatcher. In our study area Dull-blue Flycatcher was recorded at 780 m, 870 m, 1000 m elevations. They are the higher canopy cover elevations towards the upper elevations and above the 1000 m the canopy covers are less than them.

Two montane species that were not recorded at upper Issengard, are Sri Lankan wood pigeon and Dull-blue flycatcher. Their absence was discussed previously. Montane species are usually exposed to higher weather fluctuations. Therefore, in response to adverse weather conditions many montane species move to lower elevations, placing the altitudinal migration (Walther et al., 2017). But during the study period a clear altitudinal migration of avian species was not detected.

The four migrants who were not available at high elevations are Asian Paradise-Flycatcher (*Terpsiphone paradisi*), Indian Cuckoo (*Cuculus micropterus*), Indian Blue Robin (*Larvivora brunnea*), and Indian Pitta (*Pitta brachyura*). Their distributions are dry lowlands and adjoining dry lower hills, lowest to mid hills, throughout the country, and except highest hills throughout the country respectively (Warakagoda et al., 2012). These distributions align with the elevational gradients that were recorded in this study with respect to these species.

Threatened species that were not recorded at higher elevations are Banded Bay Cuckoo (*Cacomantis sonneratii*), Sri Lankan Wood Pigeon (*Columba torringtoniae*), Sri Lankan Drongo (*Dicrurus forficatus*), and Dull-blue Flycatcher (*Eumyias sordidus*). From these four except for Banded Bay Cuckoo the distributions of others were discussed above. Banded Bay Cuckoo could also be observed from lowlands to mid hills (Warakagoda et al., 2012) as we recorded.

Number of 7 Accipitriformes and 1 Falconiformes who are carnivores in foraging guild, as the birds of prey indicated at this study site. Peregrine Falcon (*Falco peregrinus*), Legg's Hawk eagle (*Nisaetus kelaarti Legge*), Black eagle (*Ictinaetus malayensis*), Rufous Bellied eagle (*Lophotriorchis kienerii*), Oriental Honey Buzzard (*Pernis ptilorhyncus*), White bellied Sea eagle (*Haliaeetus leucogaster*), Brahmini Kite (*Haliastur indus*), and Crested Serpent eagle (*Spilornis cheela*) are generally found above the 1100 m. Apart from that Crested Serpent eagle throughout the elevation gradient, Black eagle except for 480 m, throughout the elevation gradient, Brahmini Kite and Oriental Honey Buzzard at 1000 m were observed. They feed on mostly small mammals, and reptiles. In order to detect and capture prey while soaring or gliding at higher elevations their vision has evolved drastically. They possess a distinctive eye ridge, produced by a well-developed superciliary ridge, and the feathers in the region extend above and in front of the eye. Their distinctive, piercing stare is provided by this eyebrow. In Falcons, they have dark feathers in front of and below the eye; malar stripes which help them to reduce glare from the sun when searching for and pursuing fast -moving prey (Jones et al., 2007). They preferably select near the improved grassland area as their habitat (Abram &

Thomas, 2023). In this study area also, birds of prey associate the forest patches and roosting sites near the grassland.

Specifically from the conservation viewpoint it should be mentioned that even though a continuous research has not been completed yet about other animal groups, with the data we collected from our camera traps and observations we have done at the Issengard Biosphere Reserve, it is rich in mammals as, the flagship species Sri Lankan leopard (*Panthera pardus kotiya*), endemic Golden palm civet (*Paradoxy zeylonensis*), Purple-faced Langur (*Trachypithecus vetulus*), Toque macaque (*Macaca sinica*), White-spotted Mouse Deer (*Moschiola meminna*), Dusky striped squirrel (*Funambulus obscurus*), threatened species; Pangolin (*Manis crassicaudata*). Apart from that wild *Bubalus bubalis* herds contribute to maintaining the grassland of upper Issengard. From the invertebrate assemblage specific butterflies as endemic and Sri Lankan national butterfly; Sri Lankan Birdwing (*Troides darsius*), endemic; Sri Lankan Tree nymph (*Idea iasonia*), and Jewel four ring (*Ypthima singala*) Native; Large banded swift (*Pelopidas subochracea subochracea*) , Banded peacock (*Papilo crino*) , and Sawtooth (*Prioneris thestylis*) can be mentioned as some mammalian and butterfly species that we have found at Issengard Biosphere Reserve during our studies. In addition, the driving forces for the maintenance of this particular ecosystem are the contribution of the whole assemblage of fauna, and flora communities, climatic variabilities, and also the concerns and actions of humans towards this ecosystem. Human plays an important role in the conservation strategy since there are many camping sites at Haagala peak and the upper Issengard region where the misbehaviors of human can lead to damaging the sensitivity of this ecosystem.

Year-round active springs and high rainfall at Issengard Biosphere Reserve ensure that the fauna and flora assemblage receive adequate water. This helps to consider the Issengard Biosphere Reserve as a virtuous catchment area.

We have already encountered the deforestation of the forest for firewood of camping sites, and leaving polythene and bottles at Haagala peak. As a conservation action, we placed notice boards mostly at prevailing camping sites at Haagala peak and along the trails through Issengard Biosphere Reserve. Due to the revealed information from this study, more conservation actions and plans can be implemented legally in order to protect this precious ecosystem.

CHAPTER 5 – CONCLUSION

Avian community has recorded responding to the cloud line, for the first time in Sri Lanka during the present study, where the species richness has declined away from the cloud line. The bird species richness, and diversity indicated in a hump-shaped variation along the elevation gradient, adding evidence to the mid-domain effect, while a higher species evenness value had been maintained throughout the elevation gradient. Canopy cover, tree height, and habitat complexity have indicated similar hump-shaped variation as avian community parameters along the elevation gradient. Therefore, responses of avian communities along the elevational gradient are supported by vegetation and habitat topography. The leeward side forest patch at the peak has resulted to increase the tree height and contributed to enhancing the faunal and floral community assemblages, somewhat confounding the patterns observed along the elevation gradient. Along the climatic variations, the avian community was found to be more active at around 25 ºC temperature, above 80% relative humidity, less than 25kLux light intensity, less than 1.2 ms-1 wind speed, and non or less than 50% cloud cover. Obviously, the species richness and diversity of birds within the study site was recorded to increase at the peak of the migratory season. The variation of the vegetation with grasslands found at certain elevations has resulted in declining the avian community whereas, more insectivores and birds of prey were reported to associate these habitats. While endemic, migratory, and threatened species of birds exists throughout the elevation gradient, habitat specialist montane species were found prominently from mid to higher elevations. Rapid and frequent climatic variations are present at elevations above 1000 m, and it was observed that nearly 50% of species surviving above this elevation were able to withstand rapid climatic variations. Higher percentages of endemic, migratory, montane, and threatened species recorded to occupy at higher elevations, indicates the need for targeted conservation activity at upper elevations.

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APPENDIX 1: AVIAN COMMUNITY CHECK LIST

No	Family	Scientific name	Common name	Sp. status	NCS	GCS
1	Phasianidae	*Synoicus chinensis*	E: Asian Blue Quail S: Nil piriwatuwa	Resident	EN	LC
2	Phasianidae	*Galloperdix bicalcarata*	E: Sri Lanka Spurfowl S: Sri Lanka haban kukula	Endemic	VU	LC
3	Phasianidae	*Gallus lafayettii Lesson*	E: Sri Lanka Junglefowl S: Sri Lanka Wali kukula	Endemic	LC	LC
4	Phasianidae	*Pavo cristatus Linnaeus*	E: Indian Peafowl S: Monara	Resident	LC	LC
5	Falconidae	*Falco peregrinus Tunstall*	E: Peregrine Falcon S: Para ukusugoya	Resident	EN	NA
6	Accipitridae	*Pernis ptilorhyncus*	E: Oriental Honey-buzzard S: Silu bambarakussa	Resident	VU	NA
7	Accipitridae	*Haliastur indus*	E: Brahminy Kite S: Bamunu piyakussa, Ukussa	Resident	LC	LC
8	Accipitridae	*Haliaeetus leucogaster*	E: White-bellied Sea-eagle S: Kusa alli muhudukussa	Resident	LC	LC

9	Accipitridae	*Spilornis cheela*	E: Crested Serpent-eagle S: Silu sarapakussa	Resident	LC	LC
10	Accipitridae	*Ictinaetus malayensis*	E: Black Eagle S: Kalukussa, kalu rajaliya	Resident	NT	LC
11	Accipitridae	*Lophotriorchis kienerii*	E: Rufous-bellied Eagle S: Kusarath rajaliya	Resident	EN	NT
12	Accipitridae	*Nisaetus kelaarti Legge*	E: Legge's hawk-eagle (AKA: Mountain Hawk eagle) S: Hel kondakussa	Resident	EN	NE
13	Columbidae	*Columba torringtoniae*	E: Sri Lanka Woodpigeon S: Sri Lanka Maila Paraviya	Endemic	VU	VU
14	Columbidae	*Spilopelia suratensis*	E: Spotted Dove S: Alu-kobeiya	Resident	LC	LC
15	Columbidae	*Chalcophaps indica*	E: Emerald Dove S: Neela kobeiya	Resident	LC	LC
16	Columbidae	*Treron pompadora*	E: Sri Lanka Green-pigeon S: Sri Lanka Pitadam Batagoya	Endemic	LC	LC
17	Psittacidae	*Loriculus beryllinus*	E: Sri Lanka Hanging-parrot S: Sri Lanka giramaliththa	Endemic	LC	LC

18	Psittacidae	*Psittacula eupatria*	E: Alexandrine Parakeet S: Labu girawa	Resident	LC	NT
19	Psittacidae	*Psittacula krameri*	E: Rose-ringed Parakeet S: Rena girawa	Resident	LC	LC
20	Psittacidae	*Psittacula cyanocephala*	E: Plum-headed Parakeet S: Pandu girawa	Resident	NT	LC
21	Cuculidae	*Hierococcyx varius*	E: Common Hawk-cuckoo S: Ukusu kokilaya	Resident	VU	NA
22	Cuculidae	*Cuculus micropterus*	E: Indian Cuckoo S: Indu kokilaya	Migrant	NT	NA
23	Cuculidae	*Cacomantis sonneratii*	E: Banded Bay Cuckoo S: Vaira anukoha	Resident	VU	LC
24	Cuculidae	*Surniculus dicruroides*	E: Drongo Cuckoo S: Kavudukoha	Resident	LC	LC
25	Cuculidae	*Centropus sinensis*	E: Greater Coucal S: Atikukula	Resident	LC	LC
26	Strigidae	*Strix leptogrammica*	E: Brown Wood-owl S: Bora Wanabakamuna	Resident	NT	LC
27	Strigidae	*Glaucidium castanotum*	E: Sri Lanka Chestnut- backed Owlet S: Sri Lanka Pitathambala Upabassa	Endemic	EN	NT

28	Podargidae	*Batrachostomus moniliger*	E: Ceylon Frogmouth S: Madimuhuna	Resident	NT	LC
29	Apodidae	*Aerodramus unicolor*	E: Indian Swiftlet S: Indu upa-thurithaya, Wehilihiniya	Resident	VU	LC
30	Apodidae	*Cypsiurus balasiensis*	E: Asian Palm-swift S: Asia thal-thurithaya	Resident	LC	LC
31	Apodidae	*Apus affinis*	E: Little Swift S: Punchi thurithaya;	Resident	LC	LC
32	Trogonidae	*Harpactes fasciatus*	E: Malabar Trogon S: Lohavannichcha, Ginikurulla	Resident	NT	LC
33	Alcedinidae	*Ceyx erithaca*	E: Oriental Dwarf kingfisher S: Pitakalu Heen pilihuduwa / Rang Pilihuduwa	Resident	NT	LC
34	Alcedinidae	*Alcedo atthis*	E: Common Kingfisher S: Mal Pilihuduwa	Resident	LC	LC
35	Meropidae	*Merops philippinus*	E: Blue-tailed Bee-eater S: Nilpenda binguhariya	Migrant	CR	NA
36	Bucerotidae	*Ocyceros gingalensis*	E: Sri Lanka Grey Hornbill S: Sri Lanka Alu Kandetta; T: Ilankai naarai irattai chondu kuruvi	Endemic	LC	LC

37	Megalaimidae	*silopogon zeylanicus*	E: Brown-headed Barbet S: Polos Kottoruwa	Resident	LC	LC
38	Megalaimidae	*Psilopogon flavifrons*	E: Sri Lanka Yellow fronted Barbet S: Sri Lanka kahamunath Kottoruwa, Mukalang Kottoruwa;	Endemic	LC	LC
39	Megalaimidae	*Psilopogon rubricapillus*	E: Sri Lanka Barbet S: Mal Kottoruwa, Rathmunath Kottoruwa	Endemic	LC	LC
40	Megalaimidae	*Psilopogon haemacephalus*	E: Coppersmith Barbet S: Rathlaya Kottoruwa, Mal Kottoruwa	Resident	LC	LC
41	Picidae	*Dinopium psarodes*	E: Sri Lanka Lesser Flameback S: Sri Lanka Ginipita pili kerala	Endemic	LC	LC
42	Aegithinidae	*Aegithina tiphia*	E: Common Iora S: Podu iorava	Resident	LC	LC
43	Vangidae	*Tephrodornis affinis*	E: Sri Lanka Woodshrike S: Sri Lanka Podu vanasaratiththa	Endemic	LC	LC
44	Vangidae	*Hemipus picatus*	E: Bar-winged Flycatcher-shrike	Resident	LC	LC

			S:Wairapiya masisaratiththa, Panu kurulla			
45	Campephagidae	*Pericrocotus flammeus*	E: Orange Minivet S: Dilirath miniviththa	Resident	LC	LC
46	Oriolidae	*Oriolus xanthornus*	E: Black-hooded Oriole S: Kahakurulla	Resident	LC	LC
47	Dicruridae	*Dicrurus caerulescens*	E: White-bellied Drongo S: Kavuda	Resident	LC	LC
48	Dicruridae	*Dicrurus lophorinus*	E: Sri Lanka Drongo S: Sri Lanka Silu Kauvda	Endemic	EN	LC
49	Monarchidae	*Hypothymis azurea*	E: Black-naped Monarch S: Kalu gelasi radamara	Resident	LC	LC
50	Monarchidae	*Terpsiphone paradisi*	E: Asian Paradise flycatcher S: Asia rahanmara, Redi hora	Migrant	LC	NA
51	Paridae	*Parus cinereus*	E: Great Tit S: Maha tikiriththa	Resident	LC	NA
52	Hirundinidae	*Hirundo domicola*	E: Hill Swallow S: Kandu wehilihiniya	Resident	EN	LC
53	Hirundinidae	*Cecropis hyperythra*	E: Sri Lanka Swallow	Endemic	LC	LC

			S: Sri Lanka Lak Lihiniya			
54	Cisticolidae	*Cisticola juncidis*	E: Zitting Cisticola S: Iri Pavansariya	Resident	LC	LC
55	Cisticolidae	*Prinia hodgsonii*	E: Grey-breasted Prinia S: Layalu prinia	Resident	LC	LC
56	Cisticolidae	*Prinia sylvatica*	E: Jungle Prinia S: Vana prinia, Hambu kurulla	Resident	LC	LC
57	Cisticolidae	*Prinia socialis*	E: Ashy Prinia S: Alu prinia	Resident	LC	LC
58	Cisticolidae	*Prinia inornata*	E: Plain Prinia S: Sarala prinia	Resident	LC	LC
59	Cisticolidae	*Orthotomus sutorius*	E: Common Tailorbird S: Battichcha	Resident	LC	LC
60	***Pycnonotidae***	**Pycnonotus melanicterus**	***E: Sri Lanka Black capped Bulbul*** ***S: Kalu isasi kondaya***	Endemic	LC	LC
61	***Pycnonotidae***	**Pycnonotus cafer**	***E: Red-vented Bulbul*** ***S: Kondaya***	Resident	LC	LC
62	***Pycnonotidae***	**Pycnonotus penicillatus**	***E: Sri Lanka Yellow-eared Bulbul*** ***S: Sri Lanka kahakan kondaya;***	Endemic	EN	NT

63	***Pycnonotidae***	**Pycnonotus luteolus**	***E: White-browed Bulbul S: Bamasudu Kondaya***	Resident	LC	LC
64	***Pycnonotidae***	**Iole indica**	***E: Yellow-browed Bulbul S: Bamakaha Kondaya***	Resident	LC	LC
65	***Pycnonotidae***	**Hypsipetes ganeesa**	***E: Square-tailed Bulbul/ Asian Black Bulbul S: Kalu piri-kondaya***	Resident	LC	LC
66	***Pellorneidae***	**Pellorneum fuscocapillus**	***E: Sri Lanka Brown capped Babbler S: Sri Lanka Boraga piridemalichcha***	Endemic	LC	LC
67	***Timaliidae***	**Pomatorhinus melanurus**	***E: Sri Lanka Scimitar babbler S: Sri Lanka de demalichcha***	Endemic	NT	LC
68	***Timaliidae***	**Rhopocichla atriceps**	***E; Dark-fronted Babbler S: Vathaduru pandurudemalichcha***	Resident	LC	LC
69	Leiotrichidae	*Argya rufes*	E: Sri Lanka Orangebilled Babbler S: Sri Lanka rathu - demalichcha	Endemic	EN	NT
70	Leiotrichidae	**Argya affinis**	***E: Yellow-billed Babbler S: Demalichcha***	Resident	LC	LC
71	Zosteropidae	**Zosterops ceylonensis**	***E: Sri Lanka White-eye S: Sri Lanka***	Endemic	VU	LC

			sithasiya, ***Mal kurulla***			
72	Zosteropidae	**Zosterops palpebrosus**	***E: Indian White-eye/ Oriental White-eye*** ***S: Peradigu sithasiya***	Resident	LC	LC
73	***Sittidae***	**Sitta frontalis**	***E: Velvet-fronted Nuthatch*** ***S: Villuda yatikuriththa***	Resident	LC	LC
74	***Sturnidae***	**Gracula indica**	***E: Southern Hill Myna*** ***S: Salalihiniya***	Resident	LC	LC
75	***Sturnidae***	**Acridotheres tristis**	***E: Common Myna*** ***S: Myna***	Resident	LC	LC
76	***Turdidae***	**Geokichla spiloptera**	***E: Sri Lanka Spot-winged Thrush*** ***S: Sri Lanka thithpiya thirasikaya, Wal avichchiya***	Endemic	VU	NT
77	Muscicapidae	*Copsychus saularis*	E: Oriental Magpie-robin S: Polkichcha	Resident	LC	LC
78	Muscicapidae	*Copsychus malabaricus*	E: White-rumped Shama S: Vana Polkichcha	Resident	LC	LC
79	Muscicapidae	*Saxicoloides fulicatus*	E: Indian Robin	Resident	LC	LC

			S: Kalukichcha, Kalu polkichcha			
80	Muscicapidae	*Eumyias sordidus*	E: Sri Lanka Dull-blue Flycatcher S: Sri Lanka anumasimara	Endemic	VU	NT
81	Muscicapidae	*Cyornis tickelliae*	E: Tickell's Blue-flycatcher S: Layaran nil-masimara, Kopi kurulla	Resident	LC	LC
82	Stenostiridae	*Culicicapa ceylonensis*	E: Grey-headed Canary-flycatcher S: Aluhis kaha masimara	Resident	NT	LC
83	Chloropseidae	*Chloropsis jerdoni*	E: Jerdon’s Leafbird/ Blue-winged Leafbird S: Jaradan kolarisiya, Girakurulla	Resident	LC	LC
84	Chloropseidae	*Chloropsis aurifrons*	E: Golden-fronted Leafbird S: Ran nalal kolarisiya	Resident	LC	LC
85	Dicaeidae	*Dicaeum erythrorhynchos*	E: Pale-billed Flowerpecker S: Lathudu Pililichcha	Resident	LC	LC
86	Nectariniidae	*Leptocoma zeylonica*	E: Purple-rumped Sunbird S: Nithamba dam sutikka	Resident	LC	LC
87	Nectariniidae	*Cinnyris asiatica*	E: Purple Sunbird S: Dam sutikka	Resident	LC	LC
88	Motacillidae	*Anthus rufulus*	E: Paddyfield Pipit S: Keth waratichcha	Resident	LC	LC

89	Pittidae	*Pitta brachyura*	E: Indian Pitta S: Avichchiya	Migrant	LC	LC
90	Laniidae	*Lanius cristatus*	E: Brown Shrike S: Bora Sambariththa	Migrant	LC	LC
91	Hirundinidae	*Hirundo rustica*	E: Barn Swallow S: Atu wahilihiniya	Migrant	LC	LC
92	Phylloscopidae	*Phylloscopus nitidus*	E: Green Warbler S:	Migrant	LC	LC
93	Phylloscopidae	*Phylloscopus magnirostris*	E: Large-billed Leafwarbler S: Mathudu Gasraviya	Migrant	LC	LC
94	Muscicapidae	*Muscicapa dauurica*	E: Asian Brown Flycatcher S: Asia duburu masi mara/ Asia bora masi mara	Migrant	LC	LC
95	Muscicapidae	*Muscicapa muttui*	E: Brown-breasted Flycatcher S: Laya bora masimara	Migrant	LC	LC
96	Psittaculidae	*Psittacula calthropae*	E: Sri Lanka Layard's parakeet/ Sri Lanka Emerald collared Parakeet S: Sri Lanka Alu girawa	Endemic	NT	LC
97	Nectariniidae	*Cinnyris lotenius*	E: Long-billed Sunbird S: Dikthudu Sutikka	Resident	LC	LC

APPENDIX 2 – TREE CHECK LIST IDENTIFIED AT ISSENGARD BIOSPHERE RESERVE

No	Family	Species	Sinhala Names	NCS
1	Acanthaceae	*Justicia procumbens*	Maha nai	LC
2	Anacardiaceae	*Nothopegia beddomei*	Bala, Andum Teageddi	LC
3	Anacardiaceae	*Mangifera indica*	Amba	
4	Anacardiaceae	***Semecarpus sp.***	Badulla	VU/LC
5	Annonaceae	*Artabotrys zeylanicus*	Kalu Bambara Wel, Petika Wel, Yakada Wel	LC
6	Apiaceae (Umbelliferae)	*Centella asiatica*	Gotukola, Hin-Gotukola	LC

7	Apocynaceae	*Alstonia macrophylla*	Hawari-Nuga, Yakadamaran, Attoniya, Ginikuru Gas	LC
8	Apocynaceae	*Anodendron paniculatum*	Aswel, Dul, Gerandi-Dul	VU
9	Apocynaceae	*Tabernaemontana dichotoma*	Divi-Kaduru	LC
10	Aquifoliaceae	***Ilex sp.***		EN/LC/NT
11	Araliaceae	***Schefflera emarginata***	Heen Iththe Wel	VU
12	Arecaceae (Palmae)	*Caryota urens*	Kithul	LC
13	Arecaceae (Palmae)	***Oncosperma fasciculatum***	Katu-Kithul	VU
14	Aristolochiaceae	*Aristolochia indica*	Sapsanda, Sas Sanda, Sananda	LC

15	Asteraceae (Compositae)	*Chromolaena odorata*	Podi-Singno-Maran, Lokkan-Nattan	DD
16	Asteraceae (Compositae)	*Mikania cordata*	Gam-Palu, Kehel-Palu, Lokapalu, Maha-Kihimbiya, Wathu-Palu, Demala Wel	
17	Asteraceae (Compositae)	*Psiadia ceylanica*	Pupula	LC
18	Asphodelaceae	*Dianella ensifolia*	Dutu Sathuta	LC
19	Calophyllaceae	*Calophyllum bracteatum*	Walu Keena	NT
20	Calophyllaceae	*Calophyllum cuneifolium*	Keena	CR
21	Celastraceae	***Euonymus walkeri***	(Plane Plant)	LC
22	Celastraceae	***Salacia reticulata***	Kothala Hibatu	EN

23	Clusiaceae(Guttiferae)	*Garcinia morella*	Gokatu, Kanna Goraka, Kokatiya	NT
24	Convolvulaceae	***Argyreia kleiniana sp.***	Girithilla	LC
25	Dioscoreaceae	*Dioscorea pentaphylla*	Katu-Ala, Katuwala-Ala	LC
26	Ebenaceae	***Diospyros sp.***	Kalumediriya,	EN
27	Elaeocarpaceae	***Elaeocarpus glandulifer***	Mal Veralu, Gal Veralu	NT
28	Elaeocarpaceae	*Elaeocarpus serratus*	Weralu	LC
29	Euphorbiaceae	***Agrostistachys borneensis***	Beru	LC
30	Euphorbiaceae	*Macaranga indica*	Kenda	LC
31	Euphorbiaceae	***Mallotus sp.***	Hamparilla	LC
32	Euphorbiaceae	*Antidesma alexiteria*	Hin-Embilla, Heen-Embilla	LC
33	Euphorbiaceae	*Antidesma bunius*	Karawala-Kebella	LC
34	Euphorbiaceae	*Bridelia retusa*	Keta-Kela	LC

35	Euphorbiaceae	*Glochidion sp*	Hunukirilla	
36	Euphorbiaceae	*Phyllanthus sp*		
37	Euphorbiaceae	*Phyllanthus amarus*	Pitawakka	LC
38	Euphorbiaceae	*Phyllanthus emblica*	Nelli	VU
39	Fabaceae (Leguminosae)	*Albizia odoratissima*	Huriya, Suriya-Mara, Huri Mara	LC
40	Fabaceae (Leguminosae)	*Cajanus trinervius*	Wal Kollu (1200m)	LC
41	Fabaceae (Leguminosae)	*Cajanus sp.*	Wal Kollu (1200m)	EN/LC
42	Fabaceae (Leguminosae)	*Chamaecrista sp.*	Bin Siyabala	LC
43	Fabaceae (Leguminosae)	*Crotalaria mysorensis*		CR
44	Fabaceae (Leguminosae)	*Dalbergia pseudo-sissoo*	Bambara-Wel	LC

45	Fabaceae (Leguminosae)	*Derris scandens*	Kala-Wel, Ala-Wel,Bo-Kala-Wel	LC
46	Fabaceae (Leguminosae)	*Desmodium triflorum*	Heen-Undupiyaliya	LC
47	Fabaceae (Leguminosae) - Mimosaceae	*Mimosa pudica*	Nidi-Kumba	LC
48	Lamiaceae (Labiatae)	*Leucas biflora*	Geta-Thumba, Wilandawenna	LC
49	Lamiaceae (Labiatae)	*Platostoma elongatum*		VU
50	Lauraceae	***Actinodaphne stenophylla***	Nika-Daula	VU
51	Lauraceae	*Cinnamomum cassia*	Dawul-Kurundu, Kadu-Dawula, Nika-Dawula, Wal-Kurundu	LC*

52	Lauraceae	***Cinnamomum citriodorum***	Pengiri Kurundu	EN
53	Lauraceae	***Cinnamomum verum***	Kurundu	VU
54	Lauraceae	***Neolitsea cassia***	Dawul Kurudu	LC
55	Lecythidaceae	*Careya arborea*	Kahata	LC
56	Malpighiaceae	*Hiptage benghalensis*	Puwak-Gediya-Wel	LC
57	Malvaceae	*Grewia tiliifolia*	Damina	LC
58	Malvaceae	*Pterospermum suberifolium*	Velan	LC
59	Melastomataceae	***Osbeckia aspera***	Bowitiya	NT
60	Melastomataceae	***Osbeckia octandra***	Heen Bowitiya	LC
61	Melastomataceae	*Memecylon umbellatum*	Kora-Kaha	LC
62	Meliaceae	*Aphanamixis polystachya*	Hingul, Ela-Hirilla (700m)	VU
63	Menispermaceae	*Cyclea peltata*	Kehi-Pittan,	LC

64	Myristicaceae	*Myristica ceylanica*	Malaboda,	LC
65	Myrsinaceae	*Ardisia sp.*		
66	Myrtaceae	***Eugenia sp.1***		
67	Myrtaceae	*Psidium guineense*	Embul Pera	DD
68	Myrtaceae	***Syzygium hemachandrae***		
69	Myrtaceae	*Syzygium gardneri*	Damba, Dambu	LC
70	Myrtaceae	*Syzygium zeylanicum*		LC
71	Ochnaceae	*Campylospermum serratum*	Bo-Kera,	LC
72	Oleaceae	***Chionanthus albidiflorus***	Embul-Korakaha, Gal-Melta, Taccada-Gas	VU
73	Orchidaceae	*Arundina graminifolia*		EN/DD

74	Pandanaceae	*Pandanus odorifer*	Mudu Keyiya, Weta-Keiya	LC
75	Phyllanthaceae	*Antidesma alexiteria*	Heen Abilla	LC
76	Phyllanthaceae	*Bridelia retusa*	Keta-Kela	LC
77	Phyllanthaceae	*Phyllanthus amarus*	Pitawakka	LC
78	Phyllanthaceae	*Phyllanthus emblica*	Nelli	VU
79	Rhamnaceae	*Ziziphus linnaei*	Maha-Debara, Eraminia, Masan, Dabara	NT
80	Rubiaceae	*Canthium coromandelicum*	Kara	LC
81	Rubiaceae	***Diplospora erythrospora***	Gal Seru	VU
82	Rubiaceae	*Hedyotis fruticosa*	Weraniya	LC
83	Rubiaceae	***Wendlandia bicuspidata***	Wana-Idala	LC

84	Rutaceae	*Acronychia pedunculata*	Ankenda	LC
85	Rutaceae	*Chloroxylon swietania*	Burutha	VU
86	Rutaceae	*Clausena indica*	Migon-Karapincha	LC
87	Rutaceae	*Glycosmis angustifolia*	Bol-Pana	NT
88	Rutaceae	*Glycosmis pentaphylla*	Dodan-Pana	LC
89	Rutaceae	*Zanthoxylum asiaticum*	Kudu Mirisa	LC?
90	Sabiaceae	*Meliosma simplicifolia*	Albadda-Gas, Elbedda	NT
91	Sapindaceae	*Dimocarpus longan*	Gal Mora	LC
92	Sapindaceae	*Dodonaea viscosa*		LC
93	Sapindaceae	*Filicium decipiens*	Pihimbiya	LC

94	Sapindaceae	*Madhuca longifolia*	Me	LC
95	Smilacaceae	*Smilax perfoliata*	Maha-Kabarassa	LC
96	Symplocaceae	*Symplocos acuminata*	Bobu,Bombu, Wal Bombu	LC
97	Thymelaeaceae	*Gnidia glauca*	Naha	NT
98	Verbenaceae	*Stachytarpheta jamaicensis*	Balu Nakuta	LC

APPENDIX 3 – AVIAN COMMUNITY DATA SHEET

Avian Community Datasheet

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No	Date	Elevation of the cloud line	Time	Elevation (transect No)	Humidity	Light intensity	Temperature	Wind speed	Wind direction

No	Cloud cover	Visibility	Habitat type	Bird species	Special Notes